

1 Appendix H: Spectral Indicators and Advisory Invariants

1.1 Scope and relation to the Core and audit layer

This appendix develops spectral indicators, advisory invariants, warning policies, and verifier-side re-entry rules for the Technical Extension layer. It does not enlarge the Core or create an external-domain theorem.

The central rule is

Advisory information may guide verification, but it does not determine a Core verdict.

The repaired window-first persistence semantics of Appendix A, the Type IV discipline of Appendix D, the non-compensating status calculus of Appendix F, and the manifest schema of Appendix G remain controlling.

Definition 1.1.1 (Advisory sovereignty principle). Advisory data may guide where, when, and how verifier-side checks are performed. They do not determine, compensate for, repair, or mutate a Core verdict.

Definition 1.1.2 (No external-domain upgrade principle). No spectral indicator, potential, warning, ranking, AI confidence value, advisory band, collapse-energy candidate, or advisory attachment proves an arithmetic, PDE, geometric, categorical, cryptographic, or other external-domain conclusion. Such conclusions require separately proved and registered Interface or Bridge Theorems.

Definition 1.1.3 (v20 dependency baseline). The statements of this appendix are interpreted relative to:

- (i) AK-HPDST v20.0.0 Main Chapters 1–8;
- (ii) the corrected v20 Foundation Appendices A–G;
- (iii) the reviewed CM-v20 and TB-v20 mathematical statements, each only at its individually verified strength;
- (iv) the CR-v20 baseline for global governance and inherited Main/A–G claim use, while recognizing that the baseline does not yet register Technical Extension Appendices H–N.

The final normative CR-v20 is intentionally deferred until every appendix and the Companion are frozen. No joint CM/TB promotion, source-artifact hash synchronization, or final-CR binding is inferred from this mathematical dependency statement, and no provisional identifier is promoted silently.

Definition 1.1.4 (Technical-Extension claim discipline). Every formal statement in this appendix has a stable local label and a declared mathematical reading. No interim Claim Register, promotion receipt, JSON register, or per-claim sidecar is normative at this construction stage. Complete Claim UUIDs and the final Compact CR are generated only after the full v20 Appendix sequence is frozen and cross-audited. The local labels and immutable source artifact preserve the statement inventory until that final extraction. This workflow changes registration timing, not theorem strength.

1.2 Standing repaired Core regime

Fix a field k , work in the constructible one-parameter regime

$$\text{Pers}_k^{\text{cons}} \subset [(\mathbb{R}, \leq), \text{Vect}_k],$$

and use finite-type filtered objects in $\text{FiltCh}(k)$ when a filtered realization is invoked. For monitored degree i , right-open window W , and threshold $\tau > 0$, set

$$B_{i,W}(F) := W_W \mathbf{P}_i(F), \quad \text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(B_{i,W}(F)).$$

Windowing precedes hard deletion. Threshold clearance is measured on $B_{i,W}(F)$ before deletion.

The verifier-side conditions relevant to this appendix include, when invoked,

$$B\text{-Gate}^+, \quad \text{Eval}_{1,W,\tau}(F) = 0, \quad (\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0), \quad \text{val}_B(\mathfrak{d}_{\text{met}}) < \text{gap}_\tau(\mathcal{B}).$$

No advisory expression is a substitute for any of these typed records.

When detector, higher-edge, or defect information is discussed, the source stage is declared explicitly among

`windowed_prethreshold`, `repaired_postthreshold`, `kernel_defect`, `cokernel_defect`.

A finite observation is not a finite proof; nonzero soundness and zero completeness are independent; a higher Ext^n discharge requires a standard n -eligible realization and edge package; terminal Type IV cleanliness is distinct from transient-defect cleanliness; and a finite prefix does not establish a finite-to-infinite reduction.

Definition 1.2.1 (Typed advisory status). Every advisory datum has exactly one lifecycle status among

`proposed`, `computed`, `reproducible`, `stale`, `superseded`, `invalid`, `undefined`.

These are advisory-lifecycle statuses, not Core atomic statuses. In particular, `computed` and `reproducible` do not mean pass.

Proposition 1.2.2 (Unknown or unlabeled advisory output is non-verdict). *An advisory output with missing identity, scope, generator, normalization, status, or provenance is non-verdict data and cannot be consumed by any verifier-side clause.*

Proof. The missing fields prevent typed reconstruction and claim-use preflight. By the v20 weaker-safe rule, the output remains outside verifier-side evidence rather than being silently promoted. $\square$

1.3 Typed advisory objects and invariance discipline

Definition 1.3.1 (Advisory datum). A typed *advisory datum* is a record

$$A_{\text{adv}} = (\text{id}, \text{generator}, \text{inputs}, \text{scope}, \text{codomain}, \text{normalization}, \text{value}, \text{status}, \text{provenance}, \text{artifacts})$$

whose declared purpose is attention, warning, prioritization, navigation, calibration, or rechecking. It is not a verifier-side record merely because its value is mathematically defined.

Definition 1.3.2 (Advisory reading rule). The only default reading is

$$A_{\text{adv}} \rightsquigarrow \text{attention, warning, prioritization, navigation, calibration, or rechecking.}$$

The forbidden default reading is

$$A_{\text{adv}} \rightsquigarrow \text{proof, discharge, pass, reject, obstruction number, or external theorem.}$$

Definition 1.3.3 (Declared invariance relation). Let $\sim_{\mathcal{E}}$ be a declared equivalence or comparison relation on the input scope. An advisory datum is called $\mathcal{E}$ -invariant only when a proof or verified computation establishes

$$X \sim_{\mathcal{E}} Y \implies A_{\text{adv}}(X) = A_{\text{adv}}(Y)$$

with the same normalization, domain, codomain, and versioned generator.

Proposition 1.3.4 (The word invariant does not establish invariance). *Naming a score or profile an “invariant” does not establish invariance under isomorphism, interleaving, representative equivalence, window refinement, threshold change, or realization change.*

Proof. Each listed relation has distinct typed hypotheses. A name supplies none of them, so no invariance conclusion follows without the corresponding proof record. $\square$

1.4 Spectral indicators and typed integration

Definition 1.4.1 (Spectral indicator). A *spectral indicator* is an advisory datum derived from spectral, graded, filtered, barcode, energy-like, or threshold-sweep data. Its record must declare the monitored degree, window, threshold or sweep domain, source profile, normalization, and output codomain.

Definition 1.4.2 (Measurable spectral profile). For a declared filtered object F and window W , a spectral profile is an advisory map

$$\text{Spec}_{F,W} : D \longrightarrow [0, +\infty]$$

on a declared measurable scale domain $D \subseteq (0, +\infty)$. The record states whether the map is Borel measurable, whether it is finite almost everywhere, and whether the intended band integral is finite, extended-valued, or undefined.

Definition 1.4.3 (Band-limited spectral indicator). Let $B = [r_0, r_1] \subset D$. If $\text{Spec}_{F,W}$ is measurable on B , define the extended band indicator

$$\text{Sl}_B(F, W) := \int_B \text{Spec}_{F,W}(r) dr \in [0, +\infty].$$

A finite-valued claim additionally requires integrability. For a discrete declared mesh $\{r_j\}$, define

$$\text{Sl}_B^{\text{disc}}(F, W) := \sum_{r_j \in B} \text{Spec}_{F,W}(r_j) \Delta r_j$$

only when the weighting and convergence policy are declared. Otherwise the indicator status is `undefined`.

Definition 1.4.4 (Repaired-profile spectral indicator). A spectral indicator is *repaired-profile based* when its source is exactly

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F))$$

or a declared family of such profiles. This phrase does not make the indicator a Core evaluation or gate clause.

Definition 1.4.5 (Comparison-calibrated advisory family). A family of advisory values is comparison-calibrated only when all compared records share, or are connected by a proved transport of, the same input type, scale, normalization, monitored scope, window convention, threshold convention, generator version, and output semantics.

Proposition 1.4.6 (Uncalibrated advisory values are not comparison evidence). *Two advisory values computed under different undeclared normalizations, scopes, windows, thresholds, generator versions, or codomains cannot be used as evidence that the underlying Core-readable objects are equal, ordered, interleaved, or close.*

Proof. A numerical comparison without a common typed semantics does not determine a relation between the source objects. The required comparison record is absent, so the stronger conclusion is unavailable. $\square$

Definition 1.4.7 (Sampled advisory band). A sampled advisory band is a threshold interval J together with advisory evaluations at a declared finite or countable sample set $S \subset J$. It records only the observed sample behavior unless an interpolation, monotonicity, continuity, or finite-reduction theorem is supplied.

Proposition 1.4.8 (Sample stability is not band certification). *Stability of an advisory indicator on $S \subset J$ does not imply advisory stability, threshold-gap safety, or Type IV cleanliness at every point of J unless a separate theorem reduces the continuum claim to the sampled data.*

Proof. Unsampled parameters may contain discontinuities, threshold crossings, or diagnostic changes. A finite or countable sample alone supplies no universal quantifier over J . $\square$

1.5 Canonical finite-bar spectral tail calculus

Definition 1.5.1 (Finite-length spectral measure and tail data). Let B be a finite barcode profile. Write $\mathcal{B}_{\text{fin}}(B)$ for its finite bars and let $n_{\infty}(B)$ be the number of infinite bars, counted with multiplicity. For $r \in [0, +\infty)$, define the finite-length spectral measure

$$\Lambda_B := \sum_{I \in \mathcal{B}_{\text{fin}}(B)} \delta_{\ell(I)},$$

the strict tail count

$$N_B(r) := \Lambda_B((r, +\infty)) = \#\{I \in \mathcal{B}_{\text{fin}}(B) : \ell(I) > r\},$$

and the finite excess-mass function

$$E_B(r) := \int_{(0, +\infty)} (x - r)_+ d\Lambda_B(x) = \sum_{I \in \mathcal{B}_{\text{fin}}(B)} (\ell(I) - r)_+.$$

Infinite bars are recorded separately by $n_{\infty}(B)$; they are not assigned an arbitrary finite mass.

Theorem 1.5.2 (Exact spectral-tail identities). *For every finite barcode profile B :*

- (i) $(\Lambda_B, n_{\infty}(B))$ is invariant under barcode isomorphism and permutation of interval summands;
- (ii) N_B is integer-valued, nonincreasing, right-continuous, and changes only at finite bar lengths;
- (iii) E_B is finite-valued, nonincreasing, convex, continuous, and piecewise affine;
- (iv) the layer-cake identity holds:

$$E_B(r) = \int_r^{+\infty} N_B(s) ds;$$

- (v) away from the finite bar lengths, $E'_B(r) = -N_B(r)$.

Proof. All assertions follow from the finite sum of Dirac measures. The layer-cake identity is the finite-sum identity $(\ell - r)_+ = \int_r^{+\infty} \mathbf{1}_{\{s < \ell\}} ds$, summed over finite bars. Convexity and the derivative statement follow termwise from $r \mapsto (\ell - r)_+$. $\square$

Theorem 1.5.3 (Threshold action on spectral-tail data). *For every finite barcode profile B , threshold $\tau > 0$, and $r \geq 0$,*

$$n_{\infty}(T_{\tau}^{\text{bar}} B) = n_{\infty}(B), \quad N_{T_{\tau}^{\text{bar}} B}(r) = N_B(\max\{r, \tau\}),$$

and

$$E_{T_{\tau}^{\text{bar}} B}(r) = E_B(\max\{r, \tau\}) + (\tau - r)_+ N_B(\tau).$$

Thus the entire finite-length spectral tail of the repaired profile is determined exactly by the pre-threshold tail data and the threshold.

Proof. Hard deletion retains precisely the finite bars with length greater than τ and retains every infinite bar. The tail-count identity follows by imposing simultaneously $\ell > \tau$ and $\ell > r$. If $r \geq \tau$, the excess mass is $E_B(r)$. If $r < \tau$, every retained finite bar contributes $(\ell - r) = (\ell - \tau) + (\tau - r)$, so summing over the $N_B(\tau)$ retained finite bars gives the displayed formula. $\square$

Theorem 1.5.4 (Information content and incompleteness of the length spectrum). *For finite barcode profiles:*

- (i) Λ , N , E , and n_{∞} are additive under barcode direct sum;
- (ii) any one of Λ_B , the jump data of N_B , or the slope-jump data of E_B , together with $n_{\infty}(B)$, determines the multiset of finite bar lengths and the number of infinite bars;

(iii) these data do not determine the barcode up to isomorphism, because birth and death locations are forgotten.

For example, $\{[0, 1)\}$ and $\{[2, 3)\}$ have identical length-spectrum data but are not isomorphic as persistence modules.

Proof. Additivity is immediate from disjoint union of interval multisets. The atoms of Λ_B , equivalently the downward jumps of N_B or the upward jumps of the derivative of E_B , recover every finite length with multiplicity. The example has the same single length but different support, so no persistence-module isomorphism exists. $\square$

Corollary 1.5.5 (Exact threshold readout). *For every finite profile B and $\tau > 0$,*

$$\#\mathcal{B}_{\text{fin}}(T_{\tau}^{\text{bar}} B) = N_B(\tau), \quad \#\mathcal{B}_{\infty}(T_{\tau}^{\text{bar}} B) = n_{\infty}(B).$$

Consequently,

$$T_{\tau}^{\text{bar}} B = 0 \iff N_B(\tau) = 0 \text{ and } n_{\infty}(B) = 0.$$

Within Appendix H these quantities remain advisory until an exact verifier-side record is constructed; the displayed equivalence licenses such a re-entry but does not perform it automatically.

Proof. Set $r = 0$ in Theorem 1.5.3 for the retained finite count, and use the infinite-bar identity there. The zero criterion follows because a finite barcode is zero exactly when it has no finite or infinite bars. $\square$

Theorem 1.5.6 (Threshold-gap transport of retained count and excess mass). *Let B, C be finite barcode profiles with $d_B(B, C) \leq \varepsilon < +\infty$. Assume that no finite bar of either profile has length in the closed band $[\tau - \varepsilon, \tau + \varepsilon]$ and that $2\varepsilon \leq \tau$. Then*

$$n_{\infty}(B) = n_{\infty}(C), \quad N_B(\tau) = N_C(\tau) =: m,$$

and

$$|E_B(\tau) - E_C(\tau)| \leq 2m\varepsilon.$$

The operational form uses the strict reserves $\varepsilon < \text{gap}_{\tau}(B, C)$ and $2\varepsilon < \tau$.

Proof. Because the barcodes are finite and the bottleneck distance is finite, an optimal matching of cost at most ε exists. Infinite bars must match infinite bars. A retained finite bar cannot match the diagonal, because diagonal cost at most ε would force its length at most $2\varepsilon \leq \tau$. The two-sided threshold-gap hypothesis prevents a matched finite bar from crossing the threshold, so the retained finite bars are paired bijectively. Matched finite lengths differ by at most 2ε ; summing the differences of $(\ell - \tau)_+$ over the m retained pairs gives the bound. $\square$

Corollary 1.5.7 (Metric-safe spectral zero predicate). *Under the hypotheses of Theorem 1.5.6,*

$$T_{\tau}^{\text{bar}} B = 0 \iff T_{\tau}^{\text{bar}} C = 0.$$

Only this zero predicate and the displayed count/mass bounds are transported; raw sample ranks, birth sets, germ sets, certificate size, and arbitrary advisory normalizations are not.

Proof. Apply Corollary 1.5.5 and Theorem 1.5.6. $\square$

Proposition 1.5.8 (No gap-free stability of the tail count). *At a fixed threshold τ , the map $B \mapsto N_B(\tau)$ is not locally constant in the bottleneck metric without a threshold-gap hypothesis.*

Proof. For arbitrarily small $\eta > 0$, the one-bar profiles $\{[0, \tau + \eta)\}$ and $\{[0, \tau - \eta)\}$ are at bottleneck distance at most 2η , while their strict tail counts at τ are 1 and 0. $\square$

1.6 Advisory noninterference and non-compensation

Definition 1.6.1 (Verifier-side projection). For a complete certificate record C , let $\pi_{\text{ver}}(C)$ denote the sub-record containing only verifier-side objects, obligations, statuses, artifacts, ledgers, comparison records, diagnostics, and gate verdicts. Advisory attachments are omitted.

Theorem 1.6.2 (Advisory noninterference). *Let C and C' be well-typed certificate records satisfying*

$$\pi_{\text{ver}}(C) = \pi_{\text{ver}}(C').$$

Then their Core atomic status and failure route are identical, regardless of differences in advisory attachments.

Proof. By Core sovereignty and Appendix G semantics, the verdict function factors through the verifier-side projection. Equal verifier-side projections therefore produce equal verdicts and failure routes. Advisory fields are outside the function's input. $\square$

Corollary 1.6.3 (Attachment insertion and removal are verdict-neutral). *Adding, deleting, reordering, or recomputing advisory attachments without changing π_{ver} cannot turn *undefined* or *reject* into *pass*, nor can it turn *pass* into *reject*.*

Theorem 1.6.4 (Advisory non-compensation). *No finite or infinite collection of favorable advisory values can compensate for a failed or undefined required verifier-side obligation. In particular, advisory stability, low potential, high confidence, or zero advisory energy cannot repair missing eligibility, a missing tower artifact, failed threshold-gap evidence, a nonzero diagnostic, or a failed budget inequality.*

Proof. The v20 readiness calculus is conjunctive on required obligations and forbids pass by cancellation. Advisory values are not members of the required verifier-side obligation family unless separately reconstructed as new verified records. Hence they cannot compensate for a blocker. $\square$

1.7 Verifier-side re-entry calculus

Definition 1.7.1 (Re-entry candidate). A re-entry candidate is a proposal to use advisory input in the construction of a new verifier-side record. The advisory datum itself does not change type. The candidate specifies:

- (i) the target verifier-side record type;
- (ii) exact source and target scope;
- (iii) a theorem, verified computation, or artifact-backed construction;
- (iv) all hypotheses and non-scalar obligations;
- (v) the resulting immutable record identifier;
- (vi) one status among candidate, verified, rejected, or undefined.

Definition 1.7.2 (Re-entry witness). A re-entry witness is the theorem or artifact package proving that the newly constructed record satisfies the target verifier-side schema and semantics. A witness must identify its authority scope, environment, inputs, outputs, artifact hashes, and failure behavior.

Theorem 1.7.3 (No declaration-only re-entry). *Declaring an advisory datum to be verifier-side, renaming it as a gate, diagnostic, representative, or ledger component, or copying it into a verifier-side field does not establish a valid re-entry.*

Proof. A verifier-side record is typed by its mathematical construction, hypotheses, proof or artifact evidence, and schema conformance. A declaration or field move supplies none of these. Therefore the target record remains undefined until a re-entry witness is verified. $\square$

Proposition 1.7.4 (Re-entry is immutable and non-mutating). *A successful re-entry creates a new content-addressed verifier-side record linked to the advisory source by provenance. It does not mutate the source advisory datum or retroactively alter a prior verdict.*

Proof. Content-addressed identity and the no-retroactive-mutation rule require new records for changed types or evidence. Provenance links preserve ancestry without identifying the two semantic objects. $\square$

1.8 v20 detector, higher-edge, defect, and reduction re-entry routes

Definition 1.8.1 (Typed re-entry ladder). A valid v20 re-entry follows the ordered ladder

$$\begin{aligned} \text{advisory source} &\longrightarrow \text{typed candidate} \longrightarrow \text{verified construction} \longrightarrow \\ &\text{new verifier-side record} \longrightarrow \text{claim-use preflight} \end{aligned}$$

Skipping any arrow leaves the target verifier-side value undefined.

Theorem 1.8.2 (Advisory stencil to detector-certificate re-entry). *Let an advisory process propose a finite stencil T for threshold τ . A complete detector re-entry requires a new record that verifies the exact source stage, endpoint convention, monitored family, immutable input profile, and effective start domain*

$$W_\tau := \{t \in W : t + \tau \in W\}.$$

The record must also verify either exhaustive birth/germ data or retained-bar coverage. In the family-mesh route on a bounded effective domain, one must have

$$T \subseteq W_\tau, \quad h_{W_\tau}(T) < e_{\tau, W}(\mathcal{F}),$$

where the retained-bar excess lower bound $e_{\tau, W}(\mathcal{F}) > 0$ is independently proved and $h_{W_\tau}(T)$ is the augmented mesh. The newly constructed detector certificate, not the advisory proposal, may then discharge the registered detector clause.

Proof. Finite-stencil nonzero soundness requires only a valid source-stage structure map. Zero completeness additionally requires exact coverage of every retained bar’s active-start region. Exhaustive birth/germ data provide this directly; the strict family-mesh inequality provides it uniformly. The re-entry witness supplies these typed hypotheses and creates a distinct verifier-side artifact. $\square$

Proposition 1.8.3 (Translation-family no-go for finite advisory stencils). *Let $W = [a, +\infty)$, fix $\tau, e > 0$, and consider the single-bar translation family*

$$\mathcal{U}_{a, \tau, e} = \{[p, p + \tau + e) : p \geq a\}.$$

For every finite stencil $T \subseteq W_\tau$, there exists $B \in \mathcal{U}_{a, \tau, e}$ such that the stencil detector is zero while $T_\tau^{\text{bar}} B \neq 0$. Hence no finite stencil is uniformly complete on this family. Any broader no-go claim requires an explicit hypothesis ensuring the presence of such arbitrarily translated undetected profiles.

Proof. Every finite stencil is bounded above. Choose p greater than every sample point. The retained interval has active-start region $[p, p + e)$, disjoint from T , so every sampled structure-map image is zero although the bar length $\tau + e$ survives hard deletion. $\square$

Theorem 1.8.4 (Higher-Ext advisory proxy boundary). *No spectral statistic, scalar energy, detector ranking, or synthetic shifted vector space establishes a higher Extⁿ obstruction or vanishing clause. A valid re-entry requires a standard n -eligible realization, an artifact-backed identification of H^{-n} with the declared repaired edge, and the canonical field-coefficient edge. An equivalence with repaired-profile zero additionally requires a separately proved natural zero-reflection theorem on the registered class.*

Proof. The advisory quantity does not supply the realization, edge, naturality, representative compatibility, or zero-reflection arrows. These are independent non-scalar obligations. $\square$

Theorem 1.8.5 (Transient-defect proxy boundary). *An advisory proxy cannot classify a defect as terminal/cofinal, long transient, short transient, or absent. Such a classification requires the actual comparison morphism, its kernel and cokernel profiles, terminal-tameness on the declared terminal scope, and stage-specific complete defect detectors. Terminal diagnostic zero alone does not imply full isomorphism.*

Proof. Terminal diagnostics see cofinal summands, whereas transient detectors see retained noncofinal summands and a separate short-defect theorem is needed for the deleted remainder. A proxy contains none of these data by default. $\square$

Theorem 1.8.6 (Finite advisory trend is not finite-to-infinite reduction). *No finite or countable list of favorable advisory values, repeated zero samples, apparent convergence, or model-extrapolated tail establishes a colimit statement, eventual constancy, apex agreement, or Return Theorem. A finite-to-infinite conclusion requires a separately proved reduction package, such as exact eventual constancy together with compatible apex agreement, or another theorem of equal declared strength.*

Proof. Two directed systems can agree on every prescribed finite prefix and diverge immediately afterward. Advisory extrapolation does not provide the missing universal tail and comparison morphisms. $\square$

1.9 Budget conversion and metric safety

Definition 1.9.1 (Certified advisory-to-budget conversion). Let $(V, \oplus, 0_V, \preceq)$ be the declared metric-ledger quantale. A certified advisory-to-budget conversion consists of a new element $\delta_{\text{adv}} \in V$, together with:

- (i) immutable identifiers for the actual pre-threshold profiles B, C ;
- (ii) a scope-matched conversion theorem or verified computation;
- (iii) a declared scalarization $\text{val}_B : V \rightarrow [0, +\infty]$;
- (iv) reconstructible evidence that

$$d_B(B, C) \leq \text{val}_B(\delta_{\text{adv}});$$

- (v) all crop, window-edge, normalization, and approximation contributions required by the bound;
- (vi) inclusion of δ_{adv} in the declared metric ledger.

A numerical score-to-ledger map without item (iv) is advisory calibration, not certified budget conversion.

Theorem 1.9.2 (Soundness boundary for budget conversion). *A budget-converted advisory quantity may affect the Core budget condition only through the newly certified ledger element δ_{adv} and only on the scope for which the discrepancy bound is proved. The original advisory value remains non-verdict data.*

Proof. The Core budget consumes typed ledger elements whose scalarizations bound actual discrepancies. Definition 1.9.1 supplies such an element and evidence. No other semantic authority is transferred from the advisory source. $\square$

Proposition 1.9.3 (Zero advisory score does not imply zero metric defect). *The equality $A_{\text{adv}} = 0$ does not imply $\delta_{\text{adv}} = 0_V$, $d_B(B, C) = 0$, exact comparison, or positive threshold reserve unless a theorem proves the relevant implication.*

Proof. The advisory codomain and the metric-ledger quantale are distinct typed spaces. Equality to zero in the first has no default transport to the second. $\square$

1.10 Gate-adjacent quantities and warning semantics

Definition 1.10.1 (Gate-adjacent advisory quantity). An advisory datum is gate-adjacent when it is designed to correlate with possible residual persistence, categorical obstruction, Type IV defect, comparison instability, or budget sensitivity. Gate adjacency is a purpose label, not a logical relation.

Proposition 1.10.2 (Gate-adjacent is not gate-equivalent). *A gate-adjacent advisory datum is not equivalent to the corresponding $B\text{-Gate}^+$ clause unless an explicit theorem proves the equivalence on the declared scope and a valid re-entry witness constructs the verifier-side record.*

Proof. Correlation or predictive usefulness is weaker than logical equivalence. The additional theorem and re-entry witness are exactly the missing implications and typed construction. $\square$

Definition 1.10.3 (Spectral warning policy). A warning policy declares the indicator, scope, trigger condition, recommended response, severity semantics, false-positive/false-negative interpretation when known, and explicit statement that the warning is not a Core verdict.

Proposition 1.10.4 (Warning triggers inspection, not verdict). *A warning may trigger recomputation, refinement, resampling, representative review, tower-artifact review, threshold-gap verification, or escalation to formal proof. It cannot itself assign *pass*, *reject*, or a numbered failure type.*

Proof. Warnings are proposer-side routing signals. Atomic status and failure typing require the corresponding verifier-side object and evidence. $\square$

1.11 Type IV advisory proxies and stable bands

Definition 1.11.1 (Type IV advisory proxy). A Type IV advisory proxy is an advisory datum intended to correlate with a possible terminal-generic kernel or cokernel defect. It does not itself define either defect profile.

Definition 1.11.2 (Actual Type IV diagnostic). For an invoked monitored triple (i, W, τ) , an actual persistence morphism

$$\phi_{i,W,\tau} : M_{i,W,\tau} \longrightarrow N_{i,W,\tau}$$

is required. If source, target, kernel, and cokernel are terminal-tame on the same declared terminal scope, define

$$\mu_{i,W,\tau} := \text{tgdim}_W(\ker \phi_{i,W,\tau}), \quad \nu_{i,W,\tau} := \text{tgdim}_W(\text{coker } \phi_{i,W,\tau}).$$

The specialized Type IV status is exactly one of

not_invoked, undefined, certified_zero, obstructed.

No numerical pair is imputed in the first two cases.

Proposition 1.11.3 (Bounded-window cofinality caution). *Finite-shift triviality does not imply terminal-generic cleanliness on an arbitrary bounded window. For $W = [a, b)$, the interval profile $k_{[a,b)}$ is cofinal relative to W , yet after zero extension outside W its translation morphism is zero for every shift $\epsilon \geq b - a$. Therefore the finite-trivial-defect implication may be consumed only on a right-unbounded terminal scope, or under a separate theorem excluding bounded-window cofinal bars.*

Proof. The interval reaches the right boundary of W , so it contributes one cofinal summand. A shift at least the window length moves every source parameter outside the support, making the translation map zero. Thus finite-shift triviality and bounded-window cofinality can coexist. $\square$

Theorem 1.11.4 (Type IV proxy is not Type IV diagnosis). *A Type IV advisory proxy does not determine $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ or $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) \neq (0, 0)$. Such a determination requires the actual comparison morphism, terminal-tameness evidence, defect profiles, and terminal-generic computation from Definition 1.11.2, or a separately proved exact transport theorem to those data.*

Proof. The Type IV diagnostic is defined from the kernel and cokernel of a typed morphism. A proxy provides neither the morphism nor those defects by default. Therefore no diagnostic equality follows. $\square$

Proposition 1.11.5 (Missing Type IV artifact cannot be repaired by a proxy). *If Type IV is invoked but $\phi_{i,w,\tau}$ or terminal-tameness evidence is missing, the Type IV clause is *undefined*. No proxy value, including zero, changes that status.*

Proof. The missing artifact is a non-scalar proof obligation. Appendix F and the Type IV contract prohibit scalar or advisory manufacture of missing evidence. $\square$

Definition 1.11.6 (Advisory and Core stable bands). An advisory stable band is an interval on which declared advisory indicators satisfy an advisory stability policy. A Core Type IV stable band is an interval on which actual monitored Type IV diagnostics are certified zero throughout the band, supported by a universal artifact family or a theorem reducing the band claim to finite data.

Theorem 1.11.7 (Advisory stable band is not Core stable band). *An advisory stable band, including one supported by dense or extensive sampling, is not a Core Type IV stable band unless the actual diagnostic vanishing is verified throughout the band by the required artifact or reduction theorem.*

Proof. Advisory stability and diagnostic vanishing are different predicates on different typed data. Sampling does not supply the missing universal diagnostic statement without a reduction theorem. $\square$

1.12 Navigation quantities, AI outputs, and reproducibility

Definition 1.12.1 (Navigation potential). A navigation potential is an advisory map

$$\Phi : \mathcal{X} \longrightarrow \mathbb{R} \cup \{+\infty\}$$

on a declared search or parameter space, used only to rank, prioritize, or route candidate regions. Its orientation, normalization, tie policy, and incomparable cases must be declared.

Theorem 1.12.2 (AI confidence, ranking, and potential are non-verdict). *No AI confidence value, model probability, ranking, recommendation, low value of Φ , high value of Φ , or ensemble agreement establishes mathematical validity, verifier-side readiness, or Core status.*

Proof. These quantities are proposer-side estimates or orderings. They are not repaired evaluations, comparison records, representatives, eligibility records, Type IV artifacts, ledger bounds, or gate verdicts. $\square$

Definition 1.12.3 (Reproducible advisory computation). An advisory computation is reproducible only when the record supplies immutable inputs, generator or model identity, version and configuration, randomness and seed policy when applicable, normalization, environment, output artifact, and replay procedure. Reproducibility establishes reconstruction of the declared output, not mathematical truth of an interpretation.

Proposition 1.12.4 (Stale advisory outputs are non-consumable). *If any content-addressed input, generator version, normalization, or declared scope changes, the prior advisory output is *stale* for the changed run and cannot be silently reused as current guidance or re-entry input.*

Proof. The changed dependency produces a different computation identity. Reuse without a new record would violate artifact identity and repair ancestry. $\square$

1.13 Advisory attachments and manifest schema

Definition 1.13.1 (Advisory attachment record). An advisory attachment to a certificate record contains at least:

- (i) attachment ID and parent certificate ID;
- (ii) advisory datum ID and lifecycle status;
- (iii) input scope, monitored degree, window, threshold or sweep domain;
- (iv) generator, normalization, codomain, and value;
- (v) artifact and provenance references;
- (vi) advisory interpretation and prohibited stronger reading;
- (vii) any re-entry candidate ID, kept distinct from the attachment;
- (viii) an explicit `non_verdict` flag unless a separate verified re-entry record exists.

Definition 1.13.2 (Deferred final-CR policy). The mathematical appendix body is the current source of theorem statements, proofs, and stable local labels. No interim Claim Register, JSON register, per-claim card register, or promotion sidecar is emitted for Appendix H. The complete human-readable Compact CR and any final machine-readable projection are generated once, after all v20 Appendices are frozen and cross-audited. Final projections must be derived from the frozen source artifacts and must agree on UID, source binding, typing, dependencies, comparison policy, permitted use, forbidden reading, failure routes, ancestry, and review fields.

Proposition 1.13.3 (Advisory attachments are non-verdict by default). *An advisory attachment does not affect pass, reject, undefined, or failure type unless a separate verified re-entry record changes the verifier-side projection.*

Proof. This is the attachment form of advisory noninterference, Theorem 1.6.2. □

1.14 Optional quantitative collapse indicators

Definition 1.14.1 (Residual bar-mass indicator). For a finite repaired profile $\text{Eval}_{i,W,\tau}(F)$, a residual bar-mass indicator is an advisory statistic of the surviving barcode, such as a declared count, total finite length, weighted length, moment, or distributional functional. Infinite bars and divergent sums require an explicit convention; absent one, the value is `undefined`.

Definition 1.14.2 (Collapse-energy candidate). A collapse-energy candidate is an advisory functional

$$\mathcal{E}_{i,W,\tau}^{\text{adv}}(F)$$

with declared domain, normalization, range, treatment of infinite bars, and threshold-sweep semantics. The term “energy” does not imply positivity, coercivity, lower semicontinuity, Lyapunov monotonicity, or equivalence to a Core obstruction unless separately proved.

Theorem 1.14.3 (Collapse energy is not B-Gate⁺). *No collapse-energy candidate replaces the exact repaired topological clause, eligible categorical clause, Type IV artifact and diagnostic clause, comparison-mode obligations, or strict metric budget condition of B-Gate⁺.*

Proof. B-Gate⁺ is a typed conjunction of verifier-side obligations. A single advisory scalar is not definitionally equal to that conjunction, and no such equivalence is asserted here. □

1.15 Appendix H package theorem

Theorem 1.15.1 (Appendix H v20 advisory and spectral-tail package). *Within the Technical Extension layer, the following package is available.*

- (i) *Advisory objects are typed, versioned, and non-verdict by default.*
- (ii) *The term invariant requires a declared invariance theorem.*
- (iii) *Band integrals require measurable profiles and declared finite or extended-value semantics.*
- (iv) *The finite-length spectral measure, tail count, and excess mass satisfy the exact layer-cake, threshold-action, additivity, and information-content calculus of Theorems 1.5.2–1.5.4; the length spectrum is not a complete barcode invariant.*
- (v) *Under the two-sided threshold-gap and zero-separation hypotheses, retained counts and excess mass admit the transport bounds of Theorem 1.5.6; without a gap, tail count is unstable.*
- (vi) *Uncalibrated advisory values are not comparison evidence, and sample stability does not certify a continuum band.*
- (vii) *Advisory attachments do not alter a verdict when the verifier-side projection is fixed.*
- (viii) *Advisory values cannot compensate for failed or undefined required obligations.*
- (ix) *Re-entry constructs a new verifier-side record through the typed ladder and requires a theorem or artifact witness.*
- (x) *Advisory stencils become detector certificates only after stage, endpoint, coverage, family, and effective-domain verification; the explicit translation family of Proposition 1.8.3 satisfies the finite-stencil no-go.*
- (xi) *Higher-Ext, transient-defect, and finite-to-infinite conclusions require their complete v20 verifier-side packages and cannot be inferred from proxies or trends.*
- (xii) *Budget conversion requires an actual scope-matched discrepancy upper bound.*
- (xiii) *Gate-adjacent quantities and warning flags remain non-verdict.*
- (xiv) *Type IV proxies do not replace an actual comparison morphism or terminal-generic diagnostics; finite-shift triviality requires the right-unbounded terminal-scope restriction.*
- (xv) *Advisory stable bands do not become Core stable bands without universal diagnostic evidence.*
- (xvi) *AI confidence, ranking, and navigation potentials are proposer-side only.*
- (xvii) *Reproducibility proves replay of an output, not mathematical truth.*
- (xviii) *Residual mass and collapse-energy candidates remain advisory unless separately promoted by theorem-backed re-entry.*
- (xix) *Final CR generation is deferred until the complete Appendix sequence is frozen; no interim JSON or per-claim register is emitted for this Appendix.*

Proof. Typing and invariance are supplied by Definitions 1.2.1–1.4.5. The exact spectral calculus and metric-safe transport are Theorems 1.5.2–1.5.4 and 1.5.6. Noninterference and non-compensation are Theorems 1.6.2 and 1.6.4. Re-entry is governed by Theorems 1.7.3, 1.8.2, 1.8.4, 1.8.5, and 1.8.6. Metric conversion is controlled by Theorem 1.9.2. Type IV and stable-band separation follow from Proposition 1.11.3 and Theorems 1.11.4 and 1.11.7. AI and reproducibility boundaries follow from Theorem 1.12.2 and Proposition 1.12.4. □

1.16 Explicit exclusions

The following are not asserted.

- No advisory indicator is a Core gate, gate verdict, or failure type.
- No advisory score replaces $\text{Eval}_{1,W,\tau}(F) = 0$, an eligible Ext^1 -clause, an actual Type IV artifact, or the strict metric budget condition.
- No declaration, field rename, confidence score, or favorable ranking performs verifier-side re-entry.
- No zero advisory value implies zero metric defect, exact equality, Type IV cleanliness, or positive threshold reserve.
- No sampled threshold sweep proves a whole-band statement without a reduction theorem.
- No exact finite spectral-tail identity is automatically a verifier-side certificate; re-entry and claim-use preflight remain required.
- No fixed-threshold tail count is treated as bottleneck-stable without the two-sided threshold gap and the declared zero-separation condition.
- No advisory stencil is a complete detector without stage-specific coverage or family-mesh certification.
- No finite stencil is uniformly complete on the explicit right-unbounded translation family of Proposition 1.8.3; broader no-go statements require their own family hypotheses.
- No spectral statistic or synthetic duality supplies a standard n -eligible higher Ext^n realization or natural zero-reflection theorem.
- No terminal diagnostic zero excludes transient or short defects without the corresponding complete detector and short-defect theorem.
- No favorable finite advisory trend proves eventual constancy, colimit agreement, apex agreement, or a Return Theorem.
- No Type IV proxy is the actual computation of $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}})$.
- No finite-shift-triviality argument is transported from a right-unbounded terminal scope to a bounded window without a separate theorem excluding window-cofinal bars.
- No numerical pair is assigned when Type IV is `not_invoked` or `undefined`.
- No normative Latin u is used for the Type IV cokernel diagnostic; ν is canonical.
- No global exactness or global non-expansiveness of hard positive-threshold deletion is assumed.
- No global strict filtered collapse functor is assumed.
- No AI-generated output, human preference, or platform ranking changes a verifier-side verdict.
- No technical-extension theorem silently enlarges the Core.

1.17 Provenance and status

Definition 1.17.1 (Artifact closure). The mathematical appendix body, compiled PDF, audit report, correction patch, package-contents list, and content-hash list formed the Appendix H construction package. After the Main cross-audit, the authoritative Part III release binding is the integrated H–N TeX/PDF package and its checksum sidecar; the earlier Appendix-H package remains immutable repair ancestry. No JSON output is required at this stage, and artifact closure is not a substitute for mathematical proof. Final CR promotion is postponed until the complete v20 appendix and Companion sequence is frozen.

This appendix depends on:

- (i) Chapters 1–8, especially the v20 finite-certificate, higher-edge, Type IV/transient, reduction, and governance boundaries;
- (ii) Appendix A for repaired threshold-gap and repaired-zero transport;
- (iii) Appendices B and C for detector certificates, representatives, and higher-edge realization discipline;
- (iv) Appendices D, F, and G for actual Type IV artifacts, non-compensating status semantics, and manifest/proof-object requirements;
- (v) CM-v20 for reusable micro-theorems governing detector, metric, higher-edge, defect, Restart, and reduction clauses;
- (vi) TB-v20 only for declared toy counterexamples and boundary demonstrations.

Its role is to provide the advisory, spectral-tail, and verifier-side re-entry discipline without enlarging the Core or asserting an external-domain theorem.

New v20 mathematical results. The exact spectral-tail calculus, threshold action, length-spectrum information and incompleteness theorem, threshold-gap transport of retained counts and excess mass, the gap-free instability counterexample, bounded-window cofinality caution, and precisely scoped re-entry/no-go results are proved in this appendix at their displayed strengths.

Construction governance. Stable local labels are retained in the frozen TeX source. Complete Claim UUIDs and final CR projections are assigned only after all v20 Appendices have been completed and cross-audited. No interim JSON or per-claim sidecar is required.

Explicit exclusions: Section 1.16.

Final status. Document role: Technical Extension appendix. Evidence status: mixed proved internal mathematics, conditional theorem use where hypotheses are displayed, and operational governance. Formal final-CR promotion remains pending until the complete Technical Extension and downstream Appendix sequence is frozen.

2 Appendix I: Grid-to-Continuum and Discretization Stability

2.1 Scope, v20 dependency boundary, and release discipline

This appendix governs grids, meshes, samples, numerical fields, truncations, verified enclosures, and discretized observables when they approximate Core-readable persistence data. It does not assert that a discrete computation is automatically a continuum theorem. Its v20 strengthening adds same-domain filtration perturbation, sharp repaired-zero transfer, detector and higher-Ext transfer chains, morphism-level terminal/transient defect control, countable diagonal refinement, and exact finite-to-infinite reduction.

The central rule is

Discretized evidence enters a target claim only through a typed realization, a certified transfer, and a complete error ledger.

Appendix A controls threshold-gap transfer, Appendix G controls audit-readable artifacts, and Appendix H controls advisory noninterference and theorem-backed re-entry.

Definition 2.1.1 (Discretization sovereignty principle). Discretized data may propose, approximate, or support verifier-side records. They do not determine, compensate for, mutate, or retroactively repair a Core verdict unless a separately verified construction produces the required verifier-side record.

Definition 2.1.2 (No computation-only continuum upgrade principle). A grid, mesh, sample, solver output, computed barcode, refinement plot, or numerical confidence value does not prove a statement about a declared continuum target merely by being accurate-looking, stable-looking, or reproducible. A continuum conclusion requires an identified target, a target realization, and an exact or metric-certified transfer witness at the claimed scope.

Definition 2.1.3 (Frozen v20 dependency baseline). The statements of this appendix are interpreted relative to:

- (i) AK-HPDST v20.0.0 Main Chapters 1–8;
- (ii) the corrected Foundation Appendices A–G;
- (iii) the frozen Appendix H v20.0.0 body;
- (iv) the reviewed CM-v20 and TB-v20 mathematical statements, each only at its individually verified strength;
- (v) the CR-v20 baseline as global governance for inherited claims, not as registration of Appendix I.

The final compact Claim Register is generated only after all appendices and the Companion are frozen. No joint CM/TB promotion, hash synchronization, provisional CR promotion, machine register, or intermediate snapshot is treated as theorem evidence.

Definition 2.1.4 (Comparison policy and runtime-mode calculus). A target-directed comparison is consumed in exactly one of three runtime modes: *exact*, *metric-gap-certified*, or *not compared*. Exact mode requires an evidenced identity or isomorphism of the actual compared objects. Metric-gap-certified mode requires an actual discrepancy bound together with every threshold-gap and crop hypothesis used by the transfer theorem. Not-compared mode licenses statements only about the identified source or surrogate and cannot discharge a target-comparison obligation. The selected mode, scope, and witness are stated in the mathematical record itself; no absent sidecar field is supplied by convention.

Proposition 2.1.5 (Missing target or witness blocks continuum interpretation). *If the continuum target, its realization, or the selected comparison witness is missing or undefined, every continuum-level conclusion is undefined. A discrete-only result may remain valid for the identified discrete realization.*

Proof. The missing item is a typed dependency of the continuum statement. Core readiness is conjunctive and does not allow advisory or scalar compensation for a missing non-scalar obligation. $\square$

2.2 Typed targets, discretizations, and realizations

Definition 2.2.1 (Continuum target record). A continuum target record identifies an object X , the mathematical category in which it lives, the claimed parameter and spatial scope, and the exact statement intended to be transferred from discretized evidence. The symbol X alone is not a target record.

Definition 2.2.2 (Discretized object record). A discretized object record identifies X_h , its scale or resolution parameter h , the construction method, boundary and truncation conventions, and immutable input artifacts. Different meshes, seeds, tie policies, or solver tolerances define different records.

Definition 2.2.3 (Directed discretization family). A discretization family is a directed system $\{X_h\}_{h \in H}$ together with a declared refinement order. The notation $h \rightarrow 0$ is used only when a cofinal refinement map and its meaning are specified. A finite list of meshes is not by itself a convergence object.

Definition 2.2.4 (Discrete realization record). A discrete realization record supplies a declared map

$$\mathfrak{R}_h(X_h) \in \text{FiltCh}(k) \quad \text{or} \quad \mathfrak{R}_h(X_h) \in \text{Pers}_k^{\text{cons}},$$

together with all filtration, coefficient, boundary, tie, and constructibility conventions needed to form repaired profiles.

Definition 2.2.5 (Continuum realization record). A continuum realization record supplies

$$\mathfrak{R}(X) \in \text{FiltCh}(k) \quad \text{or} \quad \mathfrak{R}(X) \in \text{Pers}_k^{\text{cons}},$$

or a theorem-backed constructible replacement on the monitored scope. If only a replacement is available, the replacement error is a separate ledger component.

Definition 2.2.6 (Target identity and surrogate-only mode). A run is in surrogate-only mode when $\mathfrak{R}_h(X_h)$ is defined but no verified comparison to $\mathfrak{R}(X)$ is supplied. In that mode, all verdicts are explicitly about the surrogate. A basename, visual similarity, or shared problem label does not identify the surrogate with the continuum target.

Proposition 2.2.7 (Raw numerical output is not a Core object). *A matrix, array, point cloud, image, mesh file, or solver output is not directly a repaired persistence profile or a B-Gate⁺ record. It becomes Core-readable only through a declared realization and the corresponding evaluation artifacts.*

Proof. The raw object lacks the typed filtered or persistence structure, evaluation scope, and artifact schema required by Chapters 1–3 and Appendix G. □

2.3 Same-domain filtration perturbation and certified rounding

Definition 2.3.1 (Sublevel-set filtration record). Let Y be a fixed underlying object in a category admitting the declared sublevel construction. For a real-valued filtration function $f : Y \rightarrow \mathbb{R}$, write

$$Y_t^f := \{y \in Y : f(y) \leq t\}.$$

A sublevel-set filtration record fixes Y , the order convention, coefficient field, functorial homology construction, and every tameness hypothesis required for the monitored persistence module.

Theorem 2.3.2 (Uniform filtration perturbation gives an interleaving). *Let $f, g : Y \rightarrow \mathbb{R}$ be two filtration functions on the same recorded underlying object and suppose*

$$\|f - g\|_\infty \leq \eta.$$

Then the sublevel filtrations are η -interleaved by the natural inclusions

$$Y_t^f \subseteq Y_{t+\eta}^g, \quad Y_t^g \subseteq Y_{t+\eta}^f.$$

After applying any functorial homology theory, the resulting persistence modules are η -interleaved. In the declared constructible one-parameter regime, the corresponding global barcodes satisfy

$$d_B(\mathcal{B}(f), \mathcal{B}(g)) \leq \eta.$$

A bounded-window conclusion still requires direct comparison of the actual windowed profiles or a separately certified crop theorem.

Proof. The sup-norm bound gives $f(y) \leq t \Rightarrow g(y) \leq t + \eta$, and symmetrically. The two inclusion families commute with structure maps and their double composites are the canonical shifts by 2η , hence define an interleaving. Functoriality preserves it. The barcode bound follows from the registered one-parameter isometry theorem. The final sentence records the independent crop boundary. $\square$

Corollary 2.3.3 (Certified filtration quantization). *Let $q_h : \mathbb{R} \rightarrow \mathbb{R}$ be a recorded quantization or rounding map satisfying*

$$|q_h(r) - r| \leq \rho_h$$

for every filtration value occurring on the monitored source. Then f and $q_h \circ f$ induce ρ_h -interleaved persistence modules and global bottleneck distance at most ρ_h . The number ρ_h enters a target ledger only when the rounding policy and the bound are exact or interval-certified.

Proof. Apply Theorem 2.3.2 to f and $q_h \circ f$. $\square$

Definition 2.3.4 (Verified enclosure-to-filtration witness). A verified enclosure-to-filtration witness supplies a common underlying object, an exact or interval-certified enclosure proving $|f - f_h| \leq \eta_h$ everywhere on the monitored domain, and a proof that the discrete realization uses the enclosed filtration without unrecorded branch, tie, boundary, or interpolation changes. It yields a source-level perturbation bound through Theorem 2.3.2; window and threshold transport remain separate obligations.

2.4 Error ledger, comparison evidence, and numerical modes

Definition 2.4.1 (Pre-threshold discretization profiles). For monitored (i, W) , define the actual pre-threshold profiles

$$B_h := W_W \mathbf{P}_i(\mathfrak{R}_h(X_h)), \quad B := W_W \mathbf{P}_i(\mathfrak{R}(X)).$$

Every metric transfer statement in this appendix is bound to the immutable identities of these profiles, not merely to the names X_h and X .

Definition 2.4.2 (Discretization comparison record). A discretization comparison record contains: the source and target profile identities; comparison mode; window and degree; threshold when used; an exact witness or a bound

$$d_B(B_h, B) \leq \varepsilon_h;$$

the metric convention for infinite bars; the source theorem or verified computation; crop and model contributions; and failure behavior. In metric-gap-certified mode it also contains the two-sided closed-band evidence for the same ε_h .

Definition 2.4.3 (Typed discretization ledger). Let $(V, \oplus, 0_V, \preceq)$ be the declared metric-ledger quantale. A discretization ledger may include typed components

$$\mathfrak{d}_{\text{model}}, \mathfrak{d}_{\text{samp}}, \mathfrak{d}_{\text{num}}, \mathfrak{d}_{\text{crop}}, \mathfrak{d}_{\text{trunc}}, \mathfrak{d}_{\text{round}} \in V,$$

but only when each component has an actual scope-matched discrepancy witness. Their certified aggregate is denoted $\mathfrak{d}_{\text{disc}}$.

Definition 2.4.4 (Dependency-aware aggregation and no double counting). A ledger aggregation record contains a dependency graph identifying which components are independent bounds, nested bounds, or alternative bounds. The aggregate $\mathfrak{d}_{\text{disc}}$ must be justified by an aggregation theorem. The same discrepancy may not be counted twice merely because it appears under two labels.

Definition 2.4.5 (Scalarized discrepancy witness). A scalarized witness consists of $\mathfrak{d}_{\text{disc}} \in V$, a declared scalarization $\text{val}_B : V \rightarrow [0, +\infty]$, and evidence that

$$d_B(B_h, B) \leq \varepsilon_h := \text{val}_B(\mathfrak{d}_{\text{disc}}).$$

A scalar number without the profile identities and supporting theorem or artifact is not a witness.

Theorem 2.4.6 (Pathwise aggregation of certified profile errors). *Let*

$$B^{(0)} = B_h, B^{(1)}, \dots, B^{(m)} = B$$

be actual profiles on the same monitored degree, window, coefficient field, and metric convention. If certified records prove

$$d_B(B^{(j-1)}, B^{(j)}) \leq \eta_j \quad (1 \leq j \leq m),$$

then

$$d_B(B_h, B) \leq \sum_{j=1}^m \eta_j.$$

The same comparison edge may not be inserted twice merely under different ledger labels. Alternative paths give alternative bounds and are not added together unless a separate theorem requires that aggregation.

Proof. Iterate the triangle inequality for the bottleneck metric. The final two sentences enforce that the displayed path, rather than a list of names, is the mathematical source of the aggregate. $\square$

Proposition 2.4.7 (Actual-profile binding is mandatory). *A discrepancy estimate computed for different windows, thresholds, coefficient fields, filtrations, normalization policies, or source artifacts cannot certify the comparison of B_h and B unless a proved transport theorem connects the records.*

Proof. The estimate and the claimed comparison otherwise have different typed inputs. Reusing the number would be an identity collision or a silent default theorem. $\square$

Proposition 2.4.8 (Estimator-only values do not enter the ledger). *An empirical error estimate, confidence interval without a coverage theorem, residual plot, or learned predictor remains advisory until a reliability theorem or verified computation proves the required discrepancy upper bound on the declared scope.*

Proof. A scalar estimate cannot manufacture the non-scalar comparison evidence required by the ledger and comparison record. $\square$

Definition 2.4.9 (Numerical evidence modes). A numerical artifact is labeled exactly one of:

- (i) `exact`: symbolic, rational, integer, or otherwise exact computation with replay evidence;
- (ii) `interval_certified`: outward-rounded or theorem-backed enclosure proving the claimed bound;
- (iii) `heuristic`: floating or learned computation without a certified enclosure.

Only the first two may discharge a numerical theorem obligation, and only at the registered strength.

Definition 2.4.10 (Event-qualified probabilistic comparison certificate). For a randomized discretization on a declared probability space, an event-qualified certificate specifies a measurable good event G_h , a proved risk bound

$$\mathbb{P}(G_h^c) \leq \alpha_h,$$

and the exact deterministic conclusions valid on G_h , including the profile identities, discrepancy radius, crop obligations, and threshold-gap hypotheses. It is a probabilistic risk certificate, not a deterministic comparison record and not a posterior probability that the continuum theorem is true.

Theorem 2.4.11 (False-transfer risk bound for randomized discretization). *Let $E_h = T_\tau^{\text{bar}}(B_h)$ and $E = T_\tau^{\text{bar}}(B)$, where E is the fixed target profile. Suppose an event-qualified certificate has $\mathbb{P}(G_h^c) \leq \alpha_h$, and on G_h the hypotheses of the gap-certified transfer and $2\varepsilon_h \leq \tau$ hold. Then*

$$G_h \cap \{E_h = 0\} \subseteq \{E = 0\},$$

and hence

$$\mathbb{P}(E_h = 0 \text{ and } E \neq 0) \leq \alpha_h.$$

For finitely many monitored claims, a simultaneous risk bound requires a registered joint event or an explicit union-bound allocation. No countable or adaptive family receives a finite total risk bound by default.

Proof. On G_h , deterministic gap transfer gives $d_B(E_h, E) \leq \varepsilon_h$, and sharp zero separation gives $E_h = 0 \Rightarrow E = 0$. Therefore a false transfer can occur only on G_h^c . The probability bound follows. The final sentence records the separate multiple-comparison obligation. $\square$

2.5 Crop-aware and threshold-gap-safe transfer

Proposition 2.5.1 (No unconditional window non-amplification). *This appendix does not assert that passage from source-level persistence data to a bounded window is globally non-amplifying under every shift, end-point, crop, or extension convention. A transfer theorem must compare the actual windowed profiles or ledger the crop contribution.*

Proof. The v20 contract records window-edge and crop obligations separately and rejects unconditional crop-collapse commutation. Therefore a source-level bound alone is insufficient without the declared window theorem. $\square$

Definition 2.5.2 (Crop-aware window comparison). A crop-aware comparison either proves the bottleneck bound directly for B_h and B , or supplies source-level evidence together with a crop component $\mathbf{d}_{\text{crop}}$ and an aggregation theorem yielding the bound for the actual windowed profiles.

Proposition 2.5.3 (No global non-expansiveness of hard deletion). *There is no unconditional rule*

$$d_B(T_\tau^{\text{bar}}(B_h), T_\tau^{\text{bar}}(B)) \leq d_B(B_h, B)$$

for arbitrary barcodes at positive threshold. Any discretization transfer through hard deletion must use exact mode or the v20 threshold-gap hypotheses.

Proof. This is the registered rejection of global non-expansiveness, specialized to discretization comparisons. $\square$

Theorem 2.5.4 (Gap-certified discretization transfer). *Let B_h, B be finite barcodes, possibly containing infinite bars, under the registered bottleneck convention. Let $\tau > 0$, $\varepsilon_h \geq 0$, assume*

$$d_B(B_h, B) \leq \varepsilon_h,$$

and assume no finite bar of either profile has length in

$$[\tau - \varepsilon_h, \tau + \varepsilon_h].$$

Then

$$d_B(T_\tau^{\text{bar}}(B_h), T_\tau^{\text{bar}}(B)) \leq \varepsilon_h.$$

Proof. Apply the sharp two-sided threshold-gap-safe stability theorem of Chapter 2 and Appendix A to the actual pre-threshold profiles. $\square$

Proposition 2.5.5 (Closed threshold band is mandatory at the certified radius). *A clearance check at radius smaller than ε_h , or on only one of B_h, B , does not certify Theorem 2.5.4. Exact equality may still be recorded in exact mode, but it is not a positive-radius robustness certificate.*

Proof. The source theorem is explicitly two-sided and uses the closed band at the same discrepancy radius. Weakening either condition changes the theorem. $\square$

Corollary 2.5.6 (Post-threshold metric transfer). *Under Theorem 2.5.4,*

$$d_B(\text{Eval}_{i,W,\tau}(\mathfrak{R}_h(X_h)), \text{Eval}_{i,W,\tau}(\mathfrak{R}(X))) \leq \varepsilon_h.$$

In the declared constructible one-parameter regime, the same bound may be written with interleaving distance only under the registered barcode-isometry convention.

Proposition 2.5.7 (Exact discretization comparison). *If the actual pre-threshold profiles are evidenced equal or isomorphic under the declared barcode convention, deterministic threshold deletion gives equal repaired profiles. This is exact comparison mode and does not assert positive-radius robustness.*

Proof. Apply the same deterministic deletion rule to the identified profiles. $\square$

2.6 Gate verdict transfer and the half-threshold reserve

Definition 2.6.1 (Discrete-only verdict). A discrete-only verdict asserts a gate or diagnostic statement solely for $\mathfrak{R}_h(X_h)$. It does not contain the phrase “for X ”, “continuum”, or an equivalent target-level reading unless a transfer witness is attached.

Proposition 2.6.2 (Continuum verdict requires a transfer witness). *A continuum pass or reject cannot be inferred from a discrete verdict solely from small ε_h or adjusted-budget pass. The exact logical property being transferred must have its own theorem.*

Proof. Metric proximity is a quantitative relation, whereas vanishing, categorical eligibility, and diagnostic status are typed predicates. No general transport follows without an appropriate theorem. $\square$

Theorem 2.6.3 (Sharp threshold-separated vanishing transfer). *Let B_h, B be finite barcodes, possibly containing infinite bars, and set*

$$E_h := T_\tau^{\text{bar}}(B_h), \quad E := T_\tau^{\text{bar}}(B).$$

Assume $d_B(E_h, E) \leq \varepsilon_h$ and $2\varepsilon_h \leq \tau$. Then

$$E_h = 0 \iff E = 0.$$

The constant is sharp. Operational certificates use the strict reserve $2\varepsilon_h < \tau$ to avoid equality-sensitive numerical decisions.

Proof. Suppose $E_h = 0$. Any finite bar of E has length strictly greater than τ , so its cost to the diagonal is strictly greater than $\tau/2 \geq \varepsilon_h$. An infinite bar has infinite cost to the diagonal under the registered convention. Hence E cannot lie within bottleneck distance ε_h of the empty barcode unless $E = 0$. The converse is symmetric. If $2\varepsilon_h > \tau$, a single retained bar of length L with $\tau < L \leq 2\varepsilon_h$ is nonzero but lies within distance ε_h of zero, proving sharpness. $\square$

Proposition 2.6.4 (Robust nonvanishing witness). *For finite barcodes B_h, B , if $d_B(E_h, E) \leq \varepsilon_h$ and E_h contains an infinite bar or a finite bar of length greater than $2\varepsilon_h$, then $E \neq 0$. The symmetric statement also holds.*

Proof. Such a bar cannot be matched to the diagonal within cost ε_h , so every bottleneck matching at that cost must match it to a bar on the other side. $\square$

Corollary 2.6.5 (Zero/nonzero classification equivalence below half-threshold). *Under Corollary 2.5.6 and $2\varepsilon_h \leq \tau$, the discretized and continuum repaired profiles have the same zero/nonzero classification. Operational certificates use the strict reserve $2\varepsilon_h < \tau$.*

Proposition 2.6.6 (Budget pass alone does not transfer vanishing). *The inequality*

$$\text{val}_B(\mathbf{d}_{\text{disc}}) < \text{gap}_\tau(\{B_h, B\})$$

licenses metric transfer through hard deletion, but does not by itself imply $E_h = 0 \iff E = 0$. The half-threshold reserve or another vanishing-transfer theorem is additionally required.

Proof. A nonzero bar of length at most $2\varepsilon_h$ may be matched to the diagonal. Metric closeness alone therefore need not preserve emptiness. $\square$

Theorem 2.6.7 (Conditional target-side B-Gate transfer). *Assume: (i) an exact or gap-certified comparison record; (ii) the hypotheses of Corollary 2.6.5 for the topological clause; (iii) every invoked target-side categorical obligation is independently verified or transported by a registered bridge theorem; (iv) the actual continuum Type IV artifact and diagnostics satisfy their contract; and (v) the complete scalarized ledger passes. Then a discrete topological pass may participate in a target-side B-Gate⁺ pass. Omitting any invoked clause leaves the target verdict undefined or rejected according to its failure route.*

Proof. The topological clause transfers by Corollary 2.6.5; all other clauses pass by hypothesis. Apply the conjunctive B-Gate⁺ contract and non-compensation rule. $\square$

Proposition 2.6.8 (Persistence proximity does not transfer Ext discharge). *A bottleneck or interleaving bound between repaired persistence profiles does not by itself identify admissible representatives or transfer an Ext¹-vanishing statement. Such a transfer requires the registered representative and bridge hypotheses.*

Proof. The categorical clause depends on non-scalar realization and eligibility data not manufactured by a persistence distance. $\square$

2.7 Detector and higher-obstruction transfer chains

Theorem 2.7.1 (Complete detector to continuum repaired-zero transfer). *Assume a complete detector certificate at the exact repaired-profile stage proves $E_h = 0$. If the hypotheses of Theorem 2.5.4 hold and $2\varepsilon_h \leq \tau$, then*

$$E = 0.$$

Only the repaired-zero predicate is transferred. Detector ranks, sample locations, birth or germ sets, mesh size, and certificate cardinality are not transported.

Proof. Detector completeness supplies $E_h = 0$. The gap theorem supplies $d_B(E_h, E) \leq \varepsilon_h$, and Theorem 2.6.3 supplies $E = 0$. $\square$

Proposition 2.7.2 (Raw detector values are not discretization-stable). *A small bottleneck or interleaving distance does not in general preserve a fixed pointwise image rank, a fixed stencil vector, or its numerical value. Such quantities may change under arbitrarily small endpoint motion across a sample location. Metric transfer therefore acts on the certified repaired-zero predicate, not on raw detector coordinates.*

Proof. Move one endpoint of a one-bar interval across a fixed sample by an arbitrarily small amount. The bottleneck distance tends to zero while the sampled structure-map rank changes between zero and one. $\square$

Theorem 2.7.3 (Discretization-to-higher-Ext discharge). *Fix $n \geq 1$. Suppose:*

- (i) *a complete discrete detector and Theorem 2.7.1 establish $\text{Eval}_{n,W,\tau}(\mathfrak{R}(X)) = 0$;*
- (ii) *the continuum target has a complete standard n -eligible realization record;*
- (iii) *the higher edge realization identifies $H^{-n}(A_{n;X,W,\tau})$ with the declared repaired degree- n edge and sends the zero profile to zero.*

Then

$$\text{Ext}^n(A_{n;X,W,\tau}, k) = 0.$$

The reverse implication requires a separately proved natural zero-reflection theorem on the registered profile class.

Proof. The first clause supplies continuum repaired-profile zero. The second and third clauses identify the relevant H^{-n} with zero. Over a field, the canonical higher edge gives

$$\text{Ext}^n(A, k) \cong \text{Hom}_k(H^{-n}(A), k) = 0.$$

Nothing in this argument reflects Ext zero back to the repaired profile without the stated additional theorem. $\square$

Proposition 2.7.4 (Synthetic or surrogate duality is not a target bridge). *An Ext identity built from a discretized detector vector space or a shifted synthetic complex does not by itself define the continuum realization, establish representative independence, or prove a natural persistence-to-Ext bridge for the target.*

Proof. The synthetic identity concerns the chosen surrogate object. Target realization, naturality, zero reflection, and representative compatibility are independent non-scalar obligations. $\square$

2.8 Convergence, reconstruction, and eventual readiness

Definition 2.8.1 (Pointwise repaired-profile convergence). For fixed (i, W, τ) , a family converges to the target when there are certified comparison records with radii $\varepsilon_h \rightarrow 0$ for the actual repaired profiles. This definition is pointwise in the declared parameters.

Definition 2.8.2 (Uniform monitored-family convergence). For a declared monitored family $\mathcal{M}$ of degrees, windows, and thresholds, convergence is uniform when a certified rate $\rho(h) \rightarrow 0$ bounds every comparison in $\mathcal{M}$. Pointwise certificates do not imply a uniform certificate on an infinite family.

Proposition 2.8.3 (Discrete Cauchy evidence is not target convergence). *Pairwise convergence of B_h to one another establishes at most a Cauchy property in the declared metric. It does not identify a limit with $B = W_W \mathbf{P}_i(\mathfrak{R}(X))$, nor prove that such a target realization exists.*

Proof. Completeness, existence of a limit object, and identification with the declared target are separate obligations. $\square$

Definition 2.8.4 (Reconstruction witness). A reconstruction witness consists of: a theorem giving existence of a limit in a declared complete ambient category or metric space; a construction of that limit; and an exact or certified identification of the limit with the continuum realization on the monitored scope.

Definition 2.8.5 (Certified convergence rate). A certified rate is a function $\rho(h) \rightarrow 0$ together with replayable evidence that the actual pre-threshold or repaired-profile discrepancy is bounded by $\rho(h)$ on the stated scope. A fitted decay curve without a reliability theorem is advisory only.

Theorem 2.8.6 (Eventual two-sided gap readiness from target clearance). *Let B be a finite barcode and define*

$$g := \text{clear}_\tau(B) := \inf\{|\ell(I) - \tau| : I \text{ a finite bar of } B\},$$

with $g = +\infty$ if B has no finite bars. Assume $g > 0$ and $d_B(B_h, B) \leq \varepsilon_h \rightarrow 0$. For every sufficiently small h satisfying

$$3\varepsilon_h < \min\{g, \tau\},$$

neither B nor B_h has a finite bar in $[\tau - \varepsilon_h, \tau + \varepsilon_h]$. Hence the pair is eventually eligible for Theorem 2.5.4.

Proof. The target exclusion follows from $\varepsilon_h < g$. Suppose a finite bar J of B_h lies in the stated band. It cannot match the diagonal at cost ε_h , because $\ell(J) \geq \tau - \varepsilon_h > 2\varepsilon_h$. Thus it matches a finite target bar I , and endpoint displacement gives $|\ell(I) - \ell(J)| \leq 2\varepsilon_h$. Therefore $|\ell(I) - \tau| \leq 3\varepsilon_h < g$, a contradiction. $\square$

Proposition 2.8.7 (Refinement can eventually restore scalar budget admissibility). *Assume a fixed target gap $G > 0$, a base scalar bound $b < G$, a certified rate $\varepsilon_h \rightarrow 0$, and an aggregation witness*

$$\text{val}_B(\mathfrak{d}_{\text{base}} \oplus \mathfrak{d}_{\text{disc}}(h)) \leq b + \varepsilon_h.$$

Then the adjusted scalar budget is below G for sufficiently small h .

Proof. Choose h so that $\varepsilon_h < G - b$. The displayed inequality then gives the strict budget condition. $\square$

Theorem 2.8.8 (Diagonal refinement for a countable monitored family). *Let $\mathcal{M} = \{m_1, m_2, \dots\}$ be a countable monitored family. For each m_j , suppose a certified error function $\varepsilon_j(h) \rightarrow 0$ is available along a common directed refinement order, and fix tolerances $\delta_j > 0$. Write $h' \succeq h$ when h' is at least as refined as h . Then there exists a refinement sequence*

$$h_1 \preceq h_2 \preceq h_3 \preceq \dots$$

such that

$$\varepsilon_j(h_n) < \delta_j \quad (1 \leq j \leq n).$$

Thus every fixed monitored item is eventually within its tolerance. This does not give a uniform bound over the whole countable family at any finite stage.

Proof. For each finite set $\{m_1, \dots, m_n\}$, convergence supplies individual refinement indices meeting the first n tolerances. Directedness supplies a common upper refinement. Choose h_n inductively as a common refinement of h_{n-1} and the finite-stage witnesses. $\square$

Proposition 2.8.9 (Finite refinement prefixes do not certify countable readiness). *No finite prefix of a refinement sequence certifies a countable monitored family, a uniform convergence statement, eventual constancy, or a continuum Return conclusion unless a separate finite-reduction theorem proves that the omitted tail is determined by the prefix.*

Proof. A finite prefix leaves infinitely many monitored obligations and all later refinement behavior unchecked; in particular, it supplies neither a uniform tail bound nor evidence for the omitted items. Two refinement systems may agree through that prefix and diverge afterwards. $\square$

2.9 Sampling, aliasing, and numerical certification

Definition 2.9.1 (Sampling operator record). A sampling record identifies the operator $S_h : X \mapsto X_h$, sample locations or cells, randomness and seed policy, boundary behavior, antialiasing or quadrature rule, and the target norm or metric in which a reliability statement is claimed.

Definition 2.9.2 (Reliability-certified a posteriori estimator). An estimator η_h is reliability-certified when a theorem or verified computation proves, for the actual records,

$$d_B(B_h, B) \leq \eta_h$$

or proves a source-level bound that is validly aggregated into that profile bound. Effectivity or lower bounds are optional; reliability is mandatory for ledger use.

Proposition 2.9.3 (Aliasing warning is non-verdict). *A detected or suspected aliasing pattern is a warning, not a discrepancy bound, gate verdict, or failure type. It becomes verifier-side evidence only through a separately certified estimator, refinement theorem, or exact reconstruction record.*

Proof. The warning does not supply the typed upper bound or target comparison required by the verifier. $\square$

Definition 2.9.4 (Under-resolved status). A run is `under_resolved` when a required reliability bound is unavailable, the certified radius violates a threshold or budget reserve, or the chosen resolution cannot support the claimed monitored scope. This is a Technical Extension lifecycle status, not a Core atomic status. It blocks the target-level claim and is not silently converted to pass or reject.

Definition 2.9.5 (Floating-point and filtration-tie policy). A numerical persistence artifact records number format, rounding mode, tolerance, comparison policy for nearly equal filtration values, ordering of exact ties, treatment of bars with length exactly τ , and whether interval enclosures cover all branching decisions. Changing any item creates a new artifact identity.

Definition 2.9.6 (Adaptive refinement stopping rule). An adaptive stopping rule identifies the certified quantity being tested, its reliability theorem, the target reserve, the stopping inequality, maximum refinement, and failure behavior. A rule based only on consecutive outputs appearing similar is advisory.

Proposition 2.9.7 (Observed refinement plateau is advisory only). *Stability of computed barcodes or scores over finitely many refinements does not prove convergence, error below tolerance, or absence of unresolved features without a reduction or reliability theorem.*

Proof. Finite observations do not supply a universal tail bound and may reflect correlated aliasing or solver stagnation. $\square$

2.10 Discretized towers and Type IV transfer

Proposition 2.10.1 (Object proximity does not control morphism defects). *Even exact identification of source and target persistence objects does not determine the kernel or cokernel of a comparison morphism. For example, the identity and zero endomorphisms of a nonzero one-dimensional constant persistence module have the same source and target but different kernels and cokernels. Hence no object-level bottleneck bound transfers Type IV or transient-defect data without a morphism-level witness.*

Proof. The identity has zero kernel and cokernel, whereas the zero endomorphism has both kernel and cokernel isomorphic to the module. Source and target data are identical in both cases. $\square$

Theorem 2.10.2 (Exact conjugacy transfers full defect profiles). *Let $\phi_h : M_h \rightarrow N_h$ and $\phi : M \rightarrow N$ be actual morphisms. Suppose there are isomorphisms $a_h : M_h \xrightarrow{\sim} M$ and $b_h : N_h \xrightarrow{\sim} N$ such that*

$$b_h \circ \phi_h = \phi \circ a_h.$$

Then

$$\ker \phi_h \cong \ker \phi, \quad \text{coker } \phi_h \cong \text{coker } \phi.$$

Consequently every isomorphism-invariant terminal diagnostic agrees. A complete full- or interior-defect detector has the same zero/nonzero predicate only when it is reconstructed on the corresponding isomorphic defect stage with the same window, threshold, endpoint convention, and detector construction.

Proof. The source isomorphism restricts to an isomorphism of kernels, and the target isomorphism descends to an isomorphism of cokernels. Terminal-generic dimension is invariant under these isomorphisms. The detector statement then follows from exact-stage detector invariance under the explicitly matched construction data. $\square$

Theorem 2.10.3 (Metric-safe thresholded defect-zero transfer). *Let D_h, D be corresponding actual full-defect profiles or corresponding actual interior-defect profiles, reconstructed from kernel or cokernel at the same defect stage, window, and endpoint convention. Fix $\tau_{\text{def}} > 0$. Suppose*

$$d_B(D_h, D) \leq \epsilon_h,$$

no finite bar of either defect profile has length in $[\tau_{\text{def}} - \epsilon_h, \tau_{\text{def}} + \epsilon_h]$, and $2\epsilon_h \leq \tau_{\text{def}}$. Then

$$T_{\tau_{\text{def}}}^{\text{bar}}(D_h) = 0 \iff T_{\tau_{\text{def}}}^{\text{bar}}(D) = 0.$$

A complete stage-matched defect detector may supply either zero premise. Defects of length at most τ_{def} remain outside this conclusion.

Proof. Apply the sharp two-sided gap transfer to the actual defect profiles and then the sharp zero-separation theorem at threshold τ_{def} . $\square$

Corollary 2.10.4 (Metric-safe long-transient transfer). *If, in Theorem 2.10.3, D_h and D are the certified non-cofinal interior components of the corresponding kernel or cokernel profiles, then the displayed equivalence transfers absence of retained long transient defects. It says nothing about terminal/cofinal defects, which require the Type IV diagnostic, and nothing about short transient defects of length at most τ_{def} .*

Proposition 2.10.5 (Bounded-window finite-shift triviality is not terminal cleanliness). *On a bounded right-open window $W = [a, b)$, the window-cofinal interval module $k_{[a, b)}$ is ϵ -trivial for every $\epsilon \geq b - a$, yet it contributes one terminal-cofinal bar relative to W . Thus finite-shift triviality implies terminal-generic cleanliness only under the separately declared right-unbounded terminal scope or another theorem excluding this bounded-window phenomenon.*

Proof. Every translation by at least $b - a$ leaves the window support, so the shift map is zero. Nevertheless the interval reaches the terminal boundary of W and is window-cofinal. $\square$

Definition 2.10.6 (Discretized tower comparison morphism). For each monitored (i, W, τ, h) , a discretized Type IV record contains an actual persistence morphism

$$\phi_{i, W, \tau}^{(h)} : M_{i, W, \tau}^{(h)} \longrightarrow N_{i, W, \tau}^{(h)},$$

with immutable source and target identities. A scalar proxy or diagram caption is not the morphism.

Definition 2.10.7 (Discretized terminal-generic diagnostics). If the discretized source, target, kernel, and cokernel profiles are terminal-tame on a common declared tail, define

$$\mu_{i, W, \tau}^{(h)} := \text{tgdim}_W(\ker \phi_{i, W, \tau}^{(h)}), \quad \nu_{i, W, \tau}^{(h)} := \text{tgdim}_W(\text{coker } \phi_{i, W, \tau}^{(h)}).$$

If the morphism or terminal-tameness evidence is missing, the diagnostic status is undefined; no numerical pair is imputed.

Definition 2.10.8 (Continuum tower comparison morphism). The corresponding continuum record contains an actual morphism

$$\phi_{i,W,\tau} : M_{i,W,\tau} \longrightarrow N_{i,W,\tau}$$

and its own terminal-tameness evidence. A family of discretized morphisms does not constitute this artifact by itself.

Definition 2.10.9 (Type IV diagnostic transfer witness). A Type IV transfer witness supplies morphisms $a_h : M^{(h)} \rightarrow M$, $b_h : N^{(h)} \rightarrow N$ forming a commuting square with $\phi^{(h)}$ and ϕ , together with a common terminal tail on which the induced maps

$$\ker \phi^{(h)} \longrightarrow \ker \phi, \quad \text{coker } \phi^{(h)} \longrightarrow \text{coker } \phi$$

are isomorphisms, or an exact registered theorem giving the same terminal-fiber conclusion.

Proposition 2.10.10 (Discretized Type IV zero does not imply continuum zero). *At a fixed h , the equality*

$$(\mu_{\text{Collapse}}^{(h)}, \nu_{\text{Collapse}}^{(h)}) = (0, 0)$$

does not imply $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ for the continuum tower without a Type IV transfer witness.

Proof. Discretization may omit terminal-cofinal kernel or cokernel defects, and kernel/cokernel formation is not automatically transported by profile-level metric proximity. $\square$

Theorem 2.10.11 (Terminal-fiber diagnostic transfer). *Under Definition 2.10.9, the discretized and continuum terminal-generic dimensions agree degreeewise:*

$$\mu_{i,W,\tau}^{(h)} = \mu_{i,W,\tau}, \quad \nu_{i,W,\tau}^{(h)} = \nu_{i,W,\tau}.$$

Consequently, discretized certified zero transfers to continuum certified zero on the witnessed scope.

Proof. Terminal-generic dimension is invariant under isomorphism on a common terminal tail. Apply this to the induced kernel and cokernel isomorphisms. $\square$

Proposition 2.10.12 (Missing continuum tower artifact is undefined). *If continuum Type IV is invoked but $\phi_{i,W,\tau}$ or its terminal-tameness evidence is missing, the target-side Type IV clause is undefined. Discretized diagnostics, even if zero for all tested meshes, do not repair the missing artifact.*

Proof. The missing morphism is a non-scalar proof obligation and cannot be manufactured by numerical repetition or a proxy. $\square$

2.11 Finite-to-infinite discretization reduction

Definition 2.11.1 (Discretization reduction package). A finite-to-infinite discretization reduction package identifies a directed repaired-profile or morphism diagram, all transition maps, the declared colimit, a candidate continuum apex, the comparison cocone, and the theorem reducing the infinite conclusion to specified finite data. Numerical convergence or repeated agreement is not such a package by itself.

Theorem 2.11.2 (Eventually constant exact discretization reduction). *Let*

$$Q_0 \longrightarrow Q_1 \longrightarrow Q_2 \longrightarrow \cdots$$

be a directed diagram in a category admitting the declared colimit. Suppose that every transition from some stage N onward is an isomorphism and that the continuum apex Q_∞ is equipped with an isomorphism $Q_N \xrightarrow{\sim} Q_\infty$ compatible with the tail cocone. Then the induced colimit-to-apex comparison is an isomorphism. In an abelian profile category its kernel and cokernel vanish; terminal diagnostics are zero, and every applicable complete full- or interior-defect detector returns zero.

Proof. The colimit of an eventually constant diagram is canonically its stabilized value Q_N . Compose this canonical isomorphism with the compatible apex isomorphism. The defect conclusions follow from vanishing kernel and cokernel. $\square$

Proposition 2.11.3 (Finite agreement is not an infinite reduction). *Agreement of a discretization tower with a target through any prescribed finite stage does not determine eventual constancy, the colimit, the apex comparison, or a Return theorem. Two towers may share that finite prefix while one remains constant and the other acquires a new nonzero summand afterwards.*

Proof. Append the same finite prefix either by zero transitions to a constant tail or by introducing a nonzero summand at the next stage and keeping it thereafter. The finite observations agree and the colimits differ. $\square$

2.12 Artifacts, manifest schema, and final-register consistency

Definition 2.12.1 (Discretization artifact bundle). A reproducible bundle contains, as applicable:

- (i) target, surrogate, mesh, sample, and realization identities;
- (ii) coefficient field, numerical evidence mode, solver, libraries, environment, and seed;
- (iii) filtration, tie, boundary, window, threshold, and truncation policies;
- (iv) pre- and post-threshold profiles and barcode or module artifacts;
- (v) comparison matching or enclosure and closed-band evidence;
- (vi) component ledger, dependency graph, and aggregation witness;
- (vii) invoked Type IV morphisms and terminal-tameness artifacts;
- (viii) replay command and content digests.

Definition 2.12.2 (Discretization manifest entry). A manifest entry serializes the artifact bundle, comparison mode, target scope, statuses, source bindings, failure routes, and every theorem or computation used to convert discretized data into verifier-side evidence. Omitted fields are not supplied by convention.

Proposition 2.12.3 (Artifacts are necessary for audit-readable discretization evidence). *A target-directed certificate relying on discretized evidence is not audit-readable when an independent verifier cannot reconstruct the actual surrogate, realization, comparison profiles, numerical mode, discrepancy witness, threshold-band evidence, and ledger contribution.*

Proof. These data are required to identify the mathematical objects and replay the exact inference. Without them, the verifier cannot distinguish a valid comparison from a stale or mismatched computation. $\square$

Definition 2.12.4 (Deferred final Claim Register extraction). The frozen mathematical body, proofs, and stable local labels are the authoritative Appendix I source. No interim Claim Register, JSON projection, CSV register, or promotion receipt is emitted for this appendix. After the complete appendix sequence is frozen, the final compact Claim Register is constructed once from the frozen sources and is checked against the displayed labels, statement classes, dependencies, and release artifacts. A mismatch blocks final release, but no register is reproduced inside this mathematical appendix.

2.13 Appendix I package theorem

Theorem 2.13.1 (Appendix I discretization-stability package). *Within the Technical Extension layer:*

- (i) discretized and continuum objects are separate typed records;

- (ii) *target claims use exact, metric-gap-certified, or not-compared mode;*
- (iii) *the actual windowed pre-threshold profiles are the objects of metric comparison;*
- (iv) *crop, sampling, numerical, model, truncation, and rounding errors enter only through evidenced ledger components, with pathwise aggregation justified by the triangle inequality and no duplicate edge counting;*
- (v) *hard threshold deletion is transported only by exact comparison or the sharp two-sided closed-band theorem;*
- (vi) *post-threshold metric closeness and gate-classification transfer are distinct obligations;*
- (vii) *the sharp condition $2\varepsilon_h \leq \tau$ transfers the zero/nonzero classification after threshold deletion, with strict reserve used operationally;*
- (viii) *target B-Gate⁺ still requires every invoked non-topological clause;*
- (ix) *Cauchy behavior, refinement plateaus, and advisory convergence plots do not identify a continuum target;*
- (x) *positive target clearance plus certified convergence gives eventual two-sided gap readiness;*
- (xi) *aliasing and heuristic estimators remain non-verdict without reliability evidence; event-qualified randomized certificates bound false-transfer risk but do not become deterministic proofs;*
- (xii) *Type IV transfer requires actual morphisms and terminal-fiber comparison;*
- (xiii) *same-domain sup-norm filtration perturbations and certified rounding produce explicit interleaving bounds;*
- (xiv) *complete detector zero transfers only through the repaired-profile zero predicate, not through raw detector coordinates;*
- (xv) *continuum higher-Ext discharge additionally requires target-side standard eligibility and the canonical higher edge;*
- (xvi) *diagonal refinement gives eventual itemwise control on countable monitored families but no finite-stage uniform certificate;*
- (xvii) *object-level proximity does not control morphism defects; exact conjugacy or defect-stage comparison is required;*
- (xviii) *thresholded defect-zero predicates transfer only at matched full- or interior-defect stages; long transient, terminal, and short-defect conclusions remain separately typed;*
- (xix) *eventual exact stabilization plus apex agreement gives a valid finite-to-infinite reduction;*
- (xx) *content-addressed artifacts and final compact-CR consistency are mandatory for release.*

Proof. Typed target and realization separation is established in Definitions 2.2.1–2.2.6. Same-domain perturbation is Theorem 2.3.2. The ledger and numerical-evidence requirements are Definitions 2.4.3–2.4.9, with pathwise aggregation in Theorem 2.4.6 and probabilistic risk semantics in Theorem 2.4.11. Exact and metric-gap-certified threshold transfer are Proposition 2.5.7 and Theorem 2.5.4. Classification, detector, and higher-edge transfer are Theorems 2.6.3, 2.7.1, and 2.7.3; the full-gate boundary is Theorem 2.6.7. Convergence and countable diagonal refinement are controlled by Theorems 2.8.6 and 2.8.8. Morphism and defect transfer are Theorems 2.10.2, 2.10.3, Corollary 2.10.4, and Theorem 2.10.11. Exact finite-to-infinite reduction is Theorem 2.11.2, while audit readability is Proposition 2.12.3. $\square$

2.14 Explicit exclusions

The following are not asserted.

- No grid, mesh, point sample, solver output, barcode image, or refinement plot is a continuum proof by itself.
- No source-level discrepancy bound automatically survives bounded-window passage without direct windowed evidence or crop accounting.
- No global non-expansiveness of positive-threshold hard deletion is assumed.
- No one-sided clearance record is a two-sided comparison certificate.
- No adjusted-budget pass alone transfers a zero profile or B-Gate⁺ verdict.
- No persistence distance manufactures an admissible representative, an Ext¹ bridge, a tower morphism, or terminal-tameness evidence.
- No raw detector rank, sample vector, or mesh cardinality is transported by bottleneck proximity.
- No discretized or synthetic Ext identity becomes a natural continuum bridge without target realization, eligibility, and higher-edge evidence.
- No source/target object comparison controls kernel or cokernel defects without an actual morphism-level square.
- No terminal Type IV zero excludes long or short transient defects without the corresponding defect-stage certificate.
- No bounded-window finite-shift triviality is read as terminal cleanliness without a separate right-unbounded or exclusion theorem.
- No finite refinement prefix proves convergence or a uniform tail bound.
- No Cauchy discrete family is identified with a declared target without reconstruction evidence.
- No aliasing warning or learned error predictor is a ledger component without a reliability theorem.
- No confidence level is a deterministic continuum theorem; a randomized certificate controls only its explicitly registered false-transfer event and risk budget.
- No fixed-mesh Type IV zero, and no numerical pair, replaces the continuum comparison artifact.
- No normative Latin u denotes the cokernel diagnostic; v is canonical.
- No finite refinement prefix, convergence plot, or repeated agreement proves eventual constancy, a colimit identification, or a Return theorem.
- No package closure or successful replay proves a stronger mathematical statement than the displayed claims.
- No interim JSON, CSV, or Claim Register is required or emitted before the complete appendix sequence is frozen.

2.15 Provenance and status

Definition 2.15.1 (Artifact closure). The frozen mathematical body, standalone PDF, integration source, audit report, correction patch, and checksum list formed the Appendix I construction package. After the Main cross-audit, the authoritative Part III release binding is the integrated H–N TeX/PDF package and its checksum sidecar; the earlier Appendix-I package remains immutable repair ancestry. These artifacts establish identity

and reproducibility only, not proof beyond the displayed statements. Final Claim Register construction is deferred until the entire appendix and Companion sequence is frozen.

This appendix depends on:

- (i) Chapters 1–3, 5–7;
- (ii) Appendix A for sharp threshold-gap transfer;
- (iii) Appendices D, F, and G for diagnostics, status, and manifests;
- (iv) Appendix H for advisory noninterference, spectral-tail boundaries, and theorem-backed re-entry;
- (v) the reviewed CM-v20 and TB-v20 mathematical statements, each only at its individually verified strength; no joint promotion or hash synchronization is inferred.

Its role is to provide the grid-to-continuum and discretization-stability discipline without enlarging the Core or asserting an external-domain theorem.

Technical Extension definitions. The authoritative definition set consists of all labeled principles and definitions in this appendix, including the target/surrogate records, sublevel perturbation and enclosure records, comparison and ledger records, numerical modes, detector/higher-edge transfer requirements, convergence and refinement records, morphism/defect transfer witnesses, finite-to-infinite reduction package, artifact bundle, deferred final-CR policy, and artifact closure.

Technical Extension results. The authoritative result set consists of all labeled lemmas, propositions, theorems, and corollaries in this appendix. In particular it includes uniform filtration perturbation, sharp threshold-gap and repaired-zero transfer, complete-detector and higher-Ext discharge chains, eventual gap readiness, countable diagonal refinement, exact conjugacy, thresholded defect-zero transfer, and long-transient separation, terminal-fiber Type IV transfer, eventually constant finite-to-infinite reduction, and the package theorem.

Explicit exclusions. Section 2.14.

Final status. Document role: Technical Extension appendix. Evidence status: mixed proved, conditional where explicitly stated, and operational. Final compact-CR registration is deferred until every v20 appendix is audited and frozen. No interim JSON or per-appendix CR artifact is emitted.

3 Appendix J: Betti Integral and Measurement-Level Stability

3.1 Scope, v20 dependency boundary, and release discipline

This appendix develops Betti curves, Betti integrals, threshold-sweep mass, certified numerical integration, and measurement-level stability for repaired persistence profiles. Its purpose is to identify the exact mathematical information carried by scalar measurements and the precise conditions under which that information can re-enter a verifier-side proof. A scalar measurement remains weaker than the profile from which it is derived.

The central rule is

A scalar measurement acquires verifier authority only through an explicit theorem-backed discharge or type conversion.

Appendix H supplies the advisory and spectral-tail boundary. Appendix I supplies discretization, certified profile comparison, and finite-to-infinite discipline. The window-first semantics, sharp two-sided threshold-gap transfer, higher-Ext eligibility, Type IV/transient separation, and non-compensating audit calculus of the v20 Core remain controlling.

Definition 3.1.1 (Measurement sovereignty principle). Betti curves, Betti integrals, threshold sweeps, plots, quadrature values, confidence intervals, and aggregate persistence scores may summarize, route, or compress evidence. They do not determine, compensate for, mutate, or retroactively repair a Core verdict unless a separate verified construction produces the required verifier-side record.

Definition 3.1.2 (No scalar-to-verdict upgrade principle). A scalar measurement is not a persistence profile, detector certificate, representative, eligibility record, higher-Ext edge, Type IV artifact, transient-defect classifier, diagnostic pair, finite-to-infinite reduction, or metric-ledger element merely because it is exact, small, stable, or reproducible. Every semantic type change requires an explicit theorem or artifact-backed conversion on the claimed scope.

Definition 3.1.3 (Frozen v20 dependency baseline). The statements of this appendix are interpreted relative to:

- (i) AK-HPDST v20.0.0 Main Chapters 1–8;
- (ii) the corrected Foundation Appendices A–G;
- (iii) the frozen Appendix H and Appendix I v20.0.0 bodies;
- (iv) the reviewed CM-v20 and TB-v20 mathematical statements, each only at its individually verified strength;
- (v) the CR-v20 baseline as global governance for inherited claims, not as registration of Appendix J.

The final compact Claim Register is generated only after every appendix and the Companion are frozen. No joint CM/TB promotion, hash synchronization, provisional CR promotion, JSON register, CSV register, or intermediate snapshot is theorem evidence.

Definition 3.1.4 (Comparison policy and runtime-mode calculus). A target-directed comparison is consumed in exactly one of three runtime modes: *exact*, *metric-gap-certified*, or *not compared*. Exact mode requires an evidenced identity or isomorphism of the actual compared profiles. Metric-gap-certified mode requires an actual discrepancy bound together with every threshold-gap, crop, endpoint, and zero-separation hypothesis used by the transfer theorem. Not-compared mode licenses statements only about the identified source profile. The selected mode and witness are stated in the mathematical record itself; no absent sidecar field is supplied by convention.

Proposition 3.1.5 (Missing profile or witness blocks measurement interpretation). *If the repaired profile identity, measurement scope, numerical method, or a required discharge or conversion witness is missing or undefined, the stronger measurement-derived verifier conclusion is undefined. A raw scalar may remain an advisory observation about its identified computation.*

Proof. Each missing item is a typed dependency of the stronger statement. The v20 readiness calculus is conjunctive and forbids scalar compensation for a missing non-scalar obligation. $\square$

3.2 Canonical repaired Betti curves and exact barcode mass

Definition 3.2.1 (Repaired-profile identity record). For a filtered object F , monitored degree i , bounded right-open window $W = [u, u')$, and $\tau > 0$, set

$$M_{i,W,\tau}(F) := \text{Eval}_{i,W,\tau}(F) = T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

A measurement record binds to the immutable identity of this actual post-threshold profile, including source object, degree, window, threshold, endpoint convention, barcode convention, and comparison mode.

Definition 3.2.2 (Windowed repaired Betti curve). For a pointwise finite-dimensional repaired profile $M = M_{i,W,\tau}(F)$, define

$$\beta_M(t) := \dim_k M(t), \quad t \in W.$$

The notation $\beta_{i,W,\tau}^F$ is an alias only after the profile identity record is fixed.

Theorem 3.2.3 (Measurability and finite integrability of constructible Betti curves). *If M is pointwise finite-dimensional and constructible on the bounded window W , with a finite interval decomposition on that scope, then β_M is a nonnegative integer-valued measurable step function and belongs to $L^1(W)$.*

Proof. The dimension function is the finite sum of indicator functions of the interval summands clipped to W . It is therefore measurable, piecewise constant, bounded, and integrable on the finite-measure window. $\square$

Definition 3.2.4 (Raw Betti data). A Betti curve or aggregate computed before the fixed window-first repaired evaluation, or without an immutable repaired-profile identity, is *raw Betti data*. It is advisory by default and is not silently identified with $\beta_{M_{i,W,\tau}(F)}$.

Definition 3.2.5 (Extended Betti integral). For any measurable nonnegative Betti curve on W , define

$$\mathcal{B}_W(M) := \int_W \beta_M(t) dt \in [0, +\infty].$$

For a repaired profile, write $\mathcal{B}_{i,W,\tau}(F) := \mathcal{B}_W(M_{i,W,\tau}(F))$. A finite-valued claim requires Theorem 3.2.3 or another displayed integrability witness.

Theorem 3.2.6 (Barcode mass identity). *Let the finite barcode of M on W be the multiset $\{I_{\alpha}\}_{\alpha=1}^N$, counted with multiplicity. Then*

$$\mathcal{B}_W(M) = \sum_{\alpha=1}^N \lambda(I_{\alpha} \cap W),$$

where λ is Lebesgue length. Thus the Betti integral is exactly total clipped barcode mass.

Proof. The interval decomposition gives $\beta_M = \sum_{\alpha} \mathbf{1}_{I_{\alpha} \cap W}$. Integrating the finite nonnegative sum yields the identity. $\square$

Definition 3.2.7 (Normalized Betti integral). If $0 < \lambda(W) < \infty$, define

$$\overline{\mathcal{B}}_W(M) := \lambda(W)^{-1} \mathcal{B}_W(M).$$

Normalization changes units but does not add verifier authority.

Definition 3.2.8 (Multi-degree Betti measurement vector). For a finite declared degree set D , define

$$\mathbf{B}_{D,W,\tau}(F) := (\mathcal{B}_{i,W,\tau}(F))_{i \in D}.$$

Any scalarization declares weights, units, treatment of infinite values, and permitted use. Pairwise scalarization does not create a higher-degree bridge.

3.3 Threshold-sweep calculus and the spectral-tail interface

Definition 3.3.1 (Repaired Betti-mass sweep). Let B be a finite barcode already restricted to the bounded window W . For $s \geq 0$, define

$$A_{B,W}(s) := \mathcal{B}_W(T_s^{\text{bar}} B).$$

Write

$$N_B(s) := \#\{I \in \mathcal{B}(B) : \ell(I) > s\}, \quad E_B(s) := \sum_{I \in \mathcal{B}(B)} (\ell(I) - s)_+.$$

These are the bounded-window forms of the tail count and excess mass of Appendix H.

Theorem 3.3.2 (Exact threshold-mass identity). For every finite windowed barcode B and $s \geq 0$,

$$A_{B,W}(s) = \sum_{\ell(I) > s} \ell(I) = E_B(s) + sN_B(s).$$

Thus repaired Betti mass is determined exactly by the pre-threshold spectral tail.

Proof. Hard deletion retains exactly the bars of length greater than s . For each retained bar, $\ell = (\ell - s) + s$. Summing gives the formula. $\square$

Theorem 3.3.3 (Threshold-sweep structure and second moment). For a finite windowed barcode B :

- (i) $A_{B,W}$ is finite-valued, nonincreasing, right-continuous, and piecewise constant;
- (ii) its only possible discontinuities are the finite bar lengths;
- (iii) if m_s is the multiplicity of length $s > 0$, then

$$A_{B,W}(s^-) - A_{B,W}(s) = s m_s;$$

- (iv) the integrated sweep is the second length moment,

$$\int_0^{+\infty} A_{B,W}(s) ds = \sum_{I \in \mathcal{B}(B)} \ell(I)^2.$$

Proof. All assertions follow termwise from $A_{B,W}(s) = \sum_I \ell(I) \mathbf{1}_{\{s < \ell(I)\}}$. The integral identity uses $\int_0^{\ell(I)} \ell(I) ds = \ell(I)^2$. $\square$

Theorem 3.3.4 (Exact finite reduction of a threshold band). Let $J = [a, b] \subset [0, +\infty)$, and suppose the complete multiset of bar lengths of B lying in J is known. Then the restriction $A_{B,W}|_J$ is determined by one value on each connected component of

$$J \setminus \{\ell(I) : I \in \mathcal{B}(B)\}$$

together with the strict deletion convention at the finitely many breakpoints. In particular, if J contains no bar length, $A_{B,W}$ is constant on J .

Proof. Theorem 3.3.3 shows that the function changes only when the threshold crosses a bar length and is right-continuous under the rule retaining $\ell > s$. $\square$

3.4 Exact zero detection, the canonical threshold floor, and higher Ext

Definition 3.4.1 (Betti-faithful profile regime). A profile M is *Betti-faithful on W* if every nonzero interval summand has strictly positive Lebesgue-length intersection with W , and no nonzero ephemeral or point-supported summand invisible to Lebesgue integration is admitted.

Theorem 3.4.2 (Zero Betti integral in the faithful regime). *For a finite Betti-faithful profile M ,*

$$\mathcal{B}_W(M) = 0 \iff M = 0.$$

Proof. By Theorem 3.2.6, the integral is a finite sum of nonnegative clipped lengths. Betti-faithfulness makes every nonzero summand contribute strictly positively. $\square$

Proposition 3.4.3 (Lebesgue-invisible nonzero profiles obstruct an unconditional converse). *In an ambient persistence category admitting a point-supported module E_{t_0} , one has $E_{t_0} \neq 0$ but*

$$\int_W \dim_k E_{t_0}(t) dt = 0.$$

Hence zero Betti integral does not imply zero profile in an arbitrary ambient category.

Proof. The support is a singleton of Lebesgue measure zero, while the functor is nonzero at t_0 . $\square$

Theorem 3.4.4 (Canonical threshold mass gap). *Let B be a finite barcode on the bounded window W , let $\tau > 0$, and set $M = T_\tau^{\text{bar}} B$, using window-relative bar lengths and the rule deleting every bar of length at most τ . Then*

$$M = 0 \iff \mathcal{B}_W(M) \leq \tau.$$

Equivalently, every nonzero canonical repaired profile satisfies

$$\mathcal{B}_W(M) > \tau.$$

Proof. If $M = 0$, its mass is zero. If $M \neq 0$, it contains a retained interval of window-relative length strictly greater than τ ; the total nonnegative barcode mass is therefore strictly greater than τ . $\square$

Proposition 3.4.5 (The threshold floor is post-threshold only). *Theorem 3.4.4 cannot be applied to an arbitrary pre-threshold profile. For every $0 < a \leq \tau$, the interval module $k_{[0,a]}$ is nonzero and has Betti mass $a \leq \tau$, but it is removed by T_τ^{bar} .*

Proof. The example is a nonzero interval of length a . The threshold-floor separation is created by the strict post-threshold retention rule $\ell > \tau$, not by scalar integration alone. $\square$

Corollary 3.4.6 (Normalized threshold floor and automatic collapse on a short window). *If $0 < \lambda(W) < \infty$, then*

$$M = 0 \iff \overline{\mathcal{B}}_W(M) \leq \frac{\tau}{\lambda(W)}.$$

If $\lambda(W) \leq \tau$, every finite canonical repaired profile on W is zero.

Proof. Divide Theorem 3.4.4 by $\lambda(W)$. No interval contained in W can have length greater than τ when $\lambda(W) \leq \tau$. $\square$

Theorem 3.4.7 (Measured repaired-zero witness in arbitrary degree). *For every monitored degree $n \geq 0$ in the canonical finite repaired-barcode regime,*

$$\mathcal{B}_{n,W,\tau}(F) \leq \tau \iff \text{Eval}_{n,W,\tau}(F) = 0.$$

An exact zero integral is a special case.

Proof. Apply Theorem 3.4.4 to the identified degree- n repaired profile. $\square$

Theorem 3.4.8 (Measured higher-Ext discharge). *Let $n \geq 1$. Suppose Theorem 3.4.7 yields $\text{Eval}_{n,W,\tau}(F) = 0$, and suppose a target object $A_{n;F,W,\tau}$ is standard n -eligible with an artifact-backed identification*

$$H^{-n}(A_{n;F,W,\tau}) \cong \text{Eval}_{n,W,\tau}(F)$$

and the canonical field-coefficient edge

$$\text{Ext}^n(A_{n;F,W,\tau}, k) \cong \text{Hom}_k(H^{-n}(A_{n;F,W,\tau}), k).$$

Then

$$\text{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

The reverse implication requires a separately proved natural zero-reflection theorem on the declared family.

Proof. The measured witness gives zero repaired profile, hence zero H^{-n} through the eligible identification. Its linear dual is zero. No reverse implication follows from this one-way argument. $\square$

Proposition 3.4.9 (Positive exact Betti integral witnesses nonzero profile). *For any measurable nonnegative Betti curve,*

$$\mathcal{B}_W(M) > 0 \implies M \neq 0.$$

Proof. A zero profile has identically zero Betti curve. Take the contrapositive. $\square$

3.5 Discrete integration and numerical certification

Definition 3.5.1 (Quadrature record). A quadrature record contains a partition $P : u = t_0 < \cdots < t_m = u'$, sample points $q_j \in [t_j, t_{j+1})$, weights, endpoint convention, arithmetic mode, and

$$Q_P(\beta) := \sum_{j=0}^{m-1} \beta(q_j)(t_{j+1} - t_j),$$

together with either an exactness witness or a certified error enclosure.

Theorem 3.5.2 (Breakpoint-resolving quadrature is exact). *If the partition contains every breakpoint of the constructible Betti curve and each q_j lies in the corresponding constant half-open cell, then*

$$Q_P(\beta_M) = \mathcal{B}_W(M).$$

Proof. The integrand is constant on every partition cell, so its cell integral equals the sampled value times the cell length. $\square$

Theorem 3.5.3 (Bounded-variation quadrature error). *Let β be a finite-variation step function on W , let $\|P\| := \max_j(t_{j+1} - t_j)$, and choose arbitrary $q_j \in [t_j, t_{j+1})$. Then*

$$\left| Q_P(\beta) - \int_W \beta(t) dt \right| \leq \|P\| \text{Var}_W(\beta).$$

Proof. On each cell, the absolute error is at most the cell length times the oscillation. The oscillation is bounded by the variation on that cell. Sum and use the maximal cell length. $\square$

Theorem 3.5.4 (Cellwise enclosure integration). *Suppose verified constants $0 \leq l_j \leq u_j$ satisfy*

$$l_j \leq \beta_M(t) \leq u_j \quad (t \in [t_j, t_{j+1})).$$

Then

$$\sum_j l_j(t_{j+1} - t_j) \leq \mathcal{B}_W(M) \leq \sum_j u_j(t_{j+1} - t_j).$$

Outward-rounded versions of the two sums form a valid scalar enclosure.

Proof. Integrate the verified pointwise inequalities cell by cell and add them. $\square$

Proposition 3.5.5 (Sampled values alone do not reconstruct a Betti curve). *A finite set of sampled Betti values does not determine the intervening curve, its integral, or exact zero status unless a breakpoint, variation, cell-enclosure, monotonicity, interpolation, or finite-reduction witness is supplied.*

Proof. Two constructible step functions may agree on the sampled set and differ on an unsampled interval of positive measure. $\square$

Definition 3.5.6 (Numerical evidence modes). A Betti computation uses exactly one numerical mode:

- (i) `exact`: exact endpoints and arithmetic, or a proof-equivalent symbolic computation;
- (ii) `interval_certified`: outward-rounded or theorem-backed enclosures with replay evidence;
- (iii) `heuristic`: floating-point, learned, visual, or empirically calibrated output without a proof-producing enclosure.

Reproducibility is required in every mode but does not upgrade heuristic evidence.

Proposition 3.5.7 (Heuristic quadrature is non-verdict). *A heuristic Betti integral, even if stable across reruns or refinements, cannot discharge exact vanishing, nonvanishing, a metric budget, or a gate clause.*

Proof. It supplies neither a verified enclosure of the true scalar nor a theorem-backed semantic conversion. $\square$

3.6 Scalar enclosures and exact classification

Definition 3.6.1 (Scalar Betti enclosure record). A scalar enclosure record for $\mathcal{B}_W(M)$ contains exact or outward-rounded numbers $0 \leq L \leq U \leq +\infty$ and verified evidence that

$$L \leq \mathcal{B}_W(M) \leq U.$$

It records the profile identity, units, method, numerical environment, and replay artifacts.

Theorem 3.6.2 (Exact classification from a nonnegative scalar enclosure). *For a valid scalar enclosure:*

- (i) $U = 0$ implies $\mathcal{B}_W(M) = 0$;
- (ii) $L > 0$ implies $\mathcal{B}_W(M) > 0$ and hence $M \neq 0$;
- (iii) $L = 0 < U$ does not classify an arbitrary profile without additional separation data.

Proof. These are immediate from nonnegativity and interval inclusion, together with Proposition 3.4.9. $\square$

Theorem 3.6.3 (Canonical repaired-profile enclosure test). *Let $M = T_\tau^{\text{bar}} B$ be a finite canonical repaired profile on bounded W , and let $[L, U]$ be a valid enclosure of $\mathcal{B}_W(M)$. Then:*

- (i) if $U \leq \tau$, then $M = 0$;

(ii) if $L > 0$, then $M \neq 0$;

(iii) if $L = 0$ and $U > \tau$, the zero/nonzero classification is unresolved by this enclosure alone.

Proof. The first clause is Theorem 3.4.4; the second is Theorem 3.6.2; the third is the remaining logical case. $\square$

Definition 3.6.4 (Minimum positive Betti-mass witness). For a declared admissible family $\mathcal{M}$, a minimum-mass witness is a certified $\kappa > 0$ such that

$$M \in \mathcal{M}, M \neq 0 \implies \mathcal{B}_W(M) \geq \kappa.$$

The witness states its endpoint, boundary-clipping, essential-bar, and multiplicity assumptions.

Proposition 3.6.5 (Finite endpoint grid supplies a mass floor). *Suppose every clipped interval endpoint of every profile in $\mathcal{M}$, including the endpoints of W , lies in a fixed finite ordered grid G , and every admitted nonzero interval is nondegenerate. If*

$$\Delta_G := \min\{|x - y| : x, y \in G, x \neq y\} > 0,$$

then $\kappa = \Delta_G$ is a valid minimum-mass witness.

Proof. Every nonzero clipped interval has two distinct grid endpoints and therefore length at least Δ_G . A nonzero profile contains at least one such interval. $\square$

Theorem 3.6.6 (Separated zero certification). *Let $M \in \mathcal{M}$ have valid enclosure $[L, U]$, and let κ be a minimum-mass witness. If $U < \kappa$, then $M = 0$. For a canonical repaired profile, Theorem 3.6.3 gives the stronger concrete test $U \leq \tau$.*

Proof. If $M \neq 0$, then $\mathcal{B}_W(M) \geq \kappa$, contradicting $\mathcal{B}_W(M) \leq U < \kappa$. $\square$

Proposition 3.6.7 (Positive tolerance alone cannot certify arbitrary-profile vanishing). *Absent a minimum-mass or canonical threshold-floor witness, a bound $\mathcal{B}_W(M) \leq \eta$ with $\eta > 0$ does not imply $M = 0$. The interval modules $k_{[0,a]}$ have positive mass $a \rightarrow 0$.*

Proof. Choose $0 < a < \eta$. $\square$

Definition 3.6.8 (Scalar measurement ledger record). A scalar measurement ledger records value-space uncertainty such as $[L, U]$, an absolute error, or a quadrature bound. Its codomain is a declared ordered value space. It is not the Core metric ledger unless a separate certified conversion supplies a profile-discrepancy element.

3.7 Measurement-level stability

Definition 3.7.1 (Measurement-level stability modulus). For an identified class of repaired profiles, $\omega : [0, +\infty] \rightarrow [0, +\infty]$ is a Betti-measurement stability modulus if

$$d_B(M, N) \leq \varepsilon \implies |\mathcal{B}_W(M) - \mathcal{B}_W(N)| \leq \omega(\varepsilon)$$

under the declared finiteness and barcode conventions.

Definition 3.7.2 (Local matching complexity). Let Γ be a certified ε -matching between finite barcodes of M and N , and put $W^\varepsilon = [u - \varepsilon, u' + \varepsilon]$. Let $C_{W, \varepsilon}(\Gamma)$ be the number of:

- (i) matched pairs $(I, J) \in \Gamma$ for which I or J meets W^ε , counted once per pair;
- (ii) unmatched intervals on either side that meet W , counted once per interval.

Theorem 3.7.3 (Local L^1 -stability of Betti curves). *Under Definition 3.7.2,*

$$\|\beta_M - \beta_N\|_{L^1(W)} \leq 2\varepsilon C_{W,\varepsilon}(\Gamma).$$

Consequently,

$$|\mathcal{B}_W(M) - \mathcal{B}_W(N)| \leq 2\varepsilon C_{W,\varepsilon}(\Gamma).$$

Proof. For a matched pair, endpoint displacement at most ε makes the symmetric-difference length of the two clipped intervals at most 2ε . An unmatched interval has length at most 2ε . Pairs outside W^ε and unmatched intervals outside W contribute nothing on W . The pointwise inequality between the absolute difference of the two dimension sums and the sum of pairwise indicator differences, followed by integration, gives the first bound. The second follows from $|\int (\beta_M - \beta_N)| \leq \|\beta_M - \beta_N\|_1$. $\square$

Definition 3.7.4 (Expanded-window occurrence count). Define

$$R_{W,\varepsilon}(M, N) := \#\{I \in \mathcal{B}(M) : I \cap W^\varepsilon \neq \emptyset\} + \#\{J \in \mathcal{B}(N) : J \cap W^\varepsilon \neq \emptyset\}.$$

Then $C_{W,\varepsilon}(\Gamma) \leq R_{W,\varepsilon}(M, N)$.

Corollary 3.7.5 (Matching-free safe Betti-mass bound). *If a certified ε -matching exists, then*

$$|\mathcal{B}_W(M) - \mathcal{B}_W(N)| \leq 2\varepsilon R_{W,\varepsilon}(M, N).$$

If $0 < \lambda(W) < \infty$, divide the right-hand side by $\lambda(W)$ for normalized mass.

Theorem 3.7.6 (Gap-certified retained-mass transport). *Let B, C be finite barcodes on the same bounded window, and fix a certified ε -matching witnessing*

$$d_B(B, C) \leq \varepsilon.$$

Assume neither barcode has a bar length in $[\tau - \varepsilon, \tau + \varepsilon]$, and $2\varepsilon \leq \tau$. Then the retained bar counts agree; write the common count as m . Moreover,

$$|\mathcal{B}_W(T_\tau^{\text{bar}} B) - \mathcal{B}_W(T_\tau^{\text{bar}} C)| \leq 2m\varepsilon.$$

In particular, the two repaired profiles have equivalent zero predicates.

Proof. The two-sided gap prevents a matched bar from crossing the threshold. The condition $2\varepsilon \leq \tau$ prevents a retained bar from matching the diagonal. Hence retained bars are paired bijectively. Matched lengths differ by at most 2ε , so summing over the m retained pairs gives the bound. The common count is zero exactly when both repaired profiles are zero. $\square$

Proposition 3.7.7 (No cardinality-free stability modulus). *There is no finite-valued function ω depending only on the bottleneck radius such that*

$$|\mathcal{B}_W(M) - \mathcal{B}_W(N)| \leq \omega(d_B(M, N))$$

for all finite barcodes without a local cardinality or mass bound. In particular, there is no universal Lipschitz constant.

Proof. Fix $0 < \varepsilon < 1$, $W = [0, 1)$, let M_n be n copies of $[0, 1)$, and N_n be n copies of $[\varepsilon, 1 + \varepsilon)$. Their bottleneck distance is at most ε , while the mass difference on W is $n\varepsilon$, unbounded at the fixed radius ε . $\square$

Proposition 3.7.8 (Repaired-profile transport prerequisite). *When M, N arise by hard threshold deletion from compared pre-threshold profiles, a repaired-profile distance bound is verifier-usable only after exact comparison or the sharp two-sided threshold-gap transfer of Appendix I. Appendix J does not restore global non-expansiveness of hard deletion.*

Proof. The measurement theorems consume an actual bound between post-threshold profiles. Producing that bound is a distinct upstream obligation. $\square$

Definition 3.7.9 (Measurement stability certificate). A stability certificate contains repaired-profile identities, comparison mode, a certified matching or theorem, ε , local matching complexity or another proved mass bound, the resulting modulus, numerical mode, and reconstructible evidence.

3.8 Scalar measurements, metric ledgers, and verifier-side re-entry

Proposition 3.8.1 (Betti curve and Betti integral are not complete invariants). *On $W = [0, 1)$, the barcodes*

$$\{[0, 1)\} \quad \text{and} \quad \{[0, \tfrac{1}{2}), [\tfrac{1}{2}, 1)\}$$

have the same pointwise Betti curve, namely $\beta(t) = 1$ for every $t \in W$, and therefore the same Betti integral on every measurable subwindow. Nevertheless, they are not isomorphic persistence modules.

Proof. The first module has a nonzero structure map across $t = \frac{1}{2}$, whereas the second has zero transport across that breakpoint. Pointwise dimensions forget this morphism-level information. $\square$

Definition 3.8.2 (Value-space and metric-ledger separation). Let S be the ordered value space of scalar uncertainty and $(V, \oplus, 0_V, \preceq)$ the Core metric-ledger quantale. A scalar error in S and a profile-discrepancy element in V are different typed objects and are never combined without a displayed conversion theorem.

Proposition 3.8.3 (Scalar Betti error is not a metric budget). *A bound*

$$|\widehat{\mathcal{B}} - \mathcal{B}_W(M)| \leq \eta_{\mathcal{B}}$$

does not imply a bottleneck bound on the underlying profile and cannot enter the Core metric budget by itself.

Proof. Proposition 3.8.1 shows that scalar mass does not determine profile identity or distance. $\square$

Definition 3.8.4 (Restricted inverse-stability witness). For a declared profile class C , an inverse-stability witness is a proved nondecreasing function Ψ with $\Psi(0) = 0$ such that

$$M, N \in C \quad \implies \quad d_B(M, N) \leq \Psi(|\mathcal{B}_W(M) - \mathcal{B}_W(N)|).$$

It states every structural restriction making the scalar measurement separating on C .

Theorem 3.8.5 (Sound boundary for measurement-to-metric conversion). *A scalar comparison may create a metric-ledger element only through a restricted inverse-stability witness or another theorem directly proving the required profile bound. No global inverse modulus exists on all finite barcodes.*

Proof. The first statement is the required type conversion. The second follows from Proposition 3.8.1: equal scalar masses can occur at positive bottleneck distance, so any inverse modulus satisfying $\Psi(0) = 0$ fails globally. $\square$

Definition 3.8.6 (Certified measurement-to-metric conversion). A certified conversion creates a new content-addressed $\mathfrak{d}_{\text{meas}} \in V$, binds actual source and target profiles, and supplies reconstructible evidence that

$$d_B(M, \widehat{M}) \leq \text{val}_B(\mathfrak{d}_{\text{meas}}).$$

A scalar error formula without this profile bound is not a conversion.

Theorem 3.8.7 (Measurement-attachment noninterference). *Two certificate records with identical verifier-side projections have identical Core atomic status and failure route even if their Betti curves, integrals, plots, scalar enclosures, or measurement attachments differ.*

Proof. Measurement attachments remain outside the verdict function until a separately verified discharge or conversion record is constructed. $\square$

Theorem 3.8.8 (Discretization-to-scalar error composition). *Suppose Appendix I supplies $d_B(M_h, M) \leq \varepsilon_h$, a registered modulus gives*

$$|\mathcal{B}_W(M_h) - \mathcal{B}_W(M)| \leq \omega(\varepsilon_h),$$

and a numerical enclosure gives

$$|\widehat{\mathcal{B}}_h - \mathcal{B}_W(M_h)| \leq \eta_h.$$

Then

$$|\widehat{\mathcal{B}}_h - \mathcal{B}_W(M)| \leq \eta_h + \omega(\varepsilon_h).$$

Proof. Apply the triangle inequality in scalar value space. $\square$

Theorem 3.8.9 (Event-qualified scalar certification). *Fix a probability space and a measurable good event G_h with $\mathbb{P}(G_h^c) \leq \alpha_h$. Let D_h denote the event that the procedure emits a recorded zero/nonzero decision, and suppose that on $G_h \cap D_h$ every deterministic hypothesis of the selected scalar-enclosure route is valid. Then*

$$\mathbb{P}(D_h \cap \{\text{emitted decision is false}\}) \leq \alpha_h.$$

Proof. Deterministic soundness excludes a false emitted decision on $G_h \cap D_h$. Hence the false-emission event is contained in G_h^c , whose probability is at most α_h . $\square$

Proposition 3.8.10 (Probability semantics and conditional-error boundary). *Theorem 3.8.9 bounds an unconditional false-emission event; it is not a probability that a mathematical theorem is true. It also does not by itself imply $\mathbb{P}(\text{false} \mid D_h) \leq \alpha_h$. When $\mathbb{P}(D_h) > 0$, it yields only*

$$\mathbb{P}(\text{false} \mid D_h) \leq \frac{\alpha_h}{\mathbb{P}(D_h)}.$$

Simultaneous finite claims require a joint event or an explicit union-bound allocation; adaptive or countable claims require additional control.

Proposition 3.8.11 (Mixed-codomain error addition is forbidden). *The expression $\mathfrak{d}_{\text{disc}} \oplus \eta_{\mathcal{B}}$ is ill-typed when its operands lie in V and S . One must propagate metric error to scalar error through a modulus, or prove a scalar-to-metric conversion.*

Proof. No cross-operation is defined between the two typed value spaces. $\square$

Definition 3.8.12 (Dependency-aware measurement aggregation). A combined record separates source approximation, profile reconstruction, hard-deletion transport, matching, quadrature, rounding, and scalar reporting. Each primitive uncertainty is counted once; downstream bounds record which upstream components they already include.

3.9 Measured discharge, detectors, and the B-Gate boundary

Definition 3.9.1 (Measured repaired-zero discharge record). For monitored degree n , a measured repaired-zero record contains the repaired-profile identity and exactly one verified route:

- (i) an *exact faithful route*: an exact proof that $\mathcal{B}_W(M) = 0$ together with a Betti-faithfulness witness, invoking Theorem 3.4.2;
- (ii) a *canonical threshold-floor route*: evidence that $M = T_{\tau}^{\text{bar}} B$ is a finite canonical repaired profile on bounded W , together with a valid enclosure satisfying $U \leq \tau$;

- (iii) a *declared-family separation route*: evidence that $M \in \mathcal{M}$, a valid minimum-mass witness κ for $\mathcal{M}$, and a valid enclosure satisfying $U < \kappa$.

The resulting verifier-side statement $\text{Eval}_{n,W,\tau}(F) = 0$ is a new record distinct from the scalar attachment.

Theorem 3.9.2 (Sound measured repaired-zero discharge). *Every record satisfying Definition 3.9.1 soundly discharges*

$$\text{Eval}_{n,W,\tau}(F) = 0$$

at the recorded degree and scope.

Proof. The three routes are respectively Theorem 3.4.2, Theorem 3.6.3, and Theorem 3.6.6. $\square$

Corollary 3.9.3 (Measured higher-Ext chain). *A sound measured repaired-zero discharge in degree $n \geq 1$, followed by the complete standard n -eligible realization and higher-edge package, yields the Ext-vanishing conclusion of Theorem 3.4.8. The scalar record alone does not supply eligibility, naturality, or zero reflection.*

Theorem 3.9.4 (Measured Betti discharge does not imply full B-Gate⁺). *In degree one, a sound measured repaired-zero discharge proves only the repaired topological clause. It does not by itself establish representative eligibility, the Ext¹ clause, Type IV cleanliness, transient cleanliness, tower artifacts, comparison obligations, or the strict metric budget required by B-Gate⁺.*

Proof. These are distinct conjunctive verifier-side obligations and cannot be manufactured or compensated for by a scalar measurement. $\square$

Theorem 3.9.5 (Certified measured nonzero forces topological failure). *If a valid enclosure for a repaired degree-one profile has $L > 0$, then*

$$\text{Eval}_{1,W,\tau}(F) \neq 0,$$

so the repaired topological clause fails.

Proof. The enclosure implies positive Betti mass and Proposition 3.4.9 gives a nonzero profile. $\square$

Proposition 3.9.6 (Scalar discharge does not transport detector data). *A measured repaired-zero discharge determines the zero predicate only. It does not reconstruct a detector stencil, detector rank, birth or germ set, coverage certificate, family mesh, or minimum certificate size.*

Proof. These data contain localization and proof-compression information discarded by integration. $\square$

3.10 Defect measurements, Type IV, and transient separation

Definition 3.10.1 (Actual defect measurement record). Let $\phi : M \rightarrow N$ be an actual monitored persistence morphism. A defect measurement record binds separately to the actual profiles $\ker \phi$, $\text{coker } \phi$, and, when used, their certified terminal, noncofinal interior, or thresholded components. Source or target Betti mass is not a substitute for defect-profile mass.

Theorem 3.10.2 (Measured long-transient zero discharge). *Let D_{int} be a certified noncofinal interior kernel or cokernel defect profile on bounded W , and set*

$$D_{\text{int}}^{>\tau_{\text{def}}} := T_{\tau_{\text{def}}}^{\text{bar}} D_{\text{int}}.$$

If a valid enclosure of its Betti mass satisfies $U \leq \tau_{\text{def}}$, then

$$D_{\text{int}}^{>\tau_{\text{def}}} = 0.$$

This excludes retained long transient defects on the recorded scope but says nothing about terminal/cofinal defects or defects of length at most τ_{def} .

Proof. Apply Theorem 3.6.3 to the thresholded interior defect profile. The final sentence records the components not seen by this route. $\square$

Theorem 3.10.3 (Right-unbounded terminal-average identity for Type IV diagnostics). *Fix a monitored triple (i, W, τ) with declared right-unbounded terminal scope $W = [a, +\infty)$, and let $\phi : M \rightarrow N$ be its actual Type IV comparison morphism. Suppose $\ker \phi$ and $\operatorname{coker} \phi$ are terminal-tame and have constant fiber dimensions on $[T, +\infty)$ for some $T \geq a$. For every $L > 0$,*

$$\frac{1}{L} \int_T^{T+L} \dim_k(\ker \phi)(t) dt = \mu_{i,W,\tau}(\phi),$$

$$\frac{1}{L} \int_T^{T+L} \dim_k(\operatorname{coker} \phi)(t) dt = \nu_{i,W,\tau}(\phi).$$

Hence interval-certified upper bounds strictly below 1 for both normalized terminal averages certify $(\mu_{i,W,\tau}, \nu_{i,W,\tau}) = (0, 0)$; a positive lower bound for either average certifies the corresponding nonzero Type IV obstruction.

Proof. On the common terminal tail, the two dimension functions are constant with the nonnegative integer values defining terminal-generic dimension. Their normalized integrals equal those constants. An integer in $[0, 1)$ is zero, while a positive lower bound excludes zero. $\square$

Proposition 3.10.4 (Terminal-average authority and scope boundary). *Theorem 3.10.3 cannot be applied to source or target profiles, scalar proxies, bounded-window cofinality, or discretized morphisms lacking an exact terminal transfer witness. The actual morphism, induced defect profiles, right-unbounded terminal scope, and terminal-tameness evidence are load-bearing. In particular, a finite interval reaching the right endpoint of a bounded window is not governed by the average over $[T, +\infty)$.*

Theorem 3.10.5 (Terminal zero does not imply transient cleanliness). *Even when Theorem 3.10.3 certifies $(\mu_{i,W,\tau}, \nu_{i,W,\tau}) = (0, 0)$, long or short noncofinal kernel and cokernel defects may remain. Full defect cleanliness requires the complete transient detector and short-defect exclusion packages.*

Proof. Terminal-generic dimension ignores defect summands that die before the terminal tail. $\square$

Definition 3.10.6 (Betti Type IV proxy). A Betti Type IV proxy is a scalar intended to correlate with terminal defect but not bound to the actual kernel or cokernel profile under Theorem 3.10.3.

Theorem 3.10.7 (Betti Type IV proxy is not a diagnosis). *A Betti Type IV proxy does not determine $(\mu_{i,W,\tau}, \nu_{i,W,\tau}) = (0, 0)$ or its negation. Diagnosis requires the actual morphism, terminal-tameness evidence, and an exact terminal computation or the terminal-average theorem above.*

Proof. A proxy does not determine the kernel and cokernel of the required morphism. $\square$

3.11 Threshold bands and finite-to-infinite measurement discipline

Definition 3.11.1 (Measurement-stable threshold band). A threshold interval is measurement-stable when a universal theorem or artifact proves the prescribed scalar property for every threshold in the interval. A finite or countable sampled sweep is recorded separately.

Theorem 3.11.2 (Breakpoint-certified universal Betti-mass band). *For a fixed finite pre-threshold windowed barcode B , a complete list of bar lengths in a threshold band determines the entire restriction of $A_{B,W}$ to that band through the finite cells and breakpoint values of Theorem 3.3.4. Consequently, every predicate depending only on that restricted scalar function reduces to finitely many exact checks. No continuity interpolation is needed.*

Proof. The repaired Betti-mass sweep is a finite right-continuous step function with exactly the registered length breakpoints. $\square$

Proposition 3.11.3 (Sampled measurement band is not a universal certificate). *Stability on sampled thresholds does not imply stability at every threshold without a complete breakpoint, monotonicity, continuity, or other finite-reduction theorem covering the unsampled values.*

Proof. An unsampled threshold may cross a retained bar length. $\square$

Theorem 3.11.4 (Betti-stable band is not a Core Type IV stable band). *A measurement-stable Betti band is not a Core Type IV stable band unless actual Type IV diagnostics are certified zero throughout the band by the required morphism-level artifact family or reduction theorem.*

Proof. Betti stability is a scalar predicate; Type IV stability is a predicate on actual kernel and cokernel terminal fibers. $\square$

Theorem 3.11.5 (Exact eventual profile constancy gives measurement constancy). *Let $M_0 \rightarrow M_1 \rightarrow \dots$ be a measured profile system and suppose that for some N , every stage M_n with $n \geq N$ is identified isomorphically, on the same measurement scope, with a fixed apex profile M_∞ . Then*

$$\beta_{M_n} = \beta_{M_\infty}, \quad \mathcal{B}_W(M_n) = \mathcal{B}_W(M_\infty) \quad (n \geq N).$$

Proof. Betti curves and their integrals are invariant under profile isomorphism on the fixed scope. $\square$

Proposition 3.11.6 (Scalar eventual constancy is not profile reduction). *A constant or convergent sequence of Betti integrals does not imply eventual profile constancy, apex agreement, a colimit identification, or a Return Theorem. The two nonisomorphic profiles in Proposition 3.8.1 may alternate while the scalar mass remains constantly 1.*

3.12 Artifacts and deferred final registration

Definition 3.12.1 (Betti measurement artifact bundle). A Betti artifact bundle contains, as invoked: repaired-profile and barcode identities; exact endpoint data; Betti curves; threshold breakpoints; partitions and sample rules; exact or interval arithmetic logs; matchings; scalar enclosures; threshold-floor or minimum-mass witnesses; higher-Ext, defect, and conversion witnesses; dependency graph; generator version; environment; and source digests.

Definition 3.12.2 (Betti measurement manifest entry). A manifest entry records the parent certificate, degree, window, threshold, repaired profile, measurement kind and units, numerical mode, comparison mode, scalar enclosure, metric conversion when any, ledger destinations, artifact references, permitted use, forbidden reading, lifecycle status, and failure behavior. Omission is not repaired by convention.

Proposition 3.12.3 (Stale measurement artifacts are non-consumable). *If any content-addressed profile, barcode, window, threshold, integration rule, numerical environment, normalization, or dependency changes, the prior measurement artifact is stale for the changed run and cannot be silently reused.*

Proof. The changed dependency defines a different computation identity. $\square$

Proposition 3.12.4 (Betti artifacts are necessary for audit readability). *A certificate relying on Betti evidence is audit-readable only when an independent verifier can reconstruct the repaired profile, scalar value or enclosure, numerical method, error argument, semantic conversion, and ledger placement.*

Proof. Without these artifacts, neither the scalar assertion nor its permitted use can be checked. $\square$

Definition 3.12.5 (Deferred final Claim Register rule). Appendix J retains stable local labels and a frozen TeX source. Formal Claim UUIDs, cross-appendix dependency closure, and the compact human-readable Claim Register are generated only after the entire appendix and Companion sequence is frozen. No Appendix-J JSON, CSV, provisional promotion receipt, or intermediate CR snapshot is emitted.

3.13 Appendix J package theorem

Theorem 3.13.1 (Appendix J v20 Betti-measurement package). *Within the Technical Extension layer:*

- (i) *repaired Betti curves are typed functions of immutable post-threshold profiles;*
- (ii) *finite Betti integrals equal total clipped barcode mass;*
- (iii) *repaired Betti mass satisfies the exact threshold-tail identity and a finite breakpoint calculus;*
- (iv) *every nonzero canonical repaired profile has mass strictly greater than its deletion threshold;*
- (v) *theorem-backed scalar enclosures can therefore certify repaired zero in every monitored degree;*
- (vi) *standard n -eligibility and the higher edge convert measured repaired zero to Ext^n -vanishing, but no reverse direction is automatic;*
- (vii) *exact, bounded-variation, and cell-enclosure quadrature routes are separated from heuristic sampling;*
- (viii) *Betti curves admit a local matching-complexity L^1 stability bound and retained mass admits a sharper two-sided-gap bound;*
- (ix) *no cardinality-free forward modulus and no unrestricted scalar-to-metric inverse modulus exists;*
- (x) *discretization and measurement errors compose in scalar value space, including event-qualified risk bounds, not by mixed-codomain addition;*
- (xi) *measured zero discharges only its recorded repaired clause and does not reconstruct detector data or the remaining B-Gate⁺ obligations;*
- (xii) *on a declared right-unbounded terminal scope, actual terminal defect averages exactly recover μ and ν under terminal-tameness, while bounded-window cofinality, terminal, long-transient, and short-defect claims remain separate;*
- (xiii) *complete threshold breakpoints give finite band reduction, but scalar constancy does not give finite-to-infinite profile reduction;*
- (xiv) *content-addressed artifacts are mandatory and final CR generation is deferred until all appendices are frozen.*

Proof. Items (i)–(iii) are Theorems 3.2.6, 3.3.2, and 3.3.3. Items (iv)–(vi) are Theorems 3.4.4, 3.6.3, and 3.4.8. Numerical certification is Theorems 3.5.2–3.5.4. Stability is Theorems 3.7.3 and 3.7.6, with limitations from Propositions 3.7.7 and 3.8.1. Typed composition and discharge are Theorems 3.8.8, 3.8.9, and 3.9.2. Defect and band boundaries are Theorems 3.10.2, 3.10.3, 3.11.2, and 3.11.5, together with their explicit nonconverses. $\square$

3.14 Explicit exclusions

Appendix J asserts none of the following:

- a Betti integral, plot, scalar enclosure, or quadrature value is a Core gate by itself;
- the canonical threshold floor is applied to an arbitrary pre-threshold profile;

- zero integral implies zero profile outside a faithful or canonical repaired-barcode regime;
- a positive tolerance certifies arbitrary-profile zero without a threshold floor or minimum-mass witness;
- equal or close Betti integrals imply equal or close persistence profiles;
- scalar error is a metric-ledger element without an inverse-stability or direct profile-bound theorem;
- global non-expansiveness of hard positive-threshold deletion;
- measured repaired zero recreates detector localization or certificate-minimality data;
- measured topological discharge supplies higher-Ext eligibility, Type IV, transient, artifact, or budget clauses;
- the right-unbounded terminal-average theorem applies to bounded-window cofinality or implies that terminal-average zero excludes long or short transient defects;
- a sampled or scalar-stable threshold band is a Core Type IV stable band;
- scalar eventual constancy is a finite-to-infinite reduction or Return Theorem;
- an unconditional false-emission bound is silently read as the same conditional error rate among emitted decisions;
- heuristic, visual, AI-generated, or merely reproducible output is theorem evidence;
- package closure proves mathematical truth or an external-domain theorem.

3.15 Provenance and status

Definition 3.15.1 (Artifact closure). The appendix body, standalone PDF, source TeX, audit report, patches, and checksum sidecar formed the Appendix J construction package. After the Main cross-audit, the authoritative Part III release binding is the integrated H–N TeX/PDF package and its checksum sidecar; the earlier Appendix-J package remains immutable repair ancestry. These artifacts record and reproduce the displayed mathematics, do not replace proof, and contain no JSON register.

This appendix depends on:

- (i) Chapters 1–8, especially repaired evaluation, detector, higher-edge, Type IV/transient, reduction, and governance boundaries;
- (ii) Foundation Appendices A–G for threshold, representative, bridge, diagnostic, status, and manifest disciplines;
- (iii) Appendix H for spectral-tail identities and advisory re-entry;
- (iv) Appendix I for discretization transfer, certified comparison, probabilistic event bounds, and finite-to-infinite discipline;
- (v) the reviewed CM-v20 and TB-v20 mathematical statements, each only at its individually verified strength; no joint promotion or hash synchronization is inferred.

Its role is to provide the Betti-integral and measurement-stability discipline without enlarging the Core or asserting an external-domain theorem.

New v20 mathematical results. The exact threshold-mass identity, threshold-sweep second moment and finite breakpoint reduction, canonical threshold mass gap, arbitrary-degree measured zero and higher-Ext chain, cellwise integration enclosure, local L^1 Betti-curve stability, sharp gap-certified retained-mass transport, restricted inverse-stability boundary, event-qualified scalar certification with its conditional-error boundary, measured long-transient discharge, right-unbounded terminal-average recovery of μ and ν , and exact eventual measurement constancy are proved at their displayed strengths.

Final status. Document role: Technical Extension appendix. Evidence status: mixed proved and operational, with every conditional use expressed through explicit hypotheses. Formal CR-level registration remains pending until the complete appendix sequence is frozen.

4 Appendix K: AWFS and Controlled Commutation

4.1 Scope, release boundary, and controlling semantics

This appendix develops algebraic weak factorization systems (AWFS), typed commutation claims, defect composition, and controlled replacement for the Technical Extension layer. Its purpose is to determine when two operational orders may be compared, when an alternative path may replace the canonical path, and which additional obligations remain after such a replacement.

Controlled commutation is not Core equality. A replacement is verifier-usable only through an exact coherent comparison or through a certified defect bound together with every predicate-specific transport and ledger obligation used downstream.

The canonical repaired evaluation remains

$$\text{Eval}_{i,W,\tau}(F) := T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F)),$$

so window restriction precedes positive-threshold hard deletion. No AWFS factorization, lifting operation, approximate square, controlled 2-cell, comparison score, or diagrammatic annotation replaces the repaired topological clause, an eligible higher-Ext clause, an actual Type IV morphism, terminal-tameness evidence, a transient-defect certificate, the strict metric budget, or a separately proved Return theorem.

Definition 4.1.1 (Commutation sovereignty principle). Controlled commutation may organize a construction and may justify a scoped replacement. It does not determine, compensate for, mutate, or retroactively repair a Core verdict.

Definition 4.1.2 (No AWFS-to-Core upgrade principle). An AWFS factorization, coalgebra, algebra, lift, or functorial replacement acquires no Core authority merely by existing. Every Core use requires the exact verifier-side object, comparison, hypothesis, and artifact required by the consumed claim.

Definition 4.1.3 (Frozen v20 dependency baseline). Appendix K is interpreted relative to the frozen v20 Main Chapters 1–8, Foundation Appendices A–G, Appendices H–J, the reviewed CM-v20 and TB-v20 mathematical statements at their individually verified strengths, and the CR-v20 baseline as global governance for inherited claims. No joint CM/TB promotion or hash synchronization is inferred. Formal final Claim Register generation is deferred until the complete appendix and Companion sequence is frozen.

Definition 4.1.4 (Deferred final Claim Register rule). Appendix K retains stable local labels and a frozen TeX source. Formal Claim UUIDs, global dependency closure, and the compact human-readable Claim Register are generated only after all appendices are fixed. No Appendix-K JSON register, CSV register, provisional promotion receipt, or intermediate CR snapshot is emitted.

Remark 4.1.5 (Optional technical-extension status). The AWFS structures below are optional. They support algebraic lifting, functorial bookkeeping, and coherent comparison, but they do not restore a global exact collapse functor or the pre-v19 exact-collapse interpretation.

4.2 Typed squares, comparison modes, and defects

Definition 4.2.1 (Pipeline diagram). A pipeline diagram is a finite diagram in a declared category, bicategory, 2-category, or typed operational environment. Every vertex and arrow has a fixed semantic type, source, target, scope, and versioned construction.

Definition 4.2.2 (Strictly commuting square). A square

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & & \downarrow g' \\ C & \xrightarrow{f'} & D \end{array}$$

commutes strictly when

$$g' \circ f = f' \circ g$$

as an equality of morphisms in the declared category.

Definition 4.2.3 (Exact-isomorphic square). A square is exact-isomorphic when it is equipped with a specified invertible 2-cell or natural isomorphism

$$\alpha : g' \circ f \xrightarrow{\cong} f' \circ g$$

together with every naturality and coherence datum required by downstream composition. Exact-isomorphic is not renamed strict.

Definition 4.2.4 (Commutation claim). A commutation claim records the diagram, the two composites, their immutable identities, the comparison direction, the comparison mode, the defect semantics, the proof or artifact, the allowed downstream use, and failure behavior. Its local mode is one of

strict, exact-isomorphic, metric-controlled, two-cell-controlled, advisory-only, undefined.

Definition 4.2.5 (Commutation defect). A commutation defect is a typed discrepancy object $\Delta_{\square}$ comparing the two declared composites. Its codomain, order, zero, symmetry, composition law, scalarization, and scope are part of the record. Norm defects, bottleneck radii, interleaving radii, directed simulations, 2-cells, and ledger elements are different types.

Definition 4.2.6 (Zero-reflecting defect semantics). A defect semantics is zero-reflecting on a declared class when

$$\Delta_{\square} = 0$$

implies the exact equality or specified coherent isomorphism claimed on that class. Without zero reflection, a zero numerical cost is only a zero value in the cost codomain.

Proposition 4.2.7 (Defect semantics are load-bearing). *The phrase “commutes up to error” has no verifier-side meaning until the compared composites, defect codomain, orientation, scope, measurement rule, composition rule, and ledger destination are fixed.*

Proof. Different defect codomains admit different implications and different composition laws. Without the typed declaration, no theorem identifies the error with a Core-readable comparison. $\square$

Proposition 4.2.8 (Objectwise isomorphism is not natural commutation). *A family of chosen isomorphisms between the values of two composite functors does not establish exact-isomorphic commutation unless the family is natural and satisfies the required coherence conditions.*

Proof. Pipeline replacement is a statement about morphisms as well as objects. Objectwise choices need not intertwine structure maps or compose coherently. $\square$

Definition 4.2.9 (Projection to Core comparison modes). A K-local claim projects to a Core comparison as follows.

- (i) Strict and exact-isomorphic modes project to exact comparison only when the equality or coherent isomorphism is registered on the actual consumed objects.

- (ii) Metric-controlled mode projects to metric-gap-certified comparison only after an actual profile discrepancy bound, every required crop and threshold hypothesis, a certified ledger destination, and any predicate-specific transfer theorem are supplied.
- (iii) A controlled 2-cell has no automatic Core projection. It projects to exact only through a coherent invertible comparison accepted by the consumed semantics, or to metric-gap-certified only through a proved conversion to an actual metric bound.
- (iv) Advisory-only projects to not-compared. Undefined has no valid resolved comparison mode.

4.3 Controlled metric replacement and defect aggregation

Definition 4.3.1 (Soft commutation). A square soft-commutes at radius $\delta_\square$ when its typed defect satisfies

$$\Delta_\square \preceq \delta_\square$$

in the declared ordered codomain. This is a bounded-defect statement, not equality.

Definition 4.3.2 (Controlled commutation). A commutation claim is controlled when both composites are defined on the same scope, the defect semantics are fixed, a reconstructible exact or bounded witness is supplied, and every downstream use declares either an exact route or a certified defect-conversion route.

Definition 4.3.3 (Commutation ledger component). Let $(V, \oplus, 0_V, \preceq)$ be the active metric-ledger quantale. A commutation component is an element

$$\mathbf{d}_{\text{comm}} \in V$$

whose scalarization is proved to bound an actual profile discrepancy on the declared scope. A scalar commutation score is not such a component without a conversion theorem.

Definition 4.3.4 (Actual-profile binding). Every metric commutation certificate binds to the actual compared pre-threshold or repaired profiles, including degree, coefficient field, window, threshold, endpoint convention, normalization, and source artifacts. A bound for differently typed profiles is unusable without a proved transport.

Theorem 4.3.5 (Pathwise metric accumulation). *Let $P_0, \dots, P_m$ be actual persistence profiles and suppose*

$$d_B(P_{j-1}, P_j) \leq \varepsilon_j \quad (1 \leq j \leq m).$$

Then

$$d_B(P_0, P_m) \leq \sum_{j=1}^m \varepsilon_j.$$

The same path may be entered in a general ledger only through the declared aggregation map into V .

Proof. Iterate the triangle inequality for the bottleneck metric. The final sentence is a typing requirement: a numerical sum is not automatically the quantale aggregate consumed by the Core ledger. $\square$

Proposition 4.3.6 (Alternative paths are not sequential errors). *Bounds obtained from alternative comparison paths are not added as though all paths were traversed. A certificate selects a witnessed path, or uses a proved infimum/minimum rule, and records the selected dependency graph.*

Proof. Alternative proofs bound the same endpoint discrepancy by different routes. Adding them double-counts a discrepancy not sequentially incurred. $\square$

Proposition 4.3.7 (No double counting and no omission). *A commutation ledger is invalid if it counts one primitive discrepancy twice or omits a primitive edge actually used by the replacement path.*

Proof. Double counting may cause a false rejection; omission may cause a false pass. Both break the stated path semantics. $\square$

Theorem 4.3.8 (Advisory commutation noninterference). *If two certificate records have identical verifier-side projections after deleting advisory commutation annotations, they have the same Core atomic status and failure route.*

Proof. The verdict function consumes verifier-side objects and certified comparisons, not unconverted annotations or scores. $\square$

Proposition 4.3.9 (Controlled does not mean harmless). *A certified nonzero commutation defect may defeat the affected certificate when its aggregate consumes the available strict threshold or budget reserve.*

Proof. Certification establishes a valid bound, not a favorable bound. The complete strict inequality must still be checked. $\square$

4.4 Threshold-safe predicate transport

Definition 4.4.1 (Operational replacement). An operational replacement uses one composite output in place of another. The replacement is exact when the actual outputs are equal or coherently isomorphic, and metric-controlled when a certified discrepancy bound is supplied.

Theorem 4.4.2 (Metric replacement is not equality). *A bound $d_B(M, N) \leq \varepsilon$ does not transport equality, vanishing, representative admissibility, higher-Ext eligibility, or Type IV cleanliness unless a theorem transports that specific predicate under the stated hypotheses.*

Proof. Metric proximity and the listed predicates are different typed relations. No universal implication exists between them. $\square$

Theorem 4.4.3 (Gap-certified commutation through hard deletion). *Let B_L, B_R be finite barcodes, possibly with infinite bars, and assume*

$$d_B(B_L, B_R) \leq \varepsilon.$$

Let $\tau > 0$, and suppose neither barcode has a finite bar of length in the closed band

$$[\tau - \varepsilon, \tau + \varepsilon].$$

Then

$$d_B(T_\tau^{\text{bar}} B_L, T_\tau^{\text{bar}} B_R) \leq \varepsilon.$$

Proof. The two-sided closed band prevents an ε -matching from moving a finite interval across the strict survival threshold. Restrict an ε -matching to the retained intervals and preserve the infinite-bar convention. $\square$

Theorem 4.4.4 (Sharp post-threshold zero transport). *Let E_L and E_R both be outputs of strict hard deletion at the same threshold $\tau > 0$. If*

$$d_B(E_L, E_R) \leq \varepsilon, \quad 2\varepsilon \leq \tau,$$

then

$$E_L = 0 \iff E_R = 0.$$

Proof. Every surviving finite interval has length strictly greater than τ , so its diagonal cost is strictly greater than $\tau/2 \geq \varepsilon$. An infinite interval has infinite diagonal cost. Hence a nonzero retained interval cannot be matched to the empty barcode within radius ε . The argument is symmetric. $\square$

Remark 4.4.5 (Operational strict reserve). The mathematical boundary in Theorem 4.4.4 is $2\varepsilon \leq \tau$. A numerical or interval-certified workflow may impose the stricter reserve $2\varepsilon < \tau$ to avoid equality and rounding ambiguity.

Proposition 4.4.6 (Sharpness of the half-threshold boundary). *The condition $2\varepsilon \leq \tau$ in Theorem 4.4.4 is sharp. If $2\varepsilon > \tau$, choose L with*

$$\tau < L \leq 2\varepsilon,$$

let $E_L = 0$, and let E_R consist of one interval of length L . Both profiles are τ -reduced and

$$d_B(E_L, E_R) = \frac{L}{2} \leq \varepsilon,$$

but $E_L = 0$ and $E_R \neq 0$.

Proof. The only interval of E_R survives strict deletion because $L > \tau$, while its bottleneck distance to the empty barcode is its diagonal cost $L/2$. $\square$

Theorem 4.4.7 (Detector-certified zero transfer across a controlled commutation). *For some $\varepsilon \geq 0$, let B_L, B_R satisfy the hypotheses of Theorem 4.4.3 at threshold τ , and assume $2\varepsilon \leq \tau$. If a complete detector certificate at the declared profile stage proves*

$$T_\tau^{\text{bar}} B_L = 0,$$

then

$$T_\tau^{\text{bar}} B_R = 0.$$

The symmetric implication holds after exchanging the two paths. Only the repaired-zero predicate is transported; detector samples, ranks, birth or germ data, coverage witnesses, and certificate size remain path-specific.

Proof. Theorem 4.4.3 gives the repaired-profile distance bound, and Theorem 4.4.4 transports the zero predicate. The listed detector data are not invariants of the metric comparison and are not used by either theorem. $\square$

Proposition 4.4.8 (Budget pass alone is not predicate transport). *A commutation component satisfying the scalar budget inequality does not by itself transfer vanishing, representative compatibility, higher-Ext eligibility, or Type IV diagnostics.*

Proof. The budget controls discrepancy magnitude. It does not construct the non-scalar witness required by any listed predicate. $\square$

4.5 Weak and algebraic weak factorization systems

Definition 4.5.1 (Weak factorization system). A weak factorization system on a category C is a pair $(\mathcal{L}, \mathcal{R})$ such that every morphism factors as $r \circ l$ with $l \in \mathcal{L}$, $r \in \mathcal{R}$, and

$$\mathcal{L} = {}^\square \mathcal{R}, \quad \mathcal{R} = \mathcal{L}^\square.$$

Here $l \square r$ means every commutative square with left map l and right map r admits a diagonal filler.

Definition 4.5.2 (Functorial factorization). A functorial factorization assigns to each arrow $f : X \rightarrow Y$ an object Ef and arrows

$$X \xrightarrow{Lf} Ef \xrightarrow{Rf} Y, \quad f = Rf \circ Lf,$$

functorially with respect to commutative squares in the arrow category C^2 .

Definition 4.5.3 (Algebraic weak factorization system). An AWFS on C is a functorial factorization for which the left-arrow functor L carries a comonad structure on C^2 , the right-arrow functor R carries a monad structure, and the structures satisfy the standard distributive-law compatibility. An L -coalgebra and an R -algebra include specified algebraic structure, not only membership in retract-closed map classes.

Theorem 4.5.4 (Canonical algebraic lifting operation). *For an AWFS, every displayed square whose left map carries an L -coalgebra structure and whose right map carries an R -algebra structure has a specified diagonal filler. The filler is natural with respect to morphisms of coalgebras, algebras, and squares.*

Proof. The comonad, monad, and distributive-law data define the canonical algebraic lifting operation between L -coalgebras and R -algebras. Naturality is part of the algebraic lifting structure. $\square$

Theorem 4.5.5 (Underlying weak factorization system). *Under the standard AWFS axioms, the retract closures of the underlying maps of L -coalgebras and R -algebras form a weak factorization system.*

Proof. Functorial factorization supplies factorizations, canonical algebraic lifting supplies orthogonality, and retract closure yields the ordinary left and right classes. $\square$

Definition 4.5.6 (AWFS lifting certificate). An AWFS lifting certificate identifies the displayed square, the L -coalgebra structure, the R -algebra structure, the resulting diagonal filler, and the coherence or replay evidence. Merely drawing the square or naming the AWFS is insufficient.

Definition 4.5.7 (AWFS-controlled pipeline). An AWFS-controlled pipeline uses a declared AWFS to factor selected arrows or construct selected fillers, replacements, or comparison morphisms. The factorization data remain Technical Extension data until a downstream verifier record consumes a specific output.

Proposition 4.5.8 (Functorial factorization is not external commutation). *Functoriality of the AWFS factorization does not imply commutation with window restriction, hard deletion, realization, normalization, transfer, kernel, cokernel, terminal-tail passage, or a finite-to-infinite limit.*

Proof. Functoriality concerns morphisms in the arrow category of the declared AWFS. Compatibility with an external operation requires additional structure relating that operation to the comonad, monad, and factorization functor. $\square$

Definition 4.5.9 (Strong AWFS transport datum). Let C and $\mathcal{D}$ carry AWFSs (L_C, R_C) and $(L_{\mathcal{D}}, R_{\mathcal{D}})$. Write $E_C f$ and $E_{\mathcal{D}}(Qf)$ for the middle objects of their factorizations. A strong AWFS transport along a functor $Q : C \rightarrow \mathcal{D}$ contains a natural family of isomorphisms

$$q_f : Q(E_C f) \xrightarrow{\sim} E_{\mathcal{D}}(Qf)$$

such that

$$q_f \circ Q(L_C f) = L_{\mathcal{D}}(Qf), \quad R_{\mathcal{D}}(Qf) \circ q_f = Q(R_C f).$$

The comparison is additionally compatible with the counit and comultiplication of L , the unit and multiplication of R , the distributive law, and the induced algebraic lifting operation. Objectwise factor isomorphisms without naturality and these compatibilities are not a strong transport datum.

Theorem 4.5.10 (Transport of algebraic structures and lifts). *A strong AWFS transport sends registered L_C -coalgebras and R_C -algebras to the corresponding $L_{\mathcal{D}}$ -coalgebras and $R_{\mathcal{D}}$ -algebras and carries their canonical diagonal fillers to the transported canonical fillers. Without Definition 4.5.9, no such preservation is inferred.*

Proof. Compatibility with the comonad and monad structure maps transports coalgebra and algebra axioms. Compatibility with the distributive law and algebraic lifting operation identifies the image of the source filler with the canonical filler determined by the transported structures. $\square$

4.6 AWFS-assisted representatives and higher categorical obligations

Definition 4.6.1 (AWFS-assisted representative). An AWFS-assisted representative is a candidate $C_{\tau, W}F$ whose construction or comparison uses AWFS data.

Theorem 4.6.2 (No representative-status upgrade). *An AWFS-assisted representative is admissible only after the ordinary compatibility record*

$$\mathbf{P}_i(C_{\tau, W}F) \cong \text{Eval}_{i, W, \tau}(F)$$

is proved on the declared scope. The AWFS construction method does not strengthen this status.

Proof. The Core consumes the repaired-profile compatibility, not the method by which the candidate was constructed. $\square$

Theorem 4.6.3 (Isomorphic representatives preserve higher Ext zero predicates). *Let $\mathcal{A}$ be an abelian category with enough injectives, let $K \in \mathcal{A}$, and let $u : A \xrightarrow{\sim} A'$. For every $n \geq 0$, contravariance in the first argument gives an isomorphism*

$$\text{Ext}_{\mathcal{A}}^n(A', K) \xrightarrow{\sim} \text{Ext}_{\mathcal{A}}^n(A, K).$$

Consequently, Ext-zero is preserved by an actual representative isomorphism.

Proof. Derived functors preserve isomorphisms in each argument, with contravariant orientation in the first variable. $\square$

Proposition 4.6.4 (Ext transport does not manufacture eligibility). *Theorem 4.6.3 does not construct standard n -eligibility, a repaired edge realization, representative compatibility, or a natural zero-reflection theorem. Those remain separate obligations.*

Proof. The theorem begins after objects A, A' and their isomorphism are already defined in the required category. It supplies no persistence-to-Ext realization data. $\square$

4.7 Windowing and hard-deletion commutation

Definition 4.7.1 (Crop-collapse comparison). For a finite barcode B , a right-open window W , and $\tau > 0$, the crop-collapse comparison is between

$$W_W T_{\tau}^{\text{bar}} B \quad \text{and} \quad T_{\tau}^{\text{bar}} W_W B.$$

The second expression is the canonical window-first repaired order.

Theorem 4.7.2 (One-sided barcode inclusion). *At the barcode-multiset level, every interval occurrence retained by $T_{\tau}^{\text{bar}} W_W B$ also occurs in $W_W T_{\tau}^{\text{bar}} B$. Thus window-first deletion can remove occurrences that global deletion followed by cropping retains, but not conversely.*

Proof. If $\ell(I \cap W) > \tau$, then $\ell(I) \geq \ell(I \cap W) > \tau$. Hence any interval surviving after cropping also survives before cropping. $\square$

Theorem 4.7.3 (Exact crop-collapse criterion). *For a finite barcode B , the two crop-collapse barcodes are equal if and only if no interval occurrence I satisfies*

$$\ell(I) > \tau \quad \text{and} \quad 0 < \ell(I \cap W) \leq \tau.$$

The criterion includes infinite intervals through $\ell(I) = +\infty$.

Proof. By Theorem 4.7.2, the only possible extra occurrences on the global-delete-then-crop side are globally retained intervals whose nonempty cropped length fails the strict local threshold. Excluding exactly those occurrences is necessary and sufficient. $\square$

Theorem 4.7.4 (Universal crop-collapse defect bound). *For finite barcodes under the standard diagonal-cost convention,*

$$d_B\left(W_W T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} W_W B\right) \leq \frac{\tau}{2}.$$

Proof. Match every common interval occurrence identically. By Theorem 4.7.2, each unmatched extra occurrence has cropped length at most τ , so it matches the diagonal at cost at most $\tau/2$. $\square$

Proposition 4.7.5 (Sharpness of the universal crop-collapse bound). *The constant $\tau/2$ in Theorem 4.7.4 is sharp. For every $\tau > 0$, choose $L > \tau$, set*

$$B = \{[0, L)\}, \quad W = [0, \tau).$$

Then

$$W_W T_\tau^{\text{bar}} B = \{[0, \tau)\}, \quad T_\tau^{\text{bar}} W_W B = 0,$$

and

$$d_B\left(W_W T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} W_W B\right) = \frac{\tau}{2}.$$

Proof. The global interval survives because $L > \tau$. Its crop has length exactly τ , so the strict local deletion rule removes it. The remaining interval on the alternative-order side has exact diagonal cost $\tau/2$. $\square$

Remark 4.7.6 (Alternative-order output need not be post-threshold). The profile $W_W T_\tau^{\text{bar}} B$ may contain bars of length at most τ , because cropping occurs after deletion. Therefore Theorem 4.4.4 cannot be applied merely because the global path used threshold τ .

Theorem 4.7.7 (No exact-collapse restoration). *Theorems 4.7.3 and 4.7.4 do not make positive-threshold hard deletion a global exact functor, reflector, Serre localization, or strict filtered lift.*

Proof. They compare two barcode-level composites on a fixed class. They do not define or prove exact functorial behavior on arbitrary morphisms or short exact sequences. $\square$

4.8 Morphism-level commutation and Type IV

Definition 4.8.1 (Exact conjugacy square). In an abelian category, an exact conjugacy square consists of morphisms $\phi_h : M_h \rightarrow N_h$, $\phi : M \rightarrow N$, isomorphisms $a : M_h \xrightarrow{\sim} M$, $b : N_h \xrightarrow{\sim} N$, and the equality

$$b \circ \phi_h = \phi \circ a.$$

Theorem 4.8.2 (Kernel and cokernel transport under exact conjugacy). *Under Definition 4.8.1, there are induced isomorphisms*

$$\ker \phi_h \cong \ker \phi, \quad \text{coker } \phi_h \cong \text{coker } \phi.$$

Proof. Transport the kernel universal property along a and b , and dually transport the cokernel universal property. Isomorphisms preserve the corresponding finite limits and colimits. $\square$

Corollary 4.8.3 (Exact Type IV diagnostic transport). *If the four defect profiles in Theorem 4.8.2 are terminal-tame on the same declared Main-compatible terminal scope—either a bounded right-open window with the window-cofinal convention or a right-unbounded window—then*

$$\mu(\phi_h) = \mu(\phi), \quad \nu(\phi_h) = \nu(\phi).$$

Proof. The kernel and cokernel profiles are isomorphic on the common monitored scope. The registered window-relative terminal-generic dimension is invariant under such an isomorphism, both for bounded cofinal and right-unbounded terminal conventions. $\square$

Proposition 4.8.4 (Object agreement does not control morphism defects). *Let $M \neq 0$. The morphisms*

$$\text{id}_M : M \rightarrow M \quad \text{and} \quad 0 : M \rightarrow M$$

have identical source and target objects, but

$$\ker(\text{id}_M) = 0, \quad \text{coker}(\text{id}_M) = 0,$$

whereas

$$\ker(0) = M, \quad \text{coker}(0) = M.$$

Thus even zero object-level discrepancy does not control Type IV or transient morphism defects.

Proof. The displayed kernel and cokernel computations are immediate. $\square$

Proposition 4.8.5 (Bounded-window cofinality is not terminal-tail commutation). *A commutation statement on a bounded window does not automatically transport right-unbounded terminal-generic diagnostics. A window-cofinal interval may have finite window-relative length and may become finite-shift trivial while still reaching the bounded right edge.*

Proof. Bounded-window cofinality and eventual behavior on a right-unbounded tail are different predicates. The latter requires a separately identified terminal scope and actual morphism-level defect profiles. $\square$

Corollary 4.8.6 (Long-transient defect-zero transport). *Let D_L, D_R be certified interior defect profiles arising from two compared morphism paths. Each profile is explicitly typed as a kernel-defect or cokernel-defect profile. Suppose both are subjected to strict hard deletion at a defect threshold $\tau_{\text{def}} > 0$. If*

$$d_B\left(T_{\tau_{\text{def}}}^{\text{bar}} D_L, T_{\tau_{\text{def}}}^{\text{bar}} D_R\right) \leq \varepsilon, \quad 2\varepsilon \leq \tau_{\text{def}},$$

then

$$T_{\tau_{\text{def}}}^{\text{bar}} D_L = 0 \quad \Longleftrightarrow \quad T_{\tau_{\text{def}}}^{\text{bar}} D_R = 0.$$

This transfers only the absence of retained long interior defects. It does not identify short transient defects or terminal/cofinal Type IV defects.

Proof. Apply Theorem 4.4.4 at threshold τ_{def} to the two genuine post-threshold defect profiles. $\square$

4.9 Controlled 2-cells and coherence

Definition 4.9.1 (Controlled 2-cell). In a bicategory or 2-category, a controlled 2-cell is a 2-morphism

$$\alpha : g' \circ f \Rightarrow f' \circ g$$

with declared orientation, invertibility status, cost, source and target scope, unit/associativity conventions, and horizontal and vertical composition laws.

Proposition 4.9.2 (Invertible 2-cell versus strict equality). *An invertible 2-cell establishes exact-isomorphic commutation, not strict equality. It may be consumed as exact only by a downstream semantics that is explicitly invariant under the registered coherent isomorphism.*

Proof. Invertibility identifies the composites up to coherent equivalence. Equality of 1-morphisms is a stronger statement. $\square$

Proposition 4.9.3 (Zero cost does not imply invertibility). *A controlled 2-cell with cost zero need not be invertible or identical unless the cost semantics is zero-reflecting on the admitted 2-cells.*

Proof. A cost function may assign zero to distinct or noninvertible 2-cells. Zero reflection is precisely the additional implication required. $\square$

Definition 4.9.4 (Two-direction composition laws). A controlled 2-cell system records separate bounds for vertical and horizontal composition, for example

$$\delta(\beta \circ \alpha) \preceq \delta(\beta) \oplus_v \delta(\alpha), \quad \delta(\beta * \alpha) \preceq \delta(\beta) \oplus_h \delta(\alpha).$$

The two aggregation operations need not coincide.

Theorem 4.9.5 (Controlled pasting). *A pasted diagram has a certified defect bound only when all component scopes match, every required whiskering is defined, the horizontal and vertical aggregation laws are supplied, and the associativity, unit, and interchange coherence data are registered.*

Proof. Pasting is built from whiskering, horizontal composition, and vertical composition. The resulting bound follows by iterating the corresponding declared laws; missing coherence leaves the composite comparison undefined. $\square$

4.10 Directed normalization and transfer comparisons

Definition 4.10.1 (Normalization-commutation defect). A normalization defect compares normalization before an AK operation with the corresponding normalized operation afterward. The record specifies how thresholds, bar lengths, metric scales, and ledger scalarizations transform.

Definition 4.10.2 (Directed transfer defect). A transfer defect compares two directed composites involving a transfer operation. A witness from path A to path B has no reverse meaning unless symmetry or an inverse witness is proved.

Proposition 4.10.3 (Normalization is not harmless rescaling). *A normalization may change the numerical threshold, gap, metric radius, and scalarized ledger value. It is ignorable only under a proved invariance or conversion theorem.*

Proof. The Core budget and hard-deletion predicate are quantitative. Rescaling their inputs changes the consumed inequalities unless compatibility is established. $\square$

Proposition 4.10.4 (Direction matters). *A directed simulation, lax comparison, reconstruction map, or one-sided enclosure does not supply the reverse comparison by default.*

Proof. None of these structures is symmetric by definition. Reversibility is additional data. $\square$

4.11 Finite-to-infinite commutation boundary

Definition 4.11.1 (Eventually constant comparison tower). Let $(X_n), (Y_n)$ be directed towers with compatible comparisons $\alpha_n : X_n \rightarrow Y_n$. The comparison is exactly eventually constant from stage N when the transition maps from stage N onward are isomorphisms and transport every α_n to α_N .

Theorem 4.11.2 (Exact apex transport under eventual constancy). *Assume Definition 4.11.1, together with apex objects X_∞, Y_∞ and compatible isomorphisms*

$$X_N \xrightarrow{\sim} X_\infty, \quad Y_N \xrightarrow{\sim} Y_\infty.$$

Then the apex comparison is exactly the transport of α_N . In particular, strict commutation, invertibility, and exact conjugacy established at stage N transfer to the apex through the specified isomorphisms.

Proof. All later stages and the apex are identified with stage N by compatible isomorphisms. Transport the comparison and its equations through those identifications. $\square$

Proposition 4.11.3 (Finite prefixes do not determine infinite commutation). *Agreement or controlled commutation on every inspected finite prefix does not by itself prove eventual constancy, colimit commutation, apex agreement, or a Return theorem.*

Proof. Uninspected later stages may change, and a colimit or apex identification is additional structure not determined by a finite observation set. $\square$

4.12 Statuses and audit-readable artifacts

Definition 4.12.1 (Commutation status). A local commutation record has one status among strict, exact-isomorphic, controlled, advisory, uncontrolled, failed, and undefined. These are Technical Extension statuses, not Core atomic statuses. Failed defeats the affected path; uncontrolled and undefined cannot discharge an invoked comparison obligation.

Definition 4.12.2 (Commutation artifact bundle). An audit-readable bundle contains the diagram, actual composite identities, comparison direction and mode, exact or bounded witness, defect codomain, composition laws, metric conversion when used, dependency graph, predicate-specific transport theorem, AWFS coalgebra/algebra structures and fillers when invoked, Type IV morphism data when invoked, replay information, and content digests.

Proposition 4.12.3 (Uncontrolled commutation cannot support certification). *An uncontrolled or undefined commutation claim cannot support a Core or Core-adjacent certificate.*

Proof. The verifier cannot reconstruct an equality, coherent isomorphism, discrepancy bound, or failure behavior. $\square$

Proposition 4.12.4 (Failed replacement defeats only the dependent path). *If a replacement path fails its defect or predicate-transport condition, every certificate path depending on that replacement fails. An independent verified path remains logically available.*

Proof. Failure propagates along actual dependency edges, not to unrelated proofs. $\square$

4.13 Appendix K package theorem

Theorem 4.13.1 (Appendix K AWFS and controlled-commutation package). *Within the Technical Extension layer, Appendix K establishes the following package.*

- (i) *Strict equality, coherent isomorphism, metric control, and controlled 2-cells are separate comparison types.*
- (ii) *Defect zero has an exact meaning only under zero-reflecting semantics.*
- (iii) *Actual-profile binding, pathwise aggregation, and dependency control are mandatory for metric replacement.*
- (iv) *Two-sided threshold-gap transfer controls hard deletion, $2\varepsilon \leq \tau$ is the sharp mathematical boundary for zero transport between genuine post-threshold profiles, and detector certificates transport only the repaired-zero predicate.*
- (v) *An AWFS supplies canonical algebraic fillers between registered left coalgebras and right algebras and has an underlying weak factorization system.*

- (vi) *AWFS data do not commute with external AK operations without an exact AWFS transport or another explicit comparison theorem.*
- (vii) *AWFS-assisted representatives and Ext objects retain exactly their ordinary compatibility and eligibility requirements.*
- (viii) *Crop-first hard deletion is a barcode subprofile of global deletion followed by cropping; exact commutation has an explicit interval criterion and the sharp universal defect bound is $\tau/2$.*
- (ix) *Object-level agreement does not control morphism-level kernel or cokernel defects; exact conjugacy does.*
- (x) *Type IV diagnostics transfer only through actual morphism-level defect isomorphisms on the correct terminal scope; long-transient defect zero may instead be transported only at its explicitly thresholded interior-defect stage.*
- (xi) *Controlled 2-cell pasting requires orientation, separate horizontal and vertical cost laws, and coherence.*
- (xii) *Directed normalization and transfer comparisons are not reversible by default.*
- (xiii) *Finite-stage commutation reaches an apex only through exact eventual constancy and compatible apex agreement.*
- (xiv) *Advisory, uncontrolled, undefined, or failed commutation cannot silently support certification.*
- (xv) *No theorem in this appendix enlarges the Core, proves an external-domain result, or restores global exact hard deletion.*

Proof. Items (i)–(iii) follow from the typed comparison and metric-control sections. Item (iv) is Theorems 4.4.3, 4.4.4, and 4.4.7, together with Proposition 4.4.6. Items (v)–(vi) are Theorems 4.5.4, 4.5.5, and 4.5.10. Item (vii) is Theorems 4.6.2 and 4.6.3 together with Proposition 4.6.4. Item (viii) is Theorems 4.7.2–4.7.4 and Proposition 4.7.5. Items (ix)–(x) are Theorem 4.8.2, Corollaries 4.8.3 and 4.8.6, and Proposition 4.8.4. Item (xi) is Theorem 4.9.5. Item (xii) follows from the directed comparison section. Item (xiii) is Theorem 4.11.2. Items (xiv)–(xv) follow from the status and explicit-boundary rules. $\square$

4.14 Explicit exclusions

Appendix K asserts none of the following.

- No AWFS is required by the Core, and no AWFS factorization is a repaired collapse functor.
- No functorial factorization automatically satisfies a Beck–Chevalley, realization, window, threshold, kernel, cokernel, terminal-tail, or colimit compatibility theorem.
- No soft-commuting square is strict merely because its defect is small or numerically zero.
- No controlled commutation replaces the repaired topological clause, an eligible higher-Ext clause, Type IV artifacts, transient-defect evidence, the strict metric budget, or a Return theorem.
- No metric bound transports an unregistered predicate.
- No commutation theorem transports raw detector samples, ranks, births, germs, coverage artifacts, or certificate sizes.
- No long-transient zero certificate excludes short transient defects or terminal/cofinal Type IV defects.
- No global non-expansiveness or exactness of positive-threshold hard deletion is assumed.
- Crop–collapse commutation is not assumed. Its alternative-order output need not be post-threshold.

- No object-level isomorphism or distance controls morphism kernels and cokernels without an actual square.
- No bounded-window cofinal statement is renamed a right-unbounded terminal-tail statement.
- No zero-cost 2-cell is assumed invertible without zero reflection.
- No directed comparison is reversed without a reverse witness.
- No finite refinement prefix proves infinite commutation or apex agreement.
- No advisory diagram, AI score, visual similarity, or repeated numerical agreement is theorem evidence.

4.15 Provenance and final local status

Appendix K depends on the window-first and threshold-gap discipline of Appendix A, representative and bridge boundaries of Appendices B–C, Type IV discipline of Appendix D, status and manifest rules of Appendices F–G, advisory separation of Appendix H, discretization transfer of Appendix I, and scalar/metric separation of Appendix J. Its role is to provide algebraic and diagrammatic control without allowing commutation claims to contaminate Core verdicts.

Final local status. Document role: Technical Extension appendix. Evidence status: mixed proved, conditional where explicitly stated, and operational. Stable local labels are frozen for later final-CR extraction. Formal CR-level closure remains deferred until the complete Technical Extension sequence is audited.

5 Appendix L: Mirror/Transfer 2-cell Ledger

5.1 Scope, sovereignty, and frozen v20 baseline

This appendix specializes the controlled-commutation calculus of Appendix K to Mirror/Transfer pipelines. Its purpose is to separate exact transport, variance-sensitive dual transport, metric replacement, adjunction-based comparison, correspondence data, and advisory analogy before any of them are consumed by a Core-adjacent certificate.

The central rule is

Mirror/Transfer language acquires verifier authority only through a typed transport theorem, a scope-matched comparison record, and the complete affected ledger.

All Core-readable persistence data remain governed by the window-first evaluation

$$\text{Eval}_{i,W,\tau}(F) := T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

No Mirror/Transfer diagram, duality slogan, adjunction, kernel transform, span, correspondence, weak equivalence candidate, or controlled 2-cell replaces the repaired topological clause, higher-Ext eligibility, an admissible representative, an actual Type IV morphism, transient-defect evidence, a finite-to-infinite reduction, or the strict metric-budget condition.

Definition 5.1.1 (Mirror/Transfer sovereignty principle). Mirror/Transfer data may organize a comparison route or construct a new verifier-side record. They do not determine, compensate for, mutate, or retroactively repair a Core verdict.

Definition 5.1.2 (No Mirror/Transfer-to-Core upgrade principle). A Mirror/Transfer datum does not become a persistence-to-Ext bridge, admissible representative, tower comparison artifact, Type IV diagnosis, transient-defect classifier, or metric-ledger component merely by existing. Every Core use requires the corresponding typed record and all hypotheses of the theorem actually consumed.

Definition 5.1.3 (Frozen v20 dependency baseline). The statements of this appendix are interpreted relative to:

- (i) AK-HPDST v20.0.0 Main Chapters 1–8;
- (ii) the corrected Foundation Appendices A–G;
- (iii) the frozen v20 Appendices H–K;
- (iv) the reviewed CM-v20 and TB-v20 mathematical statements, each only at its individually verified strength;
- (v) the CR-v20 baseline as global governance for inherited claims, not as registration of Appendix L.

The final compact Claim Register is generated after the complete appendix and Companion sequence is frozen. No joint CM/TB promotion, hash synchronization, provisional CR, JSON register, CSV register, or intermediate promotion receipt is theorem evidence.

Remark 5.1.4 (Technical extension status). The structures below are optional Technical Extension tools. They compare mathematical presentations, categorical models, parameterizations, or computational routes without enlarging the Core or restoring a global exact positive-threshold collapse functor.

5.2 Typed Mirror, Transfer, and correspondence regimes

Definition 5.2.1 (Mirror/Transfer setting). A Mirror/Transfer setting consists of declared source and target environments $\mathcal{A}$ and $\mathcal{B}$, together with a comparison datum

$$\text{MT} : \mathcal{A} \dashrightarrow \mathcal{B}.$$

Its record states whether MT is covariant, contravariant, adjoint-derived, span-valued, correspondence-valued, partially defined, or advisory only. The symbol $\dashrightarrow$ asserts no total functoriality by itself.

Definition 5.2.2 (Variance and domain record). A variance and domain record contains the source and target categories, object domain, morphism domain, parameter-domain action, covariance or contravariance, endpoint convention, coefficient action, functoriality level, and failure behavior. Missing variance leaves every morphism-level or Type IV use undefined.

Definition 5.2.3 (Covariant exact-transfer regime). A covariant exact-transfer regime consists of an exact functor between declared abelian realization categories, together with any parameter and filtration transport needed to compare the actual persistence profiles. An *exact-equivalence regime* additionally supplies a quasi-inverse and unit/counit isomorphisms.

Definition 5.2.4 (Contravariant mirror regime). A contravariant mirror regime consists of an exact functor

$$\text{D} : \mathcal{A}^{\text{op}} \longrightarrow \mathcal{B}$$

and, when persistence parameters are involved, an explicit parameter reorientation and endpoint transport. An *exact anti-equivalence regime* additionally supplies a contravariant quasi-inverse and coherent unit/counit isomorphisms.

Definition 5.2.5 (Adjunction-only regime). An adjunction-only regime consists of $F : \mathcal{A} \rightleftarrows \mathcal{B} : U$ with unit $\eta : \text{id} \Rightarrow UF$ and counit $\epsilon : FU \Rightarrow \text{id}$. No equivalence, exactness, conservativity, or persistence invariance is assumed unless separately stated.

Definition 5.2.6 (Span or correspondence regime). A span or correspondence regime records

$$X \xleftarrow{p} K \xrightarrow{q} Y$$

or analogous kernel data. It is not an object-level transfer functor until the required pullback, pushforward, derived, integration, or kernel-transform operations and their domains are explicitly constructed.

Proposition 5.2.7 (Raw Mirror/Transfer output is not Core-readable). *A raw dual object, transformed complex, span, kernel, numerical mirror candidate, or transfer output is not automatically a filtered object, persistence profile, admissible representative, tower artifact, diagnostic, ledger component, or certificate field.*

Proof. Each listed use requires a declared realization and a verifier-side schema. A raw output supplies neither by notation alone. $\square$

Proposition 5.2.8 (Surrogate-only mode blocks target reading). *When only the transformed surrogate is realized, every conclusion is about that surrogate. A theorem about the original target requires an exact or metric-certified comparison between the actual target and surrogate records.*

Proof. The target and surrogate are different typed objects. Shared naming or intended interpretation does not identify them. $\square$

5.3 Comparison squares, exactness levels, and local modes

Definition 5.3.1 (Mirror/Transfer comparison square). A Mirror/Transfer comparison square compares the typed composites

$$Q'MT \quad \text{and} \quad MT_QQ.$$

Its record contains every object, arrow, variance convention, and domain. No arrow is inferred from the diagrammatic shape.

Definition 5.3.2 (Strict and exact-isomorphic comparison). A comparison is *strict* when the two composites are equal as morphisms or functors on the declared scope. It is *exact-isomorphic* when an invertible natural 2-cell

$$\alpha : Q'MT \xRightarrow{\cong} MT_QQ$$

is supplied with the required naturality and pasting coherence. Exact-isomorphic does not mean strict equality.

Definition 5.3.3 (Controlled Mirror/Transfer 2-cell). A controlled 2-cell is a directed comparison

$$\alpha : Q'MT \Rightarrow MT_QQ$$

with source and target composites, scope, orientation, naturality or pseudonaturality data, composition rule, and a typed defect or cost semantics.

Definition 5.3.4 (Zero-reflecting cost semantics). A cost semantics δ is zero-reflecting for a declared purpose when

$$\delta(\alpha) = 0 \quad \implies \quad \alpha \text{ is the required identity or coherent isomorphism on that scope.}$$

Without zero reflection, a zero numerical cost remains only a zero value in the cost codomain.

Proposition 5.3.5 (Zero cost is not exactness by default). *The equation $\delta(\alpha) = 0$ does not imply strict or exact-isomorphic commutation unless the cost semantics are zero-reflecting and the required coherence is present.*

Proof. A cost map may identify distinct 2-cells or discard categorical information. Coherence is a separate non-scalar obligation. $\square$

Definition 5.3.6 (L-local comparison modes). A Mirror/Transfer comparison uses exactly one local mode:

strict, exact-isomorphic, metric-controlled, two-cell-controlled, advisory-only, not-compared,
failed, undefined.

These are local typing tokens, not Core atomic statuses.

Definition 5.3.7 (Projection to Core comparison modes). The local modes project as follows.

- (i) strict and exact-isomorphic project to *exact* when the equality or coherent isomorphism is registered;
- (ii) metric-controlled projects to *metric-gap-certified* only after an actual profile bound and every predicate-specific transfer hypothesis are supplied;
- (iii) two-cell-controlled projects to *exact* only through a coherent exact-isomorphism theorem, and to *metric-gap-certified* only through a certified conversion to an actual metric bound;
- (iv) advisory-only and not-compared project to *not compared*;
- (v) failed rejects an invoked path;
- (vi) undefined leaves every invoked dependent obligation undefined.

Theorem 5.3.8 (Attachment noninterference). *Two certificate records with identical verifier-side projections have the same Core atomic status and failure route, regardless of differences in unconverted Mirror/Transfer diagrams, analogy scores, costs, or attachments.*

Proof. Unconverted Mirror/Transfer data are not inputs of the Core verdict function. □

5.4 Adjunctions, mates, and Beck–Chevalley boundaries

Definition 5.4.1 (Typed adjunction square). Let

$$F : \mathcal{A} \rightleftarrows \mathcal{B} : U, \quad F' : \mathcal{A}' \rightleftarrows \mathcal{B}' : U'$$

be adjunctions, and let $Q : \mathcal{A} \rightarrow \mathcal{A}'$, $R : \mathcal{B} \rightarrow \mathcal{B}'$. A left-oriented comparison is a natural transformation

$$\alpha : F'Q \Rightarrow RF.$$

Its right mate is the corresponding transformation

$$\alpha^b : QU \Rightarrow U'R$$

constructed from the two units and counits.

Theorem 5.4.2 (Mate correspondence). *For fixed adjunction data, taking mates gives a bijection between the appropriately typed left-oriented and right-oriented 2-cells. It transports identities and the standard compatible pasting operations. It does not, by itself, transport a metric cost or prove that the mate is invertible.*

Proof. The usual unit–counit formulas are inverse by the triangle identities. Cost preservation and invertibility are additional properties not contained in the set-theoretic bijection of 2-cells. □

Definition 5.4.3 (Beck–Chevalley certificate). A Beck–Chevalley certificate is a typed adjunction square whose specified mate is an invertible natural transformation, together with the naturality and coherence needed by the downstream base-change or transfer statement.

Proposition 5.4.4 (An invertible displayed cell need not have invertible mate). *Invertibility of one displayed comparison cell does not automatically prove the Beck–Chevalley condition for its mate. Mate invertibility must be verified in the orientation actually consumed.*

Proof. The mate bijection changes orientation through units and counits. It is not a functor between groupoids that automatically sends an inverse in one hom-category to the inverse required in the mated hom-category. □

Proposition 5.4.5 (Adjunction is not equivalence). *An adjunction does not identify source and target objects. If both unit and counit are isomorphisms, the adjunction is an equivalence; weaker unit- or counit-only hypotheses yield only the corresponding full-faithfulness statement.*

Proof. This is the standard characterization of equivalences by adjoint equivalences. One-sided isomorphism controls only one adjoint's full faithfulness. $\square$

Proposition 5.4.6 (A correspondence is not a push–pull theorem). *A span or kernel correspondence does not supply a transfer theorem until every pullback, pushforward, derived correction, properness or finiteness hypothesis, and comparison cell used by the push–pull route is defined and verified.*

Proof. The correspondence object alone does not determine the required functors or their domains, much less their exactness or Beck–Chevalley behavior. $\square$

5.5 Metric control, whiskering, and the Mirror/Transfer ledger

Definition 5.5.1 (Profile-Lipschitz transfer). A realized Mirror/Transfer operation Q is L_Q -Lipschitz on a declared class of profiles when

$$d_B(QM, QN) \leq L_Q d_B(M, N)$$

for every admissible pair in that class. The value L_Q and the profile class are part of the theorem record.

Theorem 5.5.2 (Whiskering amplifies metric defects by the proved modulus). *If $d_B(M, N) \leq \varepsilon$ and Q is L_Q -Lipschitz on the compared profiles, then*

$$d_B(QM, QN) \leq L_Q \varepsilon.$$

A non-expansive transfer is the special case $L_Q \leq 1$.

Proof. Apply the defining Lipschitz inequality to the certified pair. $\square$

Proposition 5.5.3 (Functoriality is not non-expansiveness). *A functorial or pseudofunctorial Mirror/Transfer operation need not be non-expansive for bottleneck or interleaving distance. A quantitative modulus requires a separate theorem.*

Proof. Functoriality preserves composition and identities, not metric scale. $\square$

Definition 5.5.4 (Mirror/Transfer ledger component). Let $(V, \oplus, 0_V, \preceq)$ be the active metric-ledger quantale. A Mirror/Transfer ledger component is an element

$$\mathfrak{d}_{\text{MT}} \in V$$

with immutable source and target profile identities and a theorem-backed scalarization bound on the actual discrepancy consumed by the certificate.

Definition 5.5.5 (Dependency-aware path aggregation). A Mirror/Transfer path record identifies a chain of actual profiles

$$P_0, P_1, \dots, P_m$$

and certified edge defects. The complete path bound is formed using the declared ledger law and a dependency graph that distinguishes sequential edges, alternative paths, nested estimates, and already included downstream errors.

Theorem 5.5.6 (Metric path accumulation). *If*

$$d_B(P_{j-1}, P_j) \leq \varepsilon_j \quad (1 \leq j \leq m),$$

then

$$d_B(P_0, P_m) \leq \sum_{j=1}^m \varepsilon_j.$$

Alternative paths are not summed unless a separate aggregation theorem requires that operation.

Proof. Iterate the triangle inequality. Alternative paths are competing witnesses rather than sequential discrepancies. $\square$

Theorem 5.5.7 (Ledger-safe pasting of Mirror/Transfer 2-cells). *Let $\alpha_1, \dots, \alpha_m$ be a composable pasting diagram of controlled Mirror/Transfer 2-cells. Suppose the declared cost semantics supplies:*

- (i) *verified vertical and horizontal subcomposition laws;*
- (ii) *every required whiskering modulus;*
- (iii) *associativity, unit, and interchange coherence for the pasted diagram;*
- (iv) *certified conversion of each primitive cost into the active ledger quantale.*

Then the pasted comparison has the bound produced by those declared laws. In the additive scalar special case with non-expansive whiskering,

$$\delta(\alpha_m * \dots * \alpha_1) \leq \sum_{j=1}^m \delta(\alpha_j).$$

Without items (i)–(iv), the pasted diagram has no verifier-side aggregate defect.

Proof. Apply the certified composition and whiskering inequalities in the order prescribed by the pasting diagram. Coherence identifies the parenthesized composites to which the inequalities apply. The final ledger conversion is valid only after every primitive cost has entered the common quantale. $\square$

Proposition 5.5.8 (Taking mates does not preserve ledger cost by default). *Even when $\alpha \leftrightarrow \alpha^b$ are mates, no equality or inequality between $\delta(\alpha)$ and $\delta(\alpha^b)$ follows without a mate-compatible cost theorem. A ledger entry for one orientation cannot be silently reused for the other.*

Proof. Mate correspondence is defined by units and counits in hom-categories. A cost function is additional quantitative structure and need not be invariant under those operations. $\square$

Proposition 5.5.9 (Mixed-codomain aggregation is ill-typed). *A scalar analogy score, a categorical 2-cell, and a metric-ledger element cannot be combined by the same $\oplus$ unless certified conversions place them in a common declared quantale.*

Proof. The operands belong to different typed codomains and have different semantics. $\square$

Definition 5.5.10 (Adjusted Mirror/Transfer budget). *For an invoked path, the adjusted ledger aggregates exactly the base, window, collapse, realization, evaluation, morphism, and Mirror/Transfer components actually used. The dependency graph controls that aggregate. Its scalarization must satisfy the strict active gap condition; no component symbol names an independent hidden ledger.*

5.6 Exact affine parameter transfer

Definition 5.6.1 (Orientation-preserving affine parameter transport). Let

$$r(t) = at + c, \quad a > 0.$$

For a barcode interval $I = [b, d)$, with the evident convention at $+\infty$, define

$$r_*I = [r(b), r(d)).$$

Extend r_* occurrencewise to finite barcodes and set $r(W) = [r(u), r(v))$ for $W = [u, v)$.

Theorem 5.6.2 (Exact affine transport of windows, thresholds, and bottleneck distance). *For finite barcodes B, C , an orientation-preserving affine map $r(t) = at + c$, $a > 0$, and $\tau > 0$,*

$$r_*W_W B = W_{r(W)} r_*B,$$

$$r_*T_\tau^{\text{bar}} B = T_{a\tau}^{\text{bar}} r_*B,$$

and

$$d_B(r_*B, r_*C) = a d_B(B, C).$$

Consequently,

$$r_*(T_\tau^{\text{bar}} W_W B) = T_{a\tau}^{\text{bar}} W_{r(W)} r_*B.$$

Proof. Increasing affine transport preserves right-open endpoints and multiplies every finite interval length, every endpoint displacement, and every diagonal cost by a . The strict inequality $\ell(I) > \tau$ is therefore equivalent to $\ell(r_*I) > a\tau$. $\square$

Corollary 5.6.3 (Affine transport of repaired zero and metric reserve). *Under the hypotheses of Theorem 5.6.2, repaired zero is preserved with the scaled threshold. Moreover,*

$$2\varepsilon \leq \tau \iff 2a\varepsilon \leq a\tau.$$

Thus affine rescaling preserves the sharp zero-transfer reserve after all thresholds and metric radii are scaled coherently.

Proposition 5.6.4 (Nonlinear reparameterization has no default constant threshold transport). *For a general increasing nonlinear map r , no single constant c_r need satisfy*

$$r_*T_\tau^{\text{bar}} = T_{c_r\tau}^{\text{bar}} r_*$$

on all intervals. Equal source lengths at different birth locations may acquire different target lengths.

Proof. On a nonnegative domain, $r(t) = t^2$ sends two intervals of equal length located at different births to intervals of different lengths. A location-independent threshold scale cannot reproduce strict deletion for both. $\square$

Proposition 5.6.5 (Order reversal requires a separate dual persistence regime). *An order-reversing parameter map does not define an endofunctor of ordinary forward persistence by relabeling alone. It reverses morphism direction and changes right-open endpoint conventions; a typed anti-equivalence and parameter reorientation are required.*

Proof. Precomposition with an order-reversing map changes the variance of the persistence functor. The image of a right-open interval is not right-open in the same orientation without an additional convention. $\square$

Proposition 5.6.6 (Endpoint decoration under reversal–dual transport). *Let $\rho(t) = c - t$, and let $M = k_{[b,d)}$ be a finite-dimensional right-open interval module. Define the variance-correct reversal–dual object by*

$$(\mathbb{D}_\rho M)(t) := M(\rho(t))^*,$$

with structure maps induced by dualizing the reversed source maps. Its set-theoretic support is

$$(c - d, c - b],$$

not a right-open interval. Consequently, reversal–dual transport does not preserve the undecorated right-open barcode convention; a decorated endpoint category or an explicit endpoint-switch equivalence is required before barcode-level comparison.

Proof. The fiber is nonzero exactly when $b \leq c - t < d$, equivalently $c - d < t \leq c - b$. The structure maps are forward after the two variance reversals, but the endpoint decoration has changed. $\square$

Proposition 5.6.7 (Right-terminal tails become left-initial tails under affine reversal). *For $\rho(t) = c - t$,*

$$\rho([T, +\infty)) = (-\infty, c - T].$$

Hence an ordinary order-reversing mirror does not transport a right-terminal Type IV scope to another right-terminal scope. A right-tail diagnostic can be transferred only after an additional tail-reorientation theorem or after replacing it by a separately typed left-initial diagnostic.

Proof. The displayed equality is immediate from order reversal. Terminal-generic dimension is defined on a designated right-unbounded tail, so changing that tail to a left-unbounded initial region changes the diagnostic scope. $\square$

5.7 Collapse, evaluation, detector, and predicate transport

Proposition 5.7.1 (No automatic collapse or evaluation transfer). *A Mirror/Transfer operation does not commute with windowing, hard deletion, or repaired evaluation by default. Every shortcut requires strict comparison, exact-isomorphic comparison, or a certified metric defect for the actual profiles.*

Proof. Mirror/Transfer may change parameter scale, interval length, endpoint convention, multiplicity, or realization data, each of which can change threshold survival. $\square$

Theorem 5.7.2 (Gap-certified pre-threshold Mirror/Transfer transport). *Let B_L, B_R be finite pre-threshold profiles with*

$$d_B(B_L, B_R) \leq \varepsilon.$$

If neither profile has a finite bar length in

$$[\tau - \varepsilon, \tau + \varepsilon],$$

then

$$d_B(T_\tau^{\text{bar}} B_L, T_\tau^{\text{bar}} B_R) \leq \varepsilon.$$

Proof. Apply the sharp two-sided threshold-gap theorem to the actual compared pre-threshold profiles. $\square$

Theorem 5.7.3 (Sharp post-threshold zero transport). *Let E_L, E_R be genuine post-threshold profiles at threshold τ . If*

$$d_B(E_L, E_R) \leq \varepsilon, \quad 2\varepsilon \leq \tau,$$

then

$$E_L = 0 \iff E_R = 0.$$

Proof. Every surviving finite bar has length strictly greater than τ , hence diagonal cost strictly greater than $\tau/2 \geq \varepsilon$. Infinite bars have infinite diagonal cost. A nonzero profile therefore cannot be matched to zero within radius ε . $\square$

Proposition 5.7.4 (Sharpness of the half-threshold boundary). *If $2\varepsilon > \tau$, zero/nonzero transport can fail. Choose L with $\tau < L \leq 2\varepsilon$, let $E_L = 0$, and let E_R contain the single bar $[0, L)$. Then*

$$d_B(E_L, E_R) = L/2 \leq \varepsilon,$$

but exactly one profile is zero.

Proof. The single bar is matched to the diagonal at cost $L/2$. $\square$

Definition 5.7.5 (Mirror/Transfer detector transport record). A detector transport record identifies the source and target pre-threshold profiles, parameter map, endpoint convention, detector scope, complete source detector certificate, and either an exact transport of the detector construction or a profile-level theorem transporting only the repaired-zero predicate.

Theorem 5.7.6 (Detector-certified zero transfers only through profile control). *Assume a complete detector certificate proves*

$$T_\tau^{\text{bar}} B_L = 0.$$

If Theorem 5.7.2 applies and $2\varepsilon \leq \tau$, then

$$T_\tau^{\text{bar}} B_R = 0.$$

The detector samples, ranks, germ sets, and coverage artifacts do not transfer unless a separate exact detector-transport theorem is supplied.

Proof. The detector proves zero on the left. Gap-certified transport gives the repaired-profile bound, and Theorem 5.7.3 transports the zero predicate. The path-specific detector data are not determined by that predicate. $\square$

Proposition 5.7.7 (Budget pass alone does not transport predicates). *A Mirror/Transfer component that passes the scalar budget does not by itself transport repaired zero, admissible representatives, higher-Ext eligibility, Type IV diagnostics, transient-defect classification, or finite-to-infinite closure.*

Proof. Those are typed non-scalar predicates with independent hypotheses. $\square$

5.8 Exact equivalences, representatives, and higher Ext

Theorem 5.8.1 (Ext transport under exact equivalence). *Let $E : \mathcal{A} \rightarrow \mathcal{B}$ be an exact equivalence of abelian categories. For every $n \geq 0$ there are natural isomorphisms*

$$\text{Ext}_{\mathcal{A}}^n(X, Y) \cong \text{Ext}_{\mathcal{B}}^n(EX, EY).$$

In particular, Ext vanishing is preserved and reflected.

Proof. Transport Yoneda n -extensions along E and its exact quasi-inverse. The constructions are mutually inverse up to the unit and counit isomorphisms. $\square$

Theorem 5.8.2 (Argument reversal under exact anti-equivalence). *Let $D : \mathcal{A}^{\text{op}} \rightarrow \mathcal{B}$ be an exact anti-equivalence. Then*

$$\text{Ext}_{\mathcal{A}}^n(X, Y) \cong \text{Ext}_{\mathcal{B}}^n(DY, DX)$$

naturally in the variance-correct sense.

Proof. An exact sequence from Y to X is reversed by D , and the anti-equivalence transports Yoneda composition with the arguments exchanged. $\square$

Proposition 5.8.3 (Exact functors without equivalence do not transport Ext vanishing). *Exactness of a functor alone neither preserves nor reflects Ext vanishing as a statement about the full target Ext group. For preservation failure, let*

$$R = k[\epsilon]/(\epsilon^2)$$

and regard k -vector spaces as R -modules through $R \twoheadrightarrow k$. The resulting exact functor satisfies

$$\text{Ext}_k^1(k, k) = 0, \quad \text{Ext}_R^1(k, k) \neq 0.$$

For reflection failure, the exact zero functor sends every source extension group to the zero target group. Thus Ext transport requires an equivalence, a fully justified derived comparison, or a separately proved source/-target theorem.

Proof. The nonsplit exact sequence

$$0 \longrightarrow (\epsilon) \longrightarrow R \longrightarrow k \longrightarrow 0$$

identifies $(\epsilon) \cong k$ as an R -module and gives a nonzero class in $\text{Ext}_R^1(k, k)$, whereas vector spaces form a semisimple category. The zero-functor example shows failure of reflection. $\square$

Proposition 5.8.4 (Duality does not create the AK bridge). *An exact equivalence or anti-equivalence transports an already defined Ext group. It does not construct the standard n -eligible representative, higher edge realization, persistence-to-Ext bridge, or natural zero-reflection theorem required by the AK higher-Ext clause.*

Proof. Those data relate repaired persistence to a derived representative before any equivalence is applied. Ext transport begins only after the categorical inputs are defined. $\square$

Proposition 5.8.5 (Coefficient placement under anti-equivalence is load-bearing). *A statement*

$$\text{Ext}_{\mathcal{A}}^n(A, k) = 0$$

transports under an anti-equivalence to

$$\text{Ext}_{\mathcal{B}}^n(Dk, DA) = 0,$$

not automatically to $\text{Ext}_{\mathcal{B}}^n(DA, k) = 0$. Any coefficient identification and argument rearrangement require separate data.

Proof. Apply Theorem 5.8.2 with $X = A$, $Y = k$. The variance reverses the arguments. $\square$

Definition 5.8.6 (Mirror/Transfer representative certificate). A representative certificate contains an actual representative $C_{\tau, W}F$, its repaired-profile compatibility, and the exact or theorem-backed transport of that compatibility through the invoked Mirror/Transfer route.

Proposition 5.8.7 (Representative support is not representative validity). *A Mirror/Transfer diagram may suggest or construct a representative, but it does not establish*

$$\mathbf{P}_i(C_{\tau, W}F) \cong \text{Eval}_{i, W, \tau}(F)$$

without the corresponding compatibility witness.

Proof. Admissibility is the displayed profile relation, not the construction method. $\square$

5.9 Morphism defects, Type IV, and transient transport

Theorem 5.9.1 (Kernel and cokernel transport under exact equivalence). *Let $E : \mathcal{A} \rightarrow \mathcal{B}$ be an exact equivalence of abelian categories and $f : X \rightarrow Y$. Then*

$$E(\ker f) \cong \ker(Ef), \quad E(\operatorname{coker} f) \cong \operatorname{coker}(Ef).$$

Proof. Exact functors preserve kernels and cokernels, and the equivalence supplies the canonical comparison isomorphisms. $\square$

Theorem 5.9.2 (Kernel–cokernel swap under exact anti-equivalence). *Let $D : \mathcal{A}^{\text{op}} \rightarrow \mathcal{B}$ be an exact anti-equivalence and $f : X \rightarrow Y$. For the reversed morphism $Df : DY \rightarrow DX$,*

$$\ker(Df) \cong D(\operatorname{coker} f), \quad \operatorname{coker}(Df) \cong D(\ker f).$$

Proof. Apply D to the exact kernel–image–cokernel sequence of f ; exact contravariance reverses the sequence. $\square$

Definition 5.9.3 (Terminal-compatible exact transport). An exact equivalence or anti-equivalence is terminal-compatible on a monitored Type IV scope only when it includes:

- (i) a variance-correct transport of the actual monitored morphism;
- (ii) a parameter and endpoint transport carrying the declared source terminal scope to the declared target terminal scope, including the bounded window-cofinal or right-unbounded convention actually used;
- (iii) preservation of terminal tameness and terminal-generic fiber dimension after the stated variance correction;
- (iv) identification of the transformed window, threshold, and diagnostic scope actually consumed.

An order-reversing parameter map does not satisfy item (ii) merely by being an anti-equivalence; a scope-reorientation theorem is required whenever the terminal orientation changes.

Theorem 5.9.4 (Covariant Type IV transport). *Let $\phi : M \rightarrow N$ be the actual monitored Type IV morphism on a declared Main-compatible terminal scope. Under a terminal-compatible exact equivalence E ,*

$$\mu(E\phi) = \mu(\phi), \quad \nu(E\phi) = \nu(\phi).$$

Hence certified zero is preserved and reflected.

Proof. Theorem 5.9.1 identifies the defect profiles, and terminal compatibility preserves their terminal-generic dimensions. $\square$

Theorem 5.9.5 (Mirror swap of Type IV diagnostics). *Under a terminal-compatible exact anti-equivalence D , the variance-correct target morphism satisfies*

$$\mu(D\phi) = \nu(\phi), \quad \nu(D\phi) = \mu(\phi).$$

Thus the unordered cleanliness predicate $(\mu, \nu) = (0, 0)$ is preserved, while the ordered defect roles are exchanged.

Proof. Theorem 5.9.2 exchanges kernel and cokernel profiles. Terminal compatibility preserves the corresponding terminal-generic dimensions. $\square$

Proposition 5.9.6 (Order-reversing duality does not preserve right-terminal Type IV by default). *The reversal–dual construction of Proposition 5.6.7 does not satisfy Theorem 5.9.5 unless an additional theorem identifies the resulting left-initial defect data with a declared right-terminal target diagnostic. In particular, kernel/cokernel exchange alone is insufficient.*

Proof. Kernel/cokernel exchange is categorical, whereas tgdim is scope-sensitive. The parameter reversal sends the monitored right tail to a left tail, so the target terminal-generic dimension is not the transported source invariant without a new tail identification. $\square$

Proposition 5.9.7 (Object agreement does not control morphism defects). *Even identical source and target objects do not determine Type IV defects. For a nonzero object M , the morphisms*

$$\text{id}_M : M \rightarrow M, \quad 0 : M \rightarrow M$$

have different kernels and cokernels. Therefore object-level isomorphism or bottleneck proximity cannot replace a morphism-level transport theorem.

Proof. The identity has zero kernel and cokernel, whereas the zero map has kernel and cokernel isomorphic to M . $\square$

Definition 5.9.8 (Interior transient-defect transport). A transient-defect transport record identifies an actual kernel or cokernel interior defect profile on each route, the defect threshold τ_{def} , and an exact or metric comparison between those same typed defect profiles. Terminal/cofinal defects are recorded separately.

Theorem 5.9.9 (Metric-safe long-transient transport). *Let D_L, D_R be corresponding interior defect profiles. If*

$$d_B(T_{\tau_{\text{def}}}^{\text{bar}} D_L, T_{\tau_{\text{def}}}^{\text{bar}} D_R) \leq \varepsilon$$

and

$$2\varepsilon \leq \tau_{\text{def}},$$

then

$$T_{\tau_{\text{def}}}^{\text{bar}} D_L = 0 \iff T_{\tau_{\text{def}}}^{\text{bar}} D_R = 0.$$

This transfers only the absence of retained long interior defect.

Proof. Apply Theorem 5.7.3 at threshold τ_{def} . $\square$

Proposition 5.9.10 (Long-transient zero is not full Type IV cleanliness). *The conclusion of Theorem 5.9.9 does not exclude short transient defects or terminal/cofinal kernel and cokernel defects, and it does not create the actual Type IV morphism.*

Proof. Hard deletion at τ_{def} discards short intervals, while terminal defects are classified on a different scope. $\square$

5.10 Tower transport and finite-to-infinite discipline

Theorem 5.10.1 (Equivalences preserve existing tower colimits). *Let $E : \mathcal{A} \rightarrow \mathcal{B}$ be an equivalence and let $Q_0 \rightarrow Q_1 \rightarrow \dots$ have a colimit in $\mathcal{A}$. Then*

$$E\left(\text{colim}_n Q_n\right) \cong \text{colim}_n E(Q_n).$$

If $\phi : \text{colim}_n Q_n \rightarrow Q_\infty$ is the actual comparison morphism, its transported comparison is canonically $E\phi$.

Proof. An equivalence preserves every existing universal construction up to canonical isomorphism, since its quasi-inverse transports the universal property. $\square$

Proposition 5.10.2 (A left adjoint preserves colimits but not Type IV defects by itself). *A left adjoint preserves existing colimits, but it need not preserve kernels, reflect zero objects, or identify the transformed apex with an independently declared target apex. Therefore colimit preservation alone does not transport Type IV cleanliness.*

Proof. Colimit preservation is formal for left adjoints. Kernel preservation, conservativity, and apex comparison are independent properties. $\square$

Proposition 5.10.3 (Anti-equivalence converts colimits to limits). *An exact anti-equivalence transports a colimit diagram to a limit diagram in the variance-correct target. It does not automatically produce a target comparison of the same form*

$$\text{colim} \longrightarrow \text{apex}.$$

A dual Return or limit-to-coapex protocol must be stated separately.

Proof. Contravariance reverses cocones to cones and arrows to oppositely directed arrows. $\square$

Theorem 5.10.4 (Eventual constancy transport). *Suppose a tower is exactly constant from stage N , the stage- N object is compatibly isomorphic to the declared apex, and an exact equivalence transports the complete tower, transition maps, and apex isomorphism. Then the transported tower is exactly constant from stage N , has transported apex agreement, and satisfies the corresponding transported colimit comparison.*

Proof. Apply the equivalence to the transition isomorphisms and apex agreement, then use Theorem 5.10.1. $\square$

Proposition 5.10.5 (Finite-prefix Mirror agreement is not a Return theorem). *Agreement on finitely many stages, numerical convergence of Mirror/Transfer scores, or repeated compatibility plots does not prove eventual constancy, colimit identification, apex agreement, or a Return theorem.*

Proof. All such conclusions quantify over an unobserved tail or require a universal property not supplied by a finite prefix. $\square$

5.11 Statuses, artifacts, deferred CR, and interface boundary

Definition 5.11.1 (Mirror/Transfer status). A Mirror/Transfer record has one status among

strict, exact, controlled, advisory, unavailable, failed, undefined.

Strict and exact may satisfy only the displayed comparison obligation. Controlled additionally requires the complete ledger and predicate-specific theorem. Advisory and unavailable cannot support the affected certified path; failed rejects it; undefined leaves it undefined.

Definition 5.11.2 (Mirror/Transfer artifact bundle). An invoked bundle contains, as applicable: source and target identities; variance and parameter transport; the displayed square; natural transformations and mates; units and counits; exactness or anti-exactness evidence; barcodes and matchings; metric conversions; ledger path and dependency graph; representative and Ext data; actual tower morphisms; kernel and cokernel profiles; terminal-scope evidence; and replay instructions.

Proposition 5.11.3 (Stale Mirror/Transfer artifacts are non-consumable). *Changing any content-addressed source, target, variance convention, parameter map, threshold, window, mate orientation, cost semantics, or realization version makes the prior artifact stale for the changed claim.*

Proof. The changed field alters the identity or semantics of the comparison record. $\square$

Definition 5.11.4 (Deferred final Claim Register rule). Appendix L retains stable local labels and frozen TeX source. Formal Claim UUIDs, cross-appendix dependency closure, and the compact human-readable Claim Register are generated only after the complete appendix sequence is frozen. No Appendix-L JSON, CSV, provisional promotion receipt, or intermediate CR snapshot is emitted.

Definition 5.11.5 (Problem-specific Mirror/Transfer implication). A problem-specific implication has the form

$$\text{Mirror/Transfer compatibility} \implies \text{a declared external-domain conclusion.}$$

It is Interface or Spec material unless separately proved under the applicable problem-interface protocol.

Theorem 5.11.6 (No external-domain upgrade). *No theorem of this appendix alone proves a problem-specific external conclusion. Exact transport may move an already proved theorem between precisely identified equivalent regimes, but it does not create the source theorem or verify an unregistered interpretation of the target.*

Proof. The appendix controls comparison and transport. External semantics and their hypotheses are additional interface data. $\square$

5.12 Appendix L package theorem

Theorem 5.12.1 (Appendix L Mirror/Transfer package). *Within the Technical Extension layer, Appendix L supplies the following package.*

- (i) *Mirror/Transfer operations are separated into covariant, contravariant, adjunction, correspondence, partial, and advisory regimes.*
- (ii) *Strict equality, coherent exact isomorphism, metric control, and controlled 2-cells are different comparison types.*
- (iii) *Zero cost is exact only under zero-reflecting semantics and coherence.*
- (iv) *Adjunction mates are typed 2-cell correspondences. Beck–Chevalley invertibility and cost compatibility are separate obligations.*
- (v) *Metric defects are whiskered only through a proved Lipschitz modulus, and pasted 2-cell costs require declared composition, coherence, and ledger conversion laws.*
- (vi) *Orientation-preserving affine parameter transport exactly scales windows, thresholds, and bottleneck distance.*
- (vii) *Nonlinear or order-reversing parameter transport has no default hard-deletion rule; reversal–dual transport changes endpoint decoration and sends right tails to left tails.*
- (viii) *Pre-threshold replacement uses the two-sided closed threshold band, and post-threshold zero transfer has the sharp boundary $2\varepsilon \leq \tau$.*
- (ix) *Detector artifacts do not transfer merely because the repaired-zero predicate transfers.*
- (x) *Exact equivalences preserve Ext groups; exact anti-equivalences reverse their arguments, while exact functors without equivalence do not transport full Ext-vanishing statements.*
- (xi) *Duality transports an existing higher-Ext statement but does not create the AK persistence-to-Ext bridge.*
- (xii) *Exact equivalences preserve kernel and cokernel defects, while exact anti-equivalences exchange them.*
- (xiii) *Type IV transport requires the actual morphism and right-tail-compatible parameter semantics; ordinary order reversal is not terminal-compatible by default.*
- (xiv) *Long-transient zero is distinct from terminal Type IV cleanliness.*
- (xv) *Equivalences preserve existing colimits, left adjoints alone do not transport Type IV defects, and anti-equivalences convert colimits to limits.*
- (xvi) *Finite-prefix agreement is not finite-to-infinite closure.*

(xvii) *Formal CR registration is deferred until all appendices are frozen.*

(xviii) *No Mirror/Transfer statement silently enlarges the Core or proves an external theorem.*

Proof. The typed regime and comparison sections prove items (i)–(iv). The metric, pasting, affine, threshold, and detector sections prove the quantitative transport items. The Ext, representative, kernel/cokernel, Type IV, and transient sections prove the categorical and morphism-level items. The tower, deferred-register, and interface-boundary sections prove the remaining closure statements. $\square$

5.13 Explicit exclusions

Appendix L asserts none of the following.

- No Mirror operation is assumed to be an equivalence, anti-equivalence, or involution without proof.
- No adjunction is treated as an equivalence unless both unit and counit are isomorphisms.
- No span, kernel, or correspondence is treated as a push–pull functor without the required constructions.
- No zero-cost 2-cell is exact without zero reflection and coherence.
- No invertible displayed cell automatically satisfies the Beck–Chevalley condition for its mate.
- No mate correspondence preserves a numerical cost or ledger entry without a mate-compatible theorem.
- No pasted 2-cell diagram has an aggregate cost without composition, whiskering, coherence, and conversion laws.
- No functorial transfer is assumed non-expansive.
- No nonlinear or order-reversing parameter change preserves a constant hard threshold by default.
- No reversal–dual transport preserves right-open endpoint decoration or right-terminal Type IV scope by default.
- No global exactness or global non-expansiveness of positive-threshold hard deletion is assumed.
- No metric budget alone transports zero, admissibility, higher Ext, Type IV, or transient classification.
- No exact equivalence constructs the persistence-to-Ext bridge it transports.
- No exact functor without an equivalence or a separately proved comparison is treated as preserving or reflecting Ext vanishing.
- No anti-equivalence preserves the order of Ext arguments, kernel/cokernel roles, or colimit direction by convention.
- No object-level proximity controls morphism-level defects.
- No long-transient zero statement excludes terminal or short defects.
- No finite-stage compatibility establishes apex agreement or Return.
- No advisory Mirror/Transfer analogy supports Core acceptance.
- No problem-specific external theorem is proved here.

5.14 Provenance and final status

This appendix depends on the v20 window-first and threshold-gap semantics, representative and higher-Ext eligibility discipline, Type IV/transient-defect separation, non-compensating audit ledger, Appendix H advisory boundary, Appendix I discretization and finite-to-infinite discipline, Appendix J measurement separation, and Appendix K controlled-commutation calculus.

Final status. Document role: Technical Extension appendix. Evidence status: proved where a theorem or proposition is displayed, conditional where hypotheses are explicit, and operational for record and artifact definitions. Formal CR-level registration remains deferred until the entire appendix sequence is frozen.

6 Appendix M: Quantale and Ledger Semantics

6.1 Scope, v20 dependency baseline, and sovereignty

This appendix supplies the common value semantics for metric discrepancies, route costs, restart costs, controlled commutation and Mirror/Transfer defects, discretization errors, and theorem-backed measurement-to-profile conversions. It governs how such quantities may be aggregated and consumed by the repaired v20 audit layer. It does not create any missing mathematical object.

A ledger is a typed upper-bound calculus for an identified comparison path. It is not a substitute for the objects, morphisms, predicates, or proofs being compared.

Definition 6.1.1 (Ledger sovereignty principle). A ledger records declared discrepancies, costs, tolerances, and controlled deviations. It does not determine, compensate for, mutate, or retroactively repair a Core verdict. In particular, no ledger value replaces

$$\text{Eval}_{1,w,\tau}(F) = 0,$$

an admissible representative, a standard higher-edge eligibility record, an actual Type IV comparison morphism, terminal-tameness evidence, detector completeness, finite-to-infinite reduction, or any required artifact.

Definition 6.1.2 (No hidden ledger default). There is no implicit global aggregation law. Additive, supremum, product, weighted, status, lexicographic, and hybrid ledgers have different semantics. Every consumed ledger declares its value domain, order, aggregation, unit, support relation, scalarization when used, protected profile family, and failure behavior before the audit result is evaluated.

Definition 6.1.3 (v20 dependency baseline). Appendix M is interpreted relative to AK-HPDST v20.0.0 Main Chapters 1–8, Foundation Appendices A–G, the frozen Technical Extension Appendices H–L, the reviewed CM-v20 and TB-v20 mathematical statements at their individually verified strengths, and the CR-v20 baseline as global governance for inherited claims. No joint CM/TB promotion or hash synchronization is inferred, and no unreviewed Problem Interface or Search/Platform defect namespace is imported.

Definition 6.1.4 (Deferred final Claim Register rule). Appendix M retains stable local labels and a frozen TeX source. Formal Claim UUIDs, complete cross-appendix dependency closure, and the compact final Claim Register are generated only after the entire appendix sequence is frozen. No Appendix-M JSON, CSV, provisional promotion receipt, or intermediate Claim Register is normative.

Proposition 6.1.5 (Exact mode is not a fabricated zero-budget mode). *A proved equality or coherent profile isomorphism may be consumed in exact mode without a positive metric ledger. It is not automatically represented by a synthetic zero-cost component, and it does not by itself certify positive-radius robustness.*

Proof. Exact identity and metric robustness are different proof objects. Encoding an exact claim by an invented metric record would add a comparison mode and support theorem that were never established. $\square$

6.2 Ordered value systems, quantales, and scalarization

Definition 6.2.1 (Ordered commutative ledger monoid). An ordered commutative ledger monoid is a tuple

$$(V, \preceq, \oplus, 0_V)$$

where $(V, \oplus, 0_V)$ is a commutative monoid, $\preceq$ is a preorder, and aggregation is monotone:

$$v \preceq v', \quad w \preceq w' \implies v \oplus w \preceq v' \oplus w'.$$

The induced preorder equivalence is

$$v \simeq_{\preceq} w \iff v \preceq w \text{ and } w \preceq v.$$

Antisymmetry is optional; content-addressed record identity remains distinct from preorder equivalence.

Definition 6.2.2 (Commutative unital quantale). A commutative unital quantale is an ordered commutative ledger monoid for which $(V, \preceq)$ is a complete lattice and

$$v \oplus \bigvee_{j \in J} w_j = \bigvee_{j \in J} (v \oplus w_j)$$

for every family $\{w_j\}_{j \in J}$, and symmetrically in the other variable. Full quantale structure is required only when an argument invokes arbitrary joins, countable aggregation, residuals, or quantale-enriched closure.

Remark 6.2.3 (Terminology discipline). A four-tuple $(V, \preceq, \oplus, 0_V)$ is called a ledger monoid unless the complete-lattice and join-distribution data are supplied. The word “quantale” is not used merely as a synonym for a collection of costs.

Remark 6.2.4 (Compatibility with the Main quantale contract). The Main Core contract continues to require the declared Core metric ledger to live in a commutative quantale. Appendices H–L often use only the finite commutative-monoid reduct of that quantale, but this does not weaken or reinterpret the Core requirement. An ordered ledger monoid introduced only for Technical Extension bookkeeping may be consumed by the Core metric budget only when it is itself the declared Core quantale or when a certified unit-preserving conversion into that quantale, together with a target-side profile-bound support record, is supplied.

Definition 6.2.5 (Bottleneck scalarization). A bottleneck scalarization on a ledger monoid is a map

$$\text{val}_B : V \longrightarrow [0, +\infty]$$

that is monotone, normalized, and subadditive:

$$v \preceq w \implies \text{val}_B(v) \leq \text{val}_B(w), \quad \text{val}_B(0_V) = 0,$$

$$\text{val}_B(v \oplus w) \leq \text{val}_B(v) + \text{val}_B(w).$$

Scalarization supplies a numerical comparison interface. It becomes an upper bound for a mathematical discrepancy only through a profile-bound support record. It is not required to be injective, additive, or zero-reflecting.

Definition 6.2.6 (Zero-reflecting scalarization). A scalarization is zero-reflecting on a declared subset $S \subseteq V$ when

$$v \in S, \quad \text{val}_B(v) = 0 \implies v \approx_{\preceq} 0_V.$$

Zero-reflection is an additional theorem; it is never inferred from normalization alone.

Proposition 6.2.7 (Zero scalar value is not automatically exactness). *The equality $\text{val}_B(v) = 0$ does not by itself prove that the underlying operational comparison is strict, coherently isomorphic, or identical as a morphism-level construction.*

Proof. A non-zero-reflecting scalarization may collapse nontrivial ledger values to zero. Even an actual bottleneck distance of zero identifies barcode profiles only in the registered metric regime; it does not manufacture an equality of source constructions or comparison morphisms. $\square$

Definition 6.2.8 (Profile-bound support record). A profile-bound support record for $v \in V$ identifies two actual barcode profiles B, C , their scope, coefficient field, window and endpoint conventions, and theorem or replayable artifact evidence that

$$d_B(B, C) \leq \text{val}_B(v).$$

The record also declares whether v is primitive, aggregated, converted, or inherited from a selected comparison route.

Definition 6.2.9 (Gate-scoped metric ledger). For one invoked metric comparison and one protected pre-threshold profile family $\mathcal{B}$, a gate-scoped metric ledger is an element

$$\mathbf{d}_{\text{met}} \in V$$

with a profile-bound support record for every comparison to which it is applied. For Core consumption, V is the declared Core commutative quantale or is connected to it by the certified conversion required in Remark 6.2.4. Its scalar budget is

$$\varepsilon_B(\mathbf{d}_{\text{met}}) := \text{val}_B(\mathbf{d}_{\text{met}}).$$

The audited threshold budget is the typed strict inequality

$$\varepsilon_B(\mathbf{d}_{\text{met}}) < \text{gap}_\tau(\mathcal{B}),$$

where gap_τ is the ledger-facing alias of the registered threshold clearance.

Proposition 6.2.10 (Budget pass is scope-bound). *A budget pass is valid only for the declared profiles, route, ledger monoid, scalarization, aggregation theorem, threshold, protected family, and artifact identities. Reusing the same displayed number under another scope does not reuse the proof.*

Proof. The support relation binds the ledger element to actual profiles and a specific inference. Changing those inputs changes the proposition whose discrepancy is bounded. $\square$

Proposition 6.2.11 (Budget pass is not B-Gate pass). *The strict metric-budget inequality is one conjunctive audit obligation. It does not imply repaired topological vanishing, higher-edge eligibility, Type IV cleanliness, detector completeness, artifact adequacy, or full B-Gate⁺ pass.*

Proof. The listed obligations are non-scalar verifier-side records and predicates. Numerical slack cannot manufacture them. $\square$

6.3 Comparison graphs and path-sound aggregation

Definition 6.3.1 (Metric comparison edge). A metric comparison edge is a directed record

$$e : B \longrightarrow C$$

with a ledger value $d_e \in V$ and support evidence

$$d_B(B, C) \leq \text{val}_B(d_e).$$

Direction records the operational route even when the bottleneck distance itself is symmetric.

Definition 6.3.2 (Selected comparison path). A selected comparison path is a composable finite sequence

$$B_0 \xrightarrow{e_1} B_1 \xrightarrow{e_2} \cdots \xrightarrow{e_m} B_m.$$

Its ledger aggregate is

$$d_\gamma := d_{e_1} \oplus \cdots \oplus d_{e_m}.$$

The path identifier and intermediate profile identities are part of the certificate.

Theorem 6.3.3 (Finite path soundness). *For a selected comparison path as above,*

$$d_B(B_0, B_m) \leq \sum_{j=1}^m \text{val}_B(d_{e_j})$$

and therefore

$$d_B(B_0, B_m) \leq \text{val}_B(d_\gamma)$$

whenever the aggregation support theorem additionally proves that $\text{val}_B(d_\gamma)$ dominates the path discrepancy.

Proof. The first inequality is the triangle inequality for d_B followed by the edge support bounds. Subadditivity alone gives $\text{val}_B(d_\gamma) \leq \sum_j \text{val}_B(d_{e_j})$, not the reverse. Hence the second inequality is not automatic and must be supplied by the declared aggregation support theorem. This distinction prevents an abstract monoid operation from being mistaken for a conservative error composition. $\square$

Corollary 6.3.4 (Canonical additive path ledger). *For*

$$V = [0, +\infty], \quad \oplus = +, \quad \text{val}_B = \text{id},$$

the path aggregate

$$d_\gamma = \sum_{j=1}^m d_{e_j}$$

is automatically a conservative bottleneck upper bound.

Proposition 6.3.5 (Abstract aggregation without a path theorem is non-verdict). *A formal expression $d_1 \oplus \dots \oplus d_m$ does not enter the Core budget unless the component records form a supported route, a supported parallel family, or another theorem-backed aggregation pattern.*

Proof. The monoid law combines values, not profile comparisons. A support theorem is required to connect the algebraic aggregate to the actual discrepancy consumed by the proof. $\square$

Definition 6.3.6 (Alternative-route family). An alternative-route family is a nonempty set of independently certified paths $\Gamma(B, C)$ with the same identified endpoints. Its mathematical route bound is

$$\varepsilon_{\text{route}}(B, C) := \inf_{\gamma \in \Gamma(B, C)} \varepsilon_\gamma,$$

where each ε_γ is an independently proved upper bound for $d_B(B, C)$. Audit consumption of the infimum additionally requires reconstructible evidence for its value or for a certified upper enclosure of it; otherwise one explicit route is selected.

Proposition 6.3.7 (Alternative routes are minimized, not accumulated). *Alternative paths between the same endpoints are not sequential defects and are not added. If every ε_γ bounds the same actual distance, then*

$$d_B(B, C) \leq \inf_{\gamma} \varepsilon_\gamma.$$

A certificate either selects one route or records a justified route infimum.

Proof. The fixed number $d_B(B, C)$ is bounded by every member of the family and hence by their infimum. Adding alternatives would count mutually exclusive constructions as if all were traversed. $\square$

Definition 6.3.8 (Cycle record). A cycle record is a selected path beginning and ending at the same profile identity. Its ledger value is only a certified upper-bound allowance for deviation accumulated along that route. A positive value does not prove route inconsistency, because the bound may be loose and $d_B(B, B) = 0$. A zero value does not prove strict coherence unless a separate zero-defect-to-coherence theorem and the required higher coherence data are supplied.

6.4 Canonical components, aliases, and dependency control

Definition 6.4.1 (Canonical metric component slots). At one gate scope, metric-admissible discrepancies are placed in the v20 canonical slots

$$\begin{array}{cccccc} \mathfrak{d}_{\text{alg}}, & \mathfrak{d}_{\text{disc}}, & \mathfrak{d}_{\text{meas}}, & \mathfrak{d}_{\text{rep,met}}, & \mathfrak{d}_{\text{real,met}}, \\ & \mathfrak{d}_{\text{cmp}}, & \mathfrak{d}_{\text{crop}}, & \mathfrak{d}_{\text{cont}}, & \mathfrak{d}_{\text{ov}}, & \mathfrak{d}_{\text{rst}}. \end{array}$$

Appendix-local names such as $\mathfrak{d}_{\text{adv}}$, $\mathfrak{d}_{\text{comm}}$, $\mathfrak{d}_{\text{MT}}$, $\mathfrak{d}_\phi$, $\mathfrak{d}_{\text{src}}$, and $\mathfrak{d}_{\text{ser}}$ are subcomponents or aliases only after an explicit placement map is declared.

Definition 6.4.2 (Component support record). A component support record declares:

- (i) the component identity and canonical slot;
- (ii) the actual comparison edge, profile pair, or supported parallel family;
- (iii) its value domain, units, normalization, and scope;
- (iv) the theorem, verified computation, or artifact supplying the bound;
- (v) every upstream component already included in the value;
- (vi) staleness conditions and failure behavior.

Definition 6.4.3 (Defect relation types). Two displayed defect names have exactly one declared relation on the consuming scope:

- (i) *alias*: the names denote the same primitive discrepancy;
- (ii) *refinement*: one name decomposes the other into supported subcomponents;
- (iii) *aggregate*: one name is the declared aggregate of listed components;
- (iv) *disjoint*: the components represent non-overlapping primitive sources;
- (v) *alternative*: the components are mutually exclusive route bounds;
- (vi) *blocked overlap*: possible overlap has not been resolved, so joint consumption is undefined.

Proposition 6.4.4 (No double counting, omission, or hidden inclusion). *A metric ledger is invalid if it counts one primitive discrepancy twice, omits an invoked primitive discrepancy, or aggregates a component together with a downstream value that already includes it without a declared inclusion correction.*

Proof. The resulting scalar would no longer be the declared upper bound for the selected comparison route. Double counting may cause false rejection; omission or hidden inclusion may cause false acceptance. $\square$

Definition 6.4.5 (Unit and normalization compatibility). Components may be aggregated only after their units, parameter scales, normalizations, and bottleneck conventions are aligned by an exact identification or a certified conversion. Numerical values with different scales do not become commensurable by sharing the codomain $[0, +\infty]$.

Proposition 6.4.6 (No favorable-error cancellation). *A Core metric ledger is an upper-bound calculus. Negative credits, favorable deviations, signed cancellation, or post-hoc rebates are forbidden unless a joint theorem directly proves a conservative upper bound for the combined discrepancy.*

Proof. Norm and bottleneck upper bounds do not cancel merely because underlying signed quantities have opposite orientation. Unproved cancellation can understate the distance used by the threshold theorem. $\square$

Definition 6.4.7 (Future namespace extension protocol). A later Problem Interface, Search/Platform, proof-store, or execution appendix may introduce a defect name only by declaring its type, canonical slot or non-metric status, support record, relation to existing components, and failure route. Appendix M does not pre-register problem-specific names before those appendices are frozen.

Proposition 6.4.8 (A delta symbol is not a metric component). *Identity, synchronization, permission, adequacy, environment, replay, claim-status, and schema-conformance failures remain qualitative governance defects unless a separate theorem converts them into an actual bottleneck discrepancy bound.*

Proof. These failures primarily determine whether a record may be consumed at all. Their notation does not supply a distance between the protected profiles. $\square$

6.5 Additive, supremum, product, weighted, and hybrid models

Definition 6.5.1 (Additive sequential ledger). The additive sequential ledger is

$$V = [0, +\infty], \quad v \oplus w = v + w, \quad 0_V = 0,$$

with identity scalarization. It is canonical for a supported chain of discrepancies composed by the triangle inequality.

Definition 6.5.2 (Supremum ledger). The supremum ledger is

$$V = [0, +\infty], \quad v \oplus w = \max\{v, w\}, \quad 0_V = 0.$$

It is admissible for a parallel or uniform family only when the target discrepancy is measured by a supremum metric or another theorem proves that the worst coordinate bounds the consumed claim.

Proposition 6.5.3 (Max aggregation is unsafe for generic sequential paths). *From edge bounds*

$$d_B(B_0, B_1) \leq a, \quad d_B(B_1, B_2) \leq b$$

one cannot generally conclude

$$d_B(B_0, B_2) \leq \max\{a, b\}.$$

Thus the max ledger does not replace additive path aggregation without an additional ultrametric or contraction theorem.

Proof. The triangle inequality gives only $d_B(B_0, B_2) \leq a + b$. Ordinary metric spaces admit triples with endpoint distance strictly larger than both edge lengths. $\square$

Definition 6.5.4 (Product ledger). For ledger monoids $V_1, \dots, V_r$, the componentwise product has

$$V = \prod_{j=1}^r V_j, \quad (v_j) \oplus (w_j) = (v_j \oplus_j w_j),$$

with componentwise order. Coordinatewise comparisons may govern an auxiliary product-valued obligation. The scalar Core threshold budget, however, is discharged only after a conservative bottleneck scalarization

$$\text{val}_B : V \longrightarrow [0, +\infty]$$

and an actual profile-bound support theorem are supplied.

Proposition 6.5.5 (Coordinatewise product pass is not the scalar Core budget). *Strict inequalities in every product coordinate do not by themselves establish*

$$\text{val}_B(\mathbf{d}_{\text{met}}) < \text{gap}_\tau(\mathcal{B}).$$

They may support a separate multi-coordinate policy, but the real threshold-clearance comparison still requires the declared scalarization and discrepancy theorem consumed by the Core.

Proof. The Core threshold gap is a real number attached to identified barcode profiles. A tuple order does not produce a real bottleneck upper bound without a projection theorem. $\square$

Definition 6.5.6 (Hybrid metric-status ledger). A hybrid ledger consists of a scalarizable, profile-supported metric coordinate and one or more status, provenance, availability, or governance coordinates. It passes only when the metric coordinate satisfies the strict scalar Core budget and every invoked qualitative coordinate is admissible. A favorable metric reserve does not compensate for a failed or undefined status coordinate.

Definition 6.5.7 (Lexicographic auxiliary ledger). A lexicographic ledger declares an ordered list of coordinates and compares values by the first differing coordinate. It is an auxiliary governance order unless a separate theorem proves that its pass predicate conservatively implies the Core metric budget and all required qualitative conditions.

Definition 6.5.8 (Weighted scalar report). A weighted scalar report has the form

$$R_w = \sum_{j=1}^m w_j \varepsilon_j, \quad w_j \geq 0,$$

for declared component bounds ε_j . It enters the Core budget only when a joint support theorem proves

$$d_B(B, C) \leq R_w$$

for the actual protected profiles and route.

Proposition 6.5.9 (Weights are not free discounts). *For a generic sequential path known only through primitive bounds ε_j , the choice $w_j < 1$ is not conservative. Down-weighting is admissible only when zero, irrelevance, inclusion, correlation, contraction, or another domination theorem justifies the reduction.*

Proof. The triangle inequality supplies the undiscounted sum. Multiplying a primitive upper bound by a number below one can produce a value smaller than the actual edge discrepancy. $\square$

Proposition 6.5.10 (Weights and priorities are predeclared). *Weights, product-to-scalar maps, lexicographic priority, and status projection are declared before the budget verdict is evaluated. Changing them after observing failure defines a new certificate rather than a repair of the old one.*

Proof. Each choice changes the proposition being checked and may change the verdict. $\square$

6.6 Conversions, refinements, and remaining reserve

Definition 6.6.1 (Ledger conversion). A ledger conversion from $(V, \oplus_V, 0_V, \preceq_V)$ to $(W, \oplus_W, 0_W, \preceq_W)$ is a declared monotone map

$$\kappa : V \longrightarrow W$$

that preserves the no-defect unit,

$$\kappa(0_V) = 0_W,$$

and satisfies the declared lax aggregation inequality

$$\kappa(v \oplus_V v') \preceq_W \kappa(v) \oplus_W \kappa(v').$$

For Core use, every converted value additionally requires a target-side profile-bound support record.

Definition 6.6.2 (Conservative scalar conversion). A conversion is scalar-conservative for an identified profile comparison when, after every unit and normalization alignment,

$$\text{val}_B^W(\kappa(v)) \geq \text{val}_B^V(v)$$

for every supported source value, or when a stronger direct theorem proves that the target scalarization bounds the same actual discrepancy. If the conversion also changes the parameter scale, threshold, or clearance, a separate gap-transport record is required. A target-side pass is sound only relative to the target-side protected profiles and transported gap.

Proposition 6.6.3 (Discrepancy conversion does not transport the gap automatically). *Conservative transport of a discrepancy bound does not by itself transport a strict threshold-budget pass when the conversion changes units, parameterization, threshold, window, or protected profile family.*

Proof. A pass compares two quantities: the transported discrepancy bound and the applicable target clearance. Controlling only the first leaves the second unidentified. $\square$

Proposition 6.6.4 (Lax monoidality alone does not preserve the metric bound). *Monotonicity and lax aggregation of κ do not by themselves show that $\text{val}_B^W(\kappa(v))$ bounds the discrepancy supported by v .*

Proof. The conversion algebra controls value composition but says nothing about the numerical scalarization or the identified profile pair. A conservative scalar relation or direct support theorem is additionally required. $\square$

Definition 6.6.5 (Ledger refinement). A refinement replaces a coarse supported component d by finer components $d_1, \dots, d_m$, declares their relation to the original primitive source, and supplies a new aggregation support theorem for the refined route. An order inequality between ledger values alone is not a replacement for the new discrepancy proof.

Proposition 6.6.6 (A smaller refined number is not automatically safer). *Even if*

$$d_1 \oplus \dots \oplus d_m \preceq d,$$

the refined aggregate is not audit-admissible unless it independently bounds the actual discrepancy consumed by the certificate.

Proof. Order comparison in the ledger does not imply that the smaller value remains an upper bound for the protected profile distance. $\square$

Definition 6.6.7 (Scalar remaining reserve). For a finite protected gap $g > 0$ and scalar budget $\varepsilon < g$, the remaining scalar reserve is

$$\text{Res}(g; \varepsilon) := g - \varepsilon > 0.$$

It is descriptive budget capacity, not a new discrepancy component and not permission to omit later defects.

Definition 6.6.8 (Quantale residual, when invoked). In a quantale, aggregation by a fixed v preserves joins and therefore has a right adjoint

$$v \multimap w := \bigvee \{r \in V : v \oplus r \preceq w\}.$$

This residual records order-theoretic remaining capacity relative to w . A strict real-valued pass still requires the bottleneck scalarization and the declared strict gap policy.

Proposition 6.6.9 (No post-hoc pass engineering). *A conversion, refinement, weight policy, route selection, or scalarization introduced after observing a failed budget produces a new auditable record. It cannot silently rewrite the prior failure as pass.*

Proof. The ledger policy and support graph are inputs to the verdict. Changing them changes the certificate identity. $\square$

6.7 Infinite aggregation, restart chains, and global closure

Definition 6.7.1 (Quantale sum of a family). For a family $\{d_j\}_{j \in J}$ in a commutative quantale, define

$$\bigoplus_{j \in J} d_j := \bigvee_{F \subseteq J \text{ finite}} \bigoplus_{j \in F} d_j.$$

This is an algebraic aggregate. Its use as a metric upper bound still requires a supported infinite-route or parallel-family theorem.

Theorem 6.7.2 (Summable Cauchy route bound). *Let $(B_n)_{n \geq 0}$ be finite barcode profiles contained in a declared complete ambient bottleneck-metric space and suppose*

$$d_B(B_n, B_{n+1}) \leq \varepsilon_n, \quad \sum_{n=0}^{\infty} \varepsilon_n < \infty.$$

Then (B_n) is Cauchy and converges to some ambient limit $\widehat{B}$. Every such limit satisfies

$$d_B(B_0, \widehat{B}) \leq \sum_{n=0}^{\infty} \varepsilon_n.$$

Identification of $\widehat{B}$ with a separately declared apex profile B_∞ requires an additional apex-agreement or reconstruction witness.

Proof. For $m > n$, the triangle inequality gives

$$d_B(B_n, B_m) \leq \sum_{j=n}^{m-1} \varepsilon_j,$$

and the tail tends to zero. Completeness supplies an ambient limit $\widehat{B}$. Passing $m \rightarrow \infty$ in the bound from B_0 yields the endpoint inequality. Completeness alone does not identify that metric limit with an independently named apex object. $\square$

Proposition 6.7.3 (Finite-prefix pass is not infinite closure). *Passing every tested finite prefix does not prove summability, existence of a limit profile, identification with a declared apex, eventual constancy, or a Return theorem.*

Proof. Each conclusion requires a universal tail or limit statement not contained in finitely many checks. $\square$

Definition 6.7.4 (Restart-chain ledger). A restart chain records local window budgets, overlap comparisons, and restart-transition edges

$$B_0 \xrightarrow{r_0} B_1 \xrightarrow{r_1} \dots$$

Local profile errors, overlap errors, and restart errors occupy distinct primitive records unless an inclusion theorem identifies them.

Theorem 6.7.5 (Finite restart-chain soundness). *For a finite restart chain with supported transition bounds ρ_j ,*

$$d_B(B_0, B_m) \leq \sum_{j=0}^{m-1} \rho_j.$$

A global certificate must also supply every invoked overlap and local-window obligation; the endpoint bound does not replace them.

Proof. The distance bound is the triangle inequality. Overlap and local obligations are separate predicates in the coverage and Restart architecture. $\square$

Proposition 6.7.6 (Windowwise pass is not global pass). *A pass on each window does not imply a global pass without a declared cover, overlap compatibility, restart-transition control, global aggregation, and the finite-to-infinite theorem required by the claimed scope.*

Proof. Local inequalities do not establish compatibility or identify the global object. $\square$

6.8 Threshold-safe consumption and sharp zero transport

Definition 6.8.1 (Pre-threshold comparison contract). A metric-gap-certified threshold comparison binds the ledger to the actual pre-threshold profiles B, C , proves

$$d_B(B, C) \leq \varepsilon_B(\mathfrak{d}_{\text{met}}),$$

and supplies the registered two-sided closed-band evidence at the same radius. No post-threshold or source-level surrogate silently replaces these profiles.

Theorem 6.8.2 (Ledgered post-threshold zero transfer). *Let E, E' be finite post-threshold barcode profiles at the same strict deletion threshold $\tau > 0$, possibly with infinite bars. Suppose*

$$d_B(E, E') \leq \varepsilon.$$

If

$$2\varepsilon \leq \tau,$$

then

$$E = 0 \iff E' = 0.$$

Proof. For a finite barcode E , the bottleneck distance to the empty barcode is the maximum diagonal cost of its bars. Every surviving finite bar has length strictly greater than τ , hence diagonal cost strictly greater than $\tau/2 \geq \varepsilon$, while an infinite bar has infinite diagonal cost. Thus $E \neq 0$ implies $d_B(E, 0) > \varepsilon$. Apply this observation to either side when the other side is empty. $\square$

Proposition 6.8.3 (Sharpness of the half-threshold boundary). *If $2\varepsilon > \tau$, choose L with*

$$\tau < L \leq 2\varepsilon.$$

Then the post-threshold profile $E = \{[0, L)\}$ and the empty profile satisfy

$$d_B(E, 0) = L/2 \leq \varepsilon, \quad E \neq 0.$$

Thus the condition $2\varepsilon \leq \tau$ is sharp.

Remark 6.8.4 (Strict operational reserve). The mathematical boundary permits equality. Numerical and interval-certified audit policies may require $2\varepsilon < \tau$ to avoid equality-sensitive rounding and representation ambiguity.

Proposition 6.8.5 (Budget pass alone does not transport predicates). *A strict ledger pass does not by itself transport representative admissibility, higher Ext eligibility or vanishing, detector completeness, morphism-level Type IV diagnostics, or finite-to-infinite closure.*

Proof. Each is a typed predicate requiring its own transport theorem and non-scalar records. $\square$

Corollary 6.8.6 (Long-transient zero transport). *Let D, D' be finite constructible actual interior kernel-defect profiles, or finite constructible actual interior cokernel-defect profiles, extracted with the same interior/terminal convention and of the same declared type. If*

$$d_B\left(T_{\tau_{\text{def}}}^{\text{bar}} D, T_{\tau_{\text{def}}}^{\text{bar}} D'\right) \leq \varepsilon$$

and $2\varepsilon \leq \tau_{\text{def}}$, then the two thresholded defect profiles have the same zero/nonzero classification. This does not control short transient or terminal/cofinal defects.

6.9 Non-scalar obligations and morphism-level boundaries

Definition 6.9.1 (Non-scalar obligation class). The following remain outside the metric ledger whenever invoked: source and target identity, representative existence, profile isomorphisms, standard n -eligibility, higher-edge realization, actual comparison morphisms, kernel and cokernel construction, terminal-tameness, detector completeness, claim-use permission, artifact adequacy, manifest consistency, and finite-to-infinite or Return witnesses.

Proposition 6.9.2 (Metric slack cannot repair missing objects). *Increasing the scalar reserve or decreasing a metric estimate cannot repair a missing non-scalar obligation.*

Proof. The missing item is not an element of the ledger value domain and is not ordered by the bottleneck scalarization. $\square$

Proposition 6.9.3 (Object distance does not control morphism defects). *Even with identical source and target objects, object-level distance does not determine kernels or cokernels. For a nonzero persistence module M , the maps*

$$\text{id}_M : M \rightarrow M \quad \text{and} \quad 0 : M \rightarrow M$$

have the same endpoint objects but different kernel and cokernel profiles.

Proof. The identity has zero kernel and cokernel, whereas the zero map has kernel and cokernel isomorphic to M . $\square$

Definition 6.9.4 (Morphism-defect ledger admission). A morphism-related component enters the metric ledger only after actual morphisms are supplied and a theorem or artifact proves a bottleneck bound between the corresponding actual kernel profiles, actual cokernel profiles, or another explicitly declared morphism invariant. A distance between source and target objects is insufficient.

Proposition 6.9.5 (Type IV ledger does not replace Type IV). *A cost associated with constructing, transporting, or comparing a tower morphism does not replace the morphism, terminal-tameness evidence, or the canonical diagnostics*

$$\mu = \text{tgdim}(\ker \phi), \quad \nu = \text{tgdim}(\text{coker } \phi).$$

Proof. The cost and the diagnostic are different typed objects. The latter is computed from the actual morphism-level defect profiles. $\square$

Proposition 6.9.6 (Measurement and advisory values remain outside until converted). *A Betti integral error, AI confidence, warning score, spectral indicator, or visual agreement does not enter $\mathfrak{d}_{\text{met}}$ without a theorem-backed conversion to an actual profile discrepancy bound.*

Proof. Scalar or advisory closeness is not an inverse stability theorem for persistence profiles. $\square$

6.10 Manifests, lifecycle, and verdict semantics

Definition 6.10.1 (Ledger manifest entry). A ledger manifest entry records:

- (i) value system, order, aggregation, unit, and scalarization;
- (ii) protected profiles, threshold, comparison mode, and support graph;
- (iii) primitive components, canonical slots, units, and relation declarations;
- (iv) selected route or supported parallel-family theorem;
- (v) conversions, refinements, weights, and residual policy when invoked;

- (vi) total metric value, scalar budget, gap, reserve, and verdict;
- (vii) artifact identities, environment, replay procedure, and staleness conditions.

Definition 6.10.2 (Ledger local verdict). A ledger budget check has exactly one local state:

pass, reject, undefined, not_invoked.

Missing profiles, unsupported components, blocked overlap, missing scalarization, missing gap data, or an unresolved conversion yield `undefined`. A fully defined false strict inequality yields `reject`.

Proposition 6.10.3 (Manifest reconstruction is mandatory). *A nontrivial ledger cannot support certification unless an independent verifier can reconstruct the component graph, support records, selected route, aggregate, scalarization, and strict comparison.*

Proof. Without those data, neither the claimed upper bound nor its relation to the protected profiles is checkable. □

Proposition 6.10.4 (Stale ledger records are non-consumable). *Changing any protected profile, route edge, component artifact, normalization, scalarization, threshold, gap, environment, or aggregation policy makes the prior ledger record stale for the changed run.*

Proof. The changed dependency defines a different supported comparison proposition and therefore a different certificate identity. □

Proposition 6.10.5 (Qualitative failure is not metric slack). *Identity, adequacy, synchronization, permission, replay, environment, and schema failures block or reject consumption according to their own status rules. They are not repaired by unused numerical reserve.*

Proof. They concern whether the record is validly available, not the distance between two protected profiles. □

6.11 Appendix M package theorem

Theorem 6.11.1 (Appendix M quantale-ledger package). *Within the Technical Extension layer, Appendix M establishes the following package.*

- (i) *Ledger semantics are local, typed, scope-bound, and non-Core by themselves.*
- (ii) *Ordered ledger monoids are distinguished from genuine commutative quantales.*
- (iii) *Core metric consumption requires a monotone, normalized, subadditive bottleneck scalarization and an actual profile-bound support record.*
- (iv) *Exact mode is not fabricated as a zero-budget metric mode.*
- (v) *Component aggregation is sound only through a selected path, supported parallel family, or another explicit aggregation theorem.*
- (vi) *Additive path aggregation is canonical for sequential bottleneck bounds; max aggregation requires a uniform or ultrametric theorem.*
- (vii) *Alternative routes are selected or minimized rather than cumulatively charged.*
- (viii) *Canonical component placement, unit compatibility, alias relations, and no-double-counting rules are mandatory.*
- (ix) *Weighted, product, lexicographic, status, and hybrid models require their own conservative projection rules; coordinatewise product pass is not the scalar Core threshold budget.*

- (x) *Conversion and refinement algebra do not replace target-side discrepancy support.*
- (xi) *Countable aggregation requires summability, completeness or limit data, and the claimed finite-to-infinite identification.*
- (xii) *The sharp post-threshold zero-transfer boundary is $2\varepsilon \leq \tau$; strict inequality may be retained as an operational reserve.*
- (xiii) *Metric budgets do not replace representative, higher-Ext, detector, Type IV, artifact, or Return obligations.*
- (xiv) *Object-level distances do not control morphism-level kernel and cokernel defects.*
- (xv) *Later defect namespaces are admitted only through the future extension protocol after their source appendices are frozen.*
- (xvi) *Reconstructible manifests and explicit local verdict states are mandatory for audit use.*

Proof. Items (i)–(iv) follow from the sovereignty, value-system, scalarization, and exact-mode sections. Items (v)–(vii) are the comparison-graph results. Items (viii)–(x) are the component, model, conversion, and refinement disciplines. Item (xi) is the infinite-route and Restart boundary. Item (xii) is Theorem 6.8.2 and Proposition 6.8.3. Items (xiii)–(xiv) are the non-scalar and morphism-level boundaries. Items (xv)–(xvi) are the namespace-extension and manifest rules. $\square$

6.12 Explicit exclusions

Appendix M asserts none of the following.

- No single aggregation law is imposed globally.
- No displayed sum, maximum, weight, tuple, or lexicographic order is Core-admissible without a support theorem.
- No scalarization is assumed injective, additive, or zero-reflecting.
- No exact comparison is automatically a positive-radius robustness certificate.
- No alternative routes are added as if all were traversed.
- No primitive discrepancy is counted twice or silently omitted.
- No component with incompatible units or normalization is aggregated without certified conversion.
- No favorable deviation, negative credit, or post-hoc rebate cancels a metric upper bound by default.
- No max ledger is used for a generic sequential route.
- No weight below one discounts a primitive edge without a domination theorem.
- No coordinatewise product pass, lexicographic order, status value, or hybrid record becomes the scalar Core bottleneck budget without an admissible projection and actual profile-bound theorem.
- No conversion, refinement, weight change, or route change silently rewrites a previous failure.
- No finite-prefix or windowwise pass proves global or infinite closure.
- No budget pass alone proves repaired zero, higher Ext zero, detector completeness, Type IV cleanliness, or B-Gate⁺ pass.
- No object-level bottleneck bound manufactures a comparison morphism or controls its kernel and cokernel.

- No advisory, measurement, source, serialization, identity, adequacy, synchronization, environment, replay, or permission defect becomes metric slack merely by notation.
- No problem-specific arithmetic, PDE, geometric, mirror, categorical, cryptographic, Search, or platform defect namespace is pre-promoted here.
- No Appendix-M JSON, CSV, provisional Claim Register, or intermediate promotion artifact is required or emitted.

6.13 Provenance and final status

Appendix M depends on the v20 metric and threshold-gap contract of Main Chapters 1–3, the coverage, overlap, Restart, audit, and closure disciplines of Chapters 6–8, Foundation Appendices A–G, and the frozen Technical Extension Appendices H–L. It supplies the common ledger semantics for later appendices without pre-registering their problem-specific defect names and without enlarging the Core.

Final status. Document role: Technical Extension appendix. Evidence status: proved where a theorem or proposition is stated, operational where a manifest or lifecycle rule is declared, and conditional only where an explicitly named support or conversion theorem is required. Formal CR-level registration remains deferred until the complete appendix sequence is frozen.

7 Appendix N: Derived, Sheaf, Stack, and Higher-Categorical Extension Discipline

7.1 Scope, v20 boundary, and Part III closure role

Appendix N closes the Technical Extension sequence H–N. Its purpose is not to place more mathematics inside the Core by terminology, but to state precisely how derived, dg, sheaf, cosheaf, stack, multiparameter, stable, and higher-categorical constructions may surround the repaired v20 certificate architecture without changing its typing.

A higher enhancement may organize a proof route. It acquires verifier authority only through a theorem-backed extraction or transport into the exact Core-readable object consumed by the certificate.

Definition 7.1.1 (Higher-enhancement sovereignty principle). A derived, sheaf-theoretic, stack-theoretic, multiparameter, dg, stable, or higher-categorical enhancement is Technical Extension infrastructure. It does not determine, compensate for, mutate, or retroactively repair a Core verdict. In particular, it does not replace

$$\text{Eval}_{n,W,\tau}(F) = T_{\tau}^{\text{bar}}(W_W \mathbf{P}_n(F)),$$

a finite detector certificate, a standard degree- n eligibility and edge record, an admissible representative, an actual Type IV comparison morphism, a transient-defect certificate, a finite-to-infinite reduction, a Return theorem, or the strict metric budget.

Definition 7.1.2 (No terminology-to-Core upgrade principle). The words derived, stack, descent, equivalence, stable, coherent, ∞ -categorical, or hypercomplete do not promote a statement. Core use requires a precise source and target, complete hypotheses, a proof at the asserted strength, a permitted invocation role, and the verifier-side artifacts required by the consuming theorem.

Definition 7.1.3 (v20 synchronization boundary). The degree- n repaired-zero and higher-obstruction edge calculus already belonging to the v20 Core is not redefined in Appendix N. Appendix N governs only extensions beyond that fixed regime, including non-field coefficients, altered amplitudes, dg or A_{∞} models, sheaf and stack realizations, multiparameter extraction, higher descent, and enhanced Return structures.

Definition 7.1.4 (v20 dependency baseline). Appendix N is interpreted relative to AK-HPDST v20.0.0 Main Chapters 1–8, Foundation Appendices A–G, the reviewed CM and TB mathematical statements only at their individually verified strengths, and the frozen Technical Extension Appendices H–M. No joint CM/TB promotion, source-artifact hash synchronization, final Claim Register binding, or unreviewed Problem Interface, Search/Platform, or Companion statement is inferred here.

Definition 7.1.5 (Deferred final Claim Register rule). Appendix N retains stable local labels and a frozen TeX source. Formal Claim UUIDs, complete H–N dependency closure, and the compact final Claim Register are generated only after the remaining appendix layers and Companion are frozen. No Appendix-N JSON, CSV, provisional promotion receipt, or intermediate Claim Register is normative.

Remark 7.1.6 (Canonical notation). The canonical Type IV cokernel diagnostic is ν , not a Latin u . The fixed Core order remains window first and hard deletion second. The mathematical post-threshold zero-transfer boundary is $2\varepsilon \leq \tau$; a numerical workflow may retain $2\varepsilon < \tau$ as an operational reserve.

7.2 Enhancement interfaces and verifier-side re-entry

Definition 7.2.1 (External enhancement). An external enhancement is a declared mathematical environment $\mathcal{E}$ together with objects, morphisms, higher cells, descent structures, or realization procedures intended to organize AK-relevant data. Examples include derived and dg categories, stable ∞ -categories, sheaf and cosheaf categories, stacks, higher stacks, enriched categories, and multiparameter persistence categories.

Definition 7.2.2 (Enhancement interface). An enhancement interface records:

- (i) the ambient category, site, model, or higher category;
- (ii) source and target classes and all variance conventions;
- (iii) the object- and morphism-level domain of definition;
- (iv) the extraction or realization route to Core-readable data;
- (v) the comparison cells, equivalences, or metric bounds used;
- (vi) the ledger treatment of every nonzero profile discrepancy;
- (vii) the non-scalar obligations and failure behavior;
- (viii) immutable source, proof, computation, and replay artifacts.

Definition 7.2.3 (Verifier-side re-entry). An enhancement re-enters a certificate only by constructing a new verifier-side record of one of the following declared types:

- (i) a constructible one-parameter persistence profile or finite barcode;
- (ii) a filtered object producing such a profile;
- (iii) a repaired evaluation or complete detector certificate;
- (iv) an admissible representative and its compatibility record;
- (v) a standard degree- n eligibility, edge, or zero-reflection record;
- (vi) an actual tower comparison morphism or actual interior defect profile;
- (vii) a metric-ledger component supported by an actual profile bound;
- (viii) a finite-to-infinite reduction or external Return record.

The enhancement source and the new verifier record remain distinct content-addressed objects.

Theorem 7.2.4 (Enhancement noninterference). *Two complete certificate records with the same verifier-side projection have the same Core atomic status and failure route even when their derived, sheaf, stack, multiparameter, or higher-categorical attachments differ.*

Proof. The Core verdict factors through repaired profiles, typed structural obligations, actual morphisms, admitted metric bounds, and the declared audit record. Enhancement attachments outside that projection are not arguments of the verdict function. $\square$

Proposition 7.2.5 (Declaration-only re-entry is invalid). *Renaming an enhancement object as a repaired profile, representative, Ext edge, tower artifact, Type IV diagnostic, or metric component does not produce a valid verifier-side record.*

Proof. The target types require mathematical constructions and support evidence. A field rename supplies neither. $\square$

Proposition 7.2.6 (Object extraction and morphism extraction are independent). *An enhancement interface that extracts objects but does not extract the morphisms consumed by a theorem cannot support a morphism-level conclusion, including Type IV kernel/cokernel claims, naturality, coherence, or tower comparison.*

Proof. A morphism-level predicate is not determined by source and target objects alone. The actual map and its compatibility data are non-scalar obligations. $\square$

7.3 The fixed v20 higher-Ext edge and its extension boundary

Definition 7.3.1 (Derived Ext convention). For a ring R and objects A, K in a declared derived category of R -modules, write

$$\mathrm{Ext}_R^n(A, K) := \mathrm{Hom}_{D(R)}(A, K[n]).$$

Every extension beyond the v20 field-coefficient regime declares the side of the module convention, boundedness or amplitude, the ambient derived category, and the coefficient object. The one-sided hyper-Ext spectral sequence below is stated for a coefficient module K concentrated in degree zero. A genuine coefficient complex requires its own two-complex hyper-Ext or t -structure spectral sequence and is not silently covered by that displayed E_2 page.

Theorem 7.3.2 (Field-coefficient synchronization). *For a bounded complex A of finite-dimensional vector spaces over a field k , there is a canonical isomorphism, natural contravariantly in A ,*

$$\mathrm{Ext}_k^n(A, k) \cong \mathrm{Hom}_k(H^{-n}(A), k).$$

Consequently,

$$\mathrm{Ext}_k^n(A, k) = 0 \iff H^{-n}(A) = 0.$$

This is a synchronization statement for the established v20 field regime; it does not construct the persistence-to-cohomology edge identification required by an AK eligibility record.

Proof. The standard t -structure gives the natural hyper-Ext spectral sequence

$$E_2^{p,q} = \mathrm{Ext}_k^p(H^{-q}(A), k) \implies \mathrm{Ext}_k^{p+q}(A, k).$$

Because finite-dimensional k -vector spaces form a semisimple abelian category, every term with $p > 0$ vanishes. In total degree n the unique possible term is

$$E_2^{0,n} = \mathrm{Hom}_k(H^{-n}(A), k),$$

so the canonical edge map is an isomorphism, natural contravariantly in A . No chosen splitting of A is used. Finite-dimensional linear duality is zero-reflecting. $\square$

Theorem 7.3.3 (Hyper-Ext spectral sequence with module coefficients). *Let R be a ring, let A be a bounded complex of R -modules, and let K be an R -module regarded as a complex concentrated in degree zero. Under the standard convergence hypotheses, there is a natural spectral sequence*

$$E_2^{p,q} = \mathrm{Ext}_R^p(H^{-q}(A), K) \implies \mathrm{Ext}_R^{p+q}(A, K).$$

An edge map is an isomorphism only under additional degeneration, vanishing, and extension-control hypotheses. For a coefficient complex not concentrated in degree zero, the displayed E_2 page is not the declared default and a separate spectral-sequence record is required.

Proof. Choose an injective resolution of the coefficient module, form the associated Hom double complex, and filter by the cohomological degree of A . Boundedness gives the standard convergent hyper-Ext spectral sequence. Nonzero higher columns, differentials, and filtration extensions can contribute to the abutment, so existence of the spectral sequence alone does not identify an edge with the abutment. $\square$

Proposition 7.3.4 (Failure of the field edge in every positive degree over a general ring). *For every $n \geq 1$, the vanishing of $H^{-n}(A)$ need not imply $\mathrm{Ext}_R^n(A, K) = 0$ over a general ring. Let*

$$R = k[\epsilon]/(\epsilon^2), \quad A = K = k$$

with k in cohomological degree zero. Then

$$H^{-n}(A) = 0 \quad \text{but} \quad \mathrm{Ext}_R^n(k, k) \cong k \neq 0$$

for every $n \geq 1$.

Proof. The periodic free resolution

$$\cdots \xrightarrow{\epsilon} R \xrightarrow{\epsilon} R \xrightarrow{\epsilon} R \longrightarrow k \longrightarrow 0$$

is exact because the kernel and image of multiplication by ϵ are both the ideal (ϵ) . Applying $\text{Hom}_R(-, k)$ makes every differential zero, since ϵ acts trivially on k . Hence $\text{Ext}_R^n(k, k) \cong k$ for every $n \geq 0$, while the complex $A = k[0]$ has no negative cohomology. $\square$

Theorem 7.3.5 (Ext-acyclic zero-reflecting coefficient criterion). *Fix $n \geq 0$ and assume the hypotheses of Theorem 7.3.3. Suppose the coefficient module K is Ext-acyclic on the declared cohomology class, so*

$$\text{Ext}_R^p(M, K) = 0 \quad (p > 0)$$

for every M appearing as $H^j(A)$. Then

$$\text{Ext}_R^n(A, K) \cong \text{Hom}_R(H^{-n}(A), K).$$

If K is additionally zero-reflecting under $\text{Hom}_R(-, K)$ on that class (for example, a cogenerator there), then

$$\text{Ext}_R^n(A, K) = 0 \iff H^{-n}(A) = 0.$$

Proof. All $p > 0$ columns of the hyper-Ext spectral sequence vanish, so the total degree n abutment has only the $E_2^{0,n}$ contribution. The zero-reflection statement is the declared property that $\text{Hom}_R(M, K) = 0$ forces $M = 0$ on the cohomology class. $\square$

Definition 7.3.6 (Admissible non-field higher edge). A non-field or altered-coefficient higher edge is admissible only after recording the ambient category, coefficient object, boundedness, spectral-sequence convergence, every vanishing or degeneration hypothesis, the exact edge map, its naturality, the AK profile extractor identified with H^{-n} , and any zero-reflection theorem used in the reverse direction.

Proposition 7.3.7 (No synthetic bridge promotion). *A formal identity obtained by placing a detector value or persistence vector space in degree $-n$ does not establish a natural persistence-to-Ext bridge for an external realization class.*

Proof. The construction supplies only a chosen algebraic model. It supplies neither a natural realization theorem nor a representative-independent edge extractor or external Return arrow. $\square$

7.4 Stable, dg, and A_∞ enhancement discipline

Definition 7.4.1 (Stable enhancement). A stable enhancement consists of a stable category, stable ∞ -category, dg category, or pretriangulated A_∞ category together with a declared homotopy category, t -structure or heart when used, and a realization/extraction route to Core-readable data.

Definition 7.4.2 (Heart-readable morphism). A stable morphism $f : X \rightarrow Y$ is heart-readable for Type IV purposes when X and Y lie in a declared abelian heart $\mathcal{A}$, the morphism is represented in that heart, and its kernel and cokernel persistence profiles are extracted by a declared exact or theorem-backed procedure.

Theorem 7.4.3 (Cofiber-to-heart kernel/cokernel calculation). *Let $\mathcal{C}$ be a stable category with a t -structure and heart $\mathcal{A}$. For a morphism $f : X \rightarrow Y$ in $\mathcal{A}$, let*

$$X \xrightarrow{f} Y \longrightarrow C_f \longrightarrow X[1]$$

be a cofiber sequence. Then

$$H^{-1}(C_f) \cong \ker_{\mathcal{A}}(f), \quad H^0(C_f) \cong \text{coker}_{\mathcal{A}}(f),$$

and $H^j(C_f) = 0$ for $j \notin \{-1, 0\}$.

Proof. Apply the cohomology functors of the t -structure to the cofiber sequence. Since X and Y lie in the heart, the resulting long exact sequence reduces to

$$0 \rightarrow H^{-1}(C_f) \rightarrow X \xrightarrow{f} Y \rightarrow H^0(C_f) \rightarrow 0,$$

with all other cohomology groups zero. $\square$

Proposition 7.4.4 (Stable fiber/cofiber is not automatically Type IV data). *A stable fiber or cofiber does not by itself equal the canonical persistence kernel or cokernel defect profile. The heart, t -exact extraction, and actual morphism identification are load-bearing.*

Proof. Kernel and cokernel are heart-level constructions. In a stable category they are recovered from specified cohomology objects only after a t -structure and a heart-readable morphism have been fixed. $\square$

Theorem 7.4.5 (t -exact equivalence transport). *Let $E : \mathcal{C} \rightarrow \mathcal{D}$ be a t -exact equivalence of stable categories. Its restriction to the hearts is an exact equivalence. For every heart morphism f ,*

$$E(\ker f) \cong \ker(Ef), \quad E(\operatorname{coker} f) \cong \operatorname{coker}(Ef).$$

When the extraction and terminal-scope conventions are also preserved, the corresponding canonical Type IV zero predicate is transported.

Proof. A t -exact equivalence preserves the truncation subcategories and hence restricts to an exact equivalence of hearts. Exact equivalences preserve kernels and cokernels. The final statement additionally uses the declared profile and terminal-scope identifications. $\square$

Proposition 7.4.6 (Derived equivalence is not heart transport by default). *An equivalence of derived or stable categories does not transport heart kernels, cokernels, or Type IV diagnostics unless it is compatible with the declared t -structures. In particular, the shift autoequivalence*

$$[1] : D^b(\mathcal{A}) \longrightarrow D^b(\mathcal{A})$$

is an exact triangulated equivalence but is not t -exact for the standard t -structure and does not restrict to an endofunctor of the heart except on the zero object.

Proof. For a nonzero $X \in \mathcal{A}$, the shifted object $X[1]$ has its nonzero standard cohomology in a different degree and therefore does not lie in the heart. Thus an abstract derived equivalence alone supplies no heart-level morphism from which the canonical kernel and cokernel profiles could be read. $\square$

Proposition 7.4.7 (Quasi-equivalence is not strict implementation identity). *A dg or A_∞ quasi-equivalence identifies the appropriate homotopy-level mathematics, but it does not identify chain-level implementations, chosen representatives, filtrations, numerical environments, or content-addressed artifacts without additional comparison data.*

Proof. Quasi-equivalence is a categorical relation at the declared homotopical strength. The listed implementation data are extra structure not contained in that relation. $\square$

Proposition 7.4.8 (Higher products are not invisible by default). *An A_∞ enhancement with the same cohomology category as another model may have different higher multiplications. Any theorem depending on those higher operations requires their own artifacts and invariance proof.*

Proof. The cohomology category records m_2 up to the induced product but does not in general determine $m_3, m_4, \dots$. A conclusion sensitive to the higher structure cannot be reconstructed from cohomology alone. $\square$

7.5 Sheaf and cosheaf extraction

Definition 7.5.1 (Sheaf or cosheaf enhancement). A sheaf or cosheaf enhancement consists of a declared space or site X , a coefficient category, a sheaf, cosheaf, or derived sheaf object, and the functors used to extract parameter-indexed linear data.

Definition 7.5.2 (One-parameter sheaf extraction). A one-parameter sheaf extraction is a declared functor

$$\mathcal{X}_i : \mathcal{S} \longrightarrow [(\mathbb{R}, \leq), \text{Vect}_k]$$

from a specified sheaf/cosheaf class $\mathcal{S}$, with its variance, support convention, section or costalk rule, parameter map, and morphism action explicitly recorded. Core readability additionally requires the image to lie in the declared constructible one-parameter regime.

Theorem 7.5.3 (Conditional Core readability of sheaf extraction). *Let $\mathcal{F} \in \mathcal{S}$. If $\mathcal{X}_i(\mathcal{F})$ is a constructible one-parameter persistence object, its object and morphism identities are reconstructible, and the window and threshold conventions agree with the Core, then*

$$T_\tau^{\text{bar}}(W_W \mathcal{X}_i(\mathcal{F}))$$

is a Core-readable repaired profile. No claim about the original sheaf follows in the reverse direction without a separate faithfulness or Return theorem.

Proof. The hypotheses place the extracted object exactly in the Core persistence category and fix the required evaluation order. The reverse statement would require the extraction to reflect the relevant sheaf property, which is not part of the hypotheses. $\square$

Proposition 7.5.4 (Derived sections are not exact by notation). *A section, pushforward, or global-cohomology construction does not preserve short exact sequences merely because it is functorial. Any exactness or edge argument must state whether ordinary or derived functors are used and which acyclicity or degeneration hypotheses apply.*

Proof. Ordinary section and pushforward functors are generally only left exact, and their derived functors record the obstruction to exactness. $\square$

Proposition 7.5.5 (Sheaf equivalence does not fix extraction conventions). *An equivalence of sheaf categories does not by itself identify parameter maps, section/costalk variance, window endpoints, filtration normalization, or the extracted persistence profile.*

Proof. Those choices are additional structure on the enhancement interface and are not determined by the abstract categorical equivalence. $\square$

7.6 Multiparameter persistence and one-parameter reduction

Definition 7.6.1 (Multiparameter enhancement). A multiparameter enhancement is a functor

$$M : (\mathbb{R}^d, \leq_{\text{prod}}) \longrightarrow \text{Vect}_k, \quad d \geq 2,$$

or a declared finite-poset analogue. It is not a one-parameter barcode object by default.

Definition 7.6.2 (Monotone slice extraction). For a monotone map $\gamma : (\mathbb{R}, \leq) \rightarrow (\mathbb{R}^d, \leq_{\text{prod}})$, the slice extraction is

$$\gamma^* M := M \circ \gamma.$$

Its record contains γ , the parameter scale, endpoint convention, normalization, and the information intentionally discarded away from $\text{im } \gamma$.

Proposition 7.6.3 (A fixed slice is not conservative). *A single monotone slice does not in general determine a multiparameter persistence module. For the diagonal slice $\gamma(t) = (t, t)$ in $\mathbb{R}^2$, let $M = 0$ and let N be the rectangle interval module supported on*

$$[0, 1) \times [2, 3).$$

Then $N \neq 0$ but

$$\gamma^* N = 0 = \gamma^* M.$$

Proof. The rectangle does not meet the diagonal, so the restriction of N to every point $\gamma(t)$ is zero, although N is nonzero on its support. $\square$

Proposition 7.6.4 (Finite slice observation is not complete without coverage). *A finite family of slice restrictions cannot be read as a complete multiparameter zero certificate without a theorem showing that the chosen family detects every admissible support region in the declared module class.*

Proof. Whenever an admissible order-convex support region is disjoint from all chosen slice images, adding its interval module changes the multiparameter object while leaving every recorded slice unchanged. $\square$

Definition 7.6.5 (Multiparameter information-loss record). A one-parameter reduction of multiparameter data records the retained family of observables, the discarded directions, any coverage or faithfulness theorem, the permitted conclusions, and a status-valued failure route when no complete reduction is known. Information loss is not automatically a bottleneck metric component.

Proposition 7.6.6 (No barcode import by dimension reduction). *The existence of interval decompositions for selected one-parameter slices does not supply a global barcode decomposition of the original multiparameter module.*

Proof. The one-parameter structure theorem applies to each restricted functor. It does not classify the higher-dimensional functor or its extension data between directions. $\square$

7.7 Descent, gluing, and certificate descent

Definition 7.7.1 (Object descent datum). For a cover $\{U_a \rightarrow U\}$, an object descent datum consists of local objects, comparison isomorphisms on overlaps, and all cocycle or higher coherence data required by the ambient sheaf, stack, or higher-stack semantics.

Definition 7.7.2 (Certificate descent datum). A certificate descent datum consists of local verifier-side records together with coherent identifications of their scopes, windows, thresholds, profile identities, detector sources, representatives, actual morphisms, ledgers, artifacts, and verdict fields on every required overlap and higher intersection.

Proposition 7.7.3 (Pairwise agreement does not imply descent). *Pairwise object isomorphisms on overlaps do not form a descent datum unless the triple and higher cocycle conditions are satisfied.*

Proof. On a triple overlap, transition maps must satisfy $g_{ab}g_{bc} = g_{ac}$, with the appropriate higher coherence in a higher category. One may choose individually valid pairwise isomorphisms that violate this identity, so no global object is represented. $\square$

Proposition 7.7.4 (Object descent and certificate descent are distinct). *Effective descent of enhancement objects does not imply descent of verifier-side certificates, and certificate descent does not imply a sheaf or stack theorem about the external objects.*

Proof. A certificate contains mathematical and governance fields not present in ordinary object descent. Conversely, gluing audit records does not establish effectivity in an external stack unless its own descent hypotheses are proved. $\square$

Definition 7.7.5 (Verifier-record stack condition). A verifier-record assignment is stack-like on a declared site when local verifier records, their actual morphisms, immutable artifact references, and all higher compatibility data satisfy effective descent, and when the verifier-side projection and failure routing commute with that descent.

Theorem 7.7.6 (Conditional certificate descent). *Assume a verifier-record assignment satisfies Definition 7.7.5. Let a cover carry local complete certificate records and an effective descent datum for every invoked verifier-side field. Then there exists a global certificate record unique up to the declared descent equivalence. Its Core verdict is computed from the descended verifier-side projection; it is not inferred from the local verdict labels alone.*

Proof. Effectivity gives a global record and coherent identifications with its local restrictions. Compatibility of the verifier projection and failure routing transports the actual descended obligations. Verdict labels are outputs and therefore cannot replace descent of the data from which they are computed. $\square$

Proposition 7.7.7 (Local pass does not imply global pass). *Even when every local certificate passes, a global certificate may remain undefined or reject if higher overlap coherence, global artifact identity, global budget aggregation, or finite-to-infinite closure is absent or fails.*

Proof. These are global conjunctive obligations not discharged by the list of local verdicts. $\square$

7.8 Stacks, hyperdescent, and higher coherence

Definition 7.8.1 (Stack or higher-stack enhancement). A stack or higher-stack enhancement records the site, topology, truncation level, descent or hyperdescent convention, local charts, transition cells, and the extraction route to Core-readable data.

Proposition 7.8.2 (Hyperdescent is not finite proof compression). *A hyperdescent theorem does not by itself give a finite AK proof certificate. A finite certificate additionally requires a finite reduction of the invoked hypercover data, complete coverage, immutable artifacts, and a theorem that the finite record determines the consumed global predicate.*

Proof. Hyperdescent may quantify over an infinite simplicial or hypercover diagram. Effectivity of that diagram does not imply that a finite prefix is complete. $\square$

Proposition 7.8.3 (Truncation level is load-bearing). *Replacing a higher stack by its homotopy category, set of isomorphism classes, or finite truncation may discard automorphisms, obstruction cells, and higher coherence. Such truncation is verifier-safe only for predicates proved invariant under the discarded levels.*

Proof. Higher morphisms can carry nontrivial descent and obstruction information not visible in the truncation. Invariance under forgetting them is an additional theorem. $\square$

Definition 7.8.4 (Higher coherence certificate). A higher coherence certificate records the complete finite or finitely reduced family of associators, unitors, modifications, cocycles, and compatibility cells actually consumed by a theorem, together with the reduction theorem justifying completeness.

Proposition 7.8.5 (Finite coherence display is not automatically complete). *A finite diagram of higher cells is not a complete coherence certificate merely because it is finite. Completeness requires a coherence theorem, presentation theorem, truncation theorem, or finite-generation result covering all unshown composites used by the certificate.*

Proof. Higher compositions can generate further coherence obligations. Finiteness of the displayed fragment does not prove that no unrecorded obligation remains. $\square$

7.9 Higher comparison cells and controlled composition

Definition 7.9.1 (Higher comparison cell). A higher comparison cell is a natural transformation, 2-cell, modification, homotopy, equivalence, or higher morphism

$$\alpha : P \Rightarrow Q$$

with declared orientation, source and target composites, scope, invertibility, coherence, and cost semantics.

Definition 7.9.2 (Higher-extension comparison mode). A higher comparison has one local mode:

strict, exact-isomorphic, metric-controlled, higher-cell-controlled,
advisory-only, unavailable, failed.

These are Technical Extension tokens and do not become Core comparison modes by name alone.

Definition 7.9.3 (Core projection of higher modes). Strict equality projects to Core exact mode. Exact-isomorphic mode projects to exact only with a coherent actual isomorphism of the compared outputs. Metric-controlled mode projects to metric-gap-certified only after an actual profile bound, all threshold-gap or predicate-transfer hypotheses, and a declared ledger destination are supplied. Higher-cell-controlled mode has no automatic projection: it requires either an exact zero-reflecting coherence theorem or a certified metric conversion. Advisory-only projects to not-compared; unavailable leaves an invoked obligation undefined; failed rejects the affected path.

Proposition 7.9.4 (Invertible cell is not strict equality). *An invertible higher cell supplies coherent equivalence at its declared categorical level, not literal equality of implementations, artifacts, or records.*

Proof. Invertibility gives an inverse cell up to the ambient composition laws. Literal identity is a stronger relation. $\square$

Proposition 7.9.5 (Zero cost requires zero reflection). *The equation $\delta(\alpha) = 0$ implies exact or coherent equality only when the cost semantics are proved zero-reflecting on the declared class of cells.*

Proof. A cost function may assign zero to distinct or noninvertible cells. Zero-reflection is therefore an independent hypothesis. $\square$

Theorem 7.9.6 (Controlled higher-cell pasting). *Let $\alpha_1, \dots, \alpha_m$ be composable higher comparison cells. A bound for their paste exists only under declared vertical and horizontal aggregation laws, whiskering moduli, and the required associativity, unit, and interchange coherence. In the additive scalar case with non-expansive whiskering,*

$$\delta(\alpha_m * \dots * \alpha_1) \leq \sum_{j=1}^m \delta(\alpha_j).$$

Proof. Compose the cell witnesses and apply the declared control law at each vertical, horizontal, and whiskering step. Without those laws the pasted diagram has no determined numerical defect. $\square$

Proposition 7.9.7 (Object equivalence does not determine morphism defects). *Equivalences $X \simeq X'$ and $Y \simeq Y'$ do not determine a comparison between arbitrary morphisms $f : X \rightarrow Y$ and $f' : X' \rightarrow Y'$. A conjugacy or coherent square is required before kernels, cokernels, fibers, cofibers, or Type IV diagnostics can be transported.*

Proof. Even for $X = Y = X' = Y' = M$, the maps id_M and 0_M have identical source and target objects but different defects. $\square$

7.10 Representatives, Type IV, and transient defects

Definition 7.10.1 (Higher functorial representative system). A higher functorial representative system consists of objects $C_{\tau, W}F$, comparison morphisms for changes of input, and coherent higher cells controlling composition. Each representative remains subject to the ordinary monitored-degree compatibility

$$\mathbf{P}_i(C_{\tau, W}F) \cong \text{Eval}_{i, W, \tau}(F).$$

Proposition 7.10.2 (Higher functoriality does not strengthen admissibility). *Coherent representative choices do not make a non-admissible representative admissible and do not strengthen the equivalence level recorded by its profile compatibility.*

Proof. Admissibility is the displayed monitored-profile condition. Coherence organizes choices but does not manufacture that condition. $\square$

Definition 7.10.3 (Higher-context Type IV record). A higher-context Type IV record contains an actual heart-readable persistence morphism

$$\phi_{i, W, \tau} : M_{i, W, \tau} \longrightarrow N_{i, W, \tau},$$

the actual kernel and cokernel profiles, a declared Main-compatible terminal scope—either a bounded right-open window with the window-cofinal convention or a right-unbounded window—terminal-tameness, and any enhancement comparison needed to identify these data. A theorem based on averages over $[T, +\infty)$ or on tail reversal may impose the narrower right-unbounded scope explicitly; that extra restriction is theorem-specific rather than the definition of Type IV itself.

Proposition 7.10.4 (Higher Type IV proxy is not diagnosis). *A fiber, cofiber, mapping cone, descent obstruction, higher automorphism, spectral value, or stack-theoretic proxy does not determine*

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}})$$

unless a theorem identifies it with the actual kernel and cokernel profiles of the declared morphism under the same terminal convention.

Proof. The canonical diagnostics are defined from those actual morphism-level profiles. A proxy has a different source type until the identification theorem is supplied. $\square$

Definition 7.10.5 (Higher transient-defect record). A higher transient-defect record identifies an actual interior kernel-defect or cokernel-defect profile, its extraction from the enhancement, its own threshold and detector stage, and a complete certificate for the retained long-interior-defect predicate.

Proposition 7.10.6 (Terminal and transient conclusions remain separate). *Terminal Type IV zero does not imply absence of finite interior defect, and a long-transient zero certificate does not imply terminal cleanliness.*

Proof. The two predicates are computed on different restrictions of the actual defect profiles and have independent completeness requirements. $\square$

7.11 Limits, colimits, and finite-to-infinite reduction

Theorem 7.11.1 (Adjoint preservation boundary). *A left adjoint preserves colimits that exist in its domain, and a right adjoint preserves limits. Neither statement supplies preservation in the opposite direction, preservation of kernels or cokernels, eventual constancy, or identification with an independently declared apex.*

Proof. The first two assertions are the universal properties of adjoints. The remaining properties concern different universal constructions or independent objects and require separate hypotheses. $\square$

Theorem 7.11.2 (Equivalence preservation). *An equivalence preserves every limit and colimit that exists, up to canonical isomorphism. An anti-equivalence exchanges limits and colimits. These facts do not identify a metric limit or a descended object with a declared AK apex unless the apex comparison is supplied.*

Proof. Transport the relevant universal property through a quasi-inverse. Contravariance reverses the cones. Apex identity is extra data not contained in the universal property alone. $\square$

Definition 7.11.3 (Enhanced finite-to-infinite reduction). An enhanced finite-to-infinite reduction contains:

- (i) the finite-stage enhancement objects and actual transition morphisms;
- (ii) the extracted Core-readable stage profiles;
- (iii) exact eventual constancy or another proved finite reduction;
- (iv) compatible transport of detector, representative, edge, and defect records;
- (v) an actual stage-to-apex identification;
- (vi) every descent, limit, or colimit comparison used in constructing that identification.

Proposition 7.11.4 (Finite prefix and categorical convergence are not Return). *A finite prefix, a Cauchy estimate, a homotopy colimit, a derived limit, or effective descent does not by itself prove apex agreement or an external Return theorem.*

Proof. Each supplies at most one arrow or one ambient universal object. The declared apex and external conclusion require separately proved identification and soundness arrows. $\square$

Proposition 7.11.5 (Derived-limit obstructions must be recorded). *When a passage to inverse limits uses a derived category or a nonexact limit functor, derived-limit terms and Mittag–Leffler-type hypotheses are independent obligations. They may not be suppressed by a finite-stage stabilization plot or scalar ledger slack.*

Proof. Failure of exactness of inverse limit is measured by its derived functors. Numerical closeness of unrelated profile objects does not prove their vanishing. $\square$

7.12 Ledger, artifacts, and import governance

Definition 7.12.1 (Higher-extension defect record). A higher-extension defect record classifies a discrepancy as one of:

- (i) an actual metric profile discrepancy;
- (ii) a morphism/heart defect requiring its own profile bound;
- (iii) a status or coherence blocker;
- (iv) a provenance, serialization, or replay defect;
- (v) an advisory value.

Only the first two enter the strict Core metric budget, and only through an Appendix-M-compatible support theorem.

Definition 7.12.2 (Higher-extension ledger namespace). Local names such as

$$\mathfrak{d}_{\text{der}}, \quad \mathfrak{d}_{\text{sh}}, \quad \mathfrak{d}_{\text{desc}}, \quad \mathfrak{d}_{\text{stack}}, \quad \mathfrak{d}_{\text{higher}}$$

are gate-scoped aliases, not independent Core ledgers. Every consumed metric contribution is mapped to a canonical Appendix M slot and is bound to the actual profile pair it upper-bounds. Status, coherence, identity, and replay failures remain nonmetric blockers unless a separate theorem constructs an actual metric bound.

Proposition 7.12.3 (Higher-extension budget is path-bound). *A nonzero higher-extension metric contribution may support an audited path only when its dependency graph, actual profile path, aggregation theorem, and total strict gap comparison are all declared. A small local cost does not repair a missing extraction, morphism, descent, or apex record.*

Proof. The metric budget bounds an identified composed discrepancy. The listed missing records are non-scalar obligations outside that bound. $\square$

Definition 7.12.4 (Enhancement artifact bundle). An enhancement artifact bundle contains, as invoked, the ambient model, coefficient and variance conventions, extraction code or proof, dg/A_∞ operations, sheaf or stack objects, covers and hypercovers, comparison cells, coherence data, actual morphisms, profile identities, detector records, ledger support, environment, and source hashes.

Definition 7.12.5 (Enhancement manifest entry). An enhancement manifest entry records the enhancement class, exact scope, realization and extraction route, local comparison mode, non-scalar obligations, actual produced verifier records, metric support, dependency graph, permitted use, forbidden stronger reading, artifact references, and failure behavior.

Proposition 7.12.6 (Stale enhancement artifacts are non-consumable). *Changing the model, site, cover, coefficient object, parameter extraction, t -structure, representative, comparison cell, environment, or source dependency makes the prior enhancement artifact stale for the changed run.*

Proof. The changed data define a different mathematical or computational identity. Reuse requires a new compatibility theorem and a new content-addressed record. $\square$

Definition 7.12.7 (Admissible Core import). A Technical Extension result can enter a future Core only through a new release-level action containing its exact statement, proof, complete hypotheses, source binding, conservative-extension analysis, permitted and forbidden readings, failure routes, artifacts, and final Claim Register promotion. Appearance in Appendix N is not such an import.

7.13 Enhanced Return structures and external conclusions

Definition 7.13.1 (Enhanced Problem-Bridge chain). An enhanced Problem-Bridge chain separates the following arrows:

$$\begin{aligned} \text{external object} &\longrightarrow \text{enhancement object} \longrightarrow \text{Core-readable extraction} \longrightarrow \text{AK certificate}, \\ \text{AK certificate} &\longrightarrow \text{enhanced interpretation} \longrightarrow \text{external conclusion}. \end{aligned}$$

Every arrow records its direction, hypotheses, faithfulness or soundness strength, artifacts, and failure route.

Theorem 7.13.2 (Enhanced Return non-composition boundary). *An AK certificate together with an enhancement equivalence does not imply an external conclusion. Such a conclusion requires separate proofs of the realization from the external object to the enhancement, the extraction from the enhancement to the Core, every invoked finite-to-infinite reduction, and the Return or soundness theorem back to the external semantics.*

Proof. The composite implication exists only when every arrow in the displayed chain exists and its codomain matches the next domain. A missing or weaker arrow leaves the external conclusion unlicensed. $\square$

Proposition 7.13.3 (Problem-specific higher implications are not internal theorems). *An arithmetic, PDE, geometric, motivic, mirror-symmetric, physical, or cryptographic conclusion formulated using higher enhancements remains an Interface, Bridge Program, specification, or non-claim unless a complete problem-specific theorem chain is supplied.*

Proof. The higher enhancement does not supply the external semantics or Return arrow by itself. $\square$

7.14 Part III closure and package theorem

Theorem 7.14.1 (Part III conservative closure). *The frozen Technical Extension Appendices H–N form a conservative infrastructure layer around the v20 Core. They provide advisory calculus, discretization transfer, Betti measurement, controlled commutation, Mirror/Transfer comparison, ledger semantics, and higher-enhancement discipline. They do not weaken a Core hypothesis, convert a non-scalar obligation into metric slack, replace an actual morphism, collapse transient and terminal defects, infer a finite certificate from finite observation, or license an external conclusion without a Return theorem.*

Proof. All seven appendices retain the same re-entry, comparison, noncompensation, profile-support, and exclusion rules. Appendix N adds no exception. It therefore closes the interface from enhancements into the Core and the separate interface from Core certificates to external conclusions. $\square$

Theorem 7.14.2 (Appendix N higher-extension package). *Within the Technical Extension layer, Appendix N establishes the following package.*

- (i) *v20 higher Ext edges already in the Core are not redefined by higher-enhancement language.*
- (ii) *External enhancements affect verdicts only through immutable verifier-side re-entry.*
- (iii) *The field-coefficient higher Ext identity is separated from non-field hyper-Ext extensions.*
- (iv) *Over a general ring, $H^{-n} = 0$ alone need not imply $\text{Ext}^n = 0$, and the failure occurs in every positive degree; Ext-acyclic zero-reflecting coefficient hypotheses give a valid replacement regime.*
- (v) *Stable cofibers recover heart kernels and cokernels only through a declared t -structure and heart-readable morphism; a derived equivalence must additionally be t -exact to transport those heart data.*
- (vi) *Sheaf/cosheaf objects require a constructible one-parameter extraction before repaired evaluation.*
- (vii) *A fixed multiparameter slice is nonconservative, and finite slice families require a coverage theorem.*
- (viii) *Pairwise overlap agreement is weaker than descent, and object descent is weaker than certificate descent.*
- (ix) *Hyperdescent and finite proof compression are independent obligations.*
- (x) *Higher cells require orientation, coherence, zero-reflection or metric conversion, and controlled pasting.*
- (xi) *Object equivalence does not determine morphism defects or Type IV diagnostics.*
- (xii) *Terminal Type IV and transient-defect conclusions remain separate.*
- (xiii) *Adjoint, equivalence, limit, colimit, descent, and metric-limit results do not by themselves identify the AK apex or prove Return.*
- (xiv) *Higher-extension costs enter the Core budget only as path-bound actual profile discrepancies under Appendix M semantics.*
- (xv) *Technical Extension results enter a future Core only through explicit release-level import and final registration.*
- (xvi) *Part III closes as conservative non-Core infrastructure.*

Proof. Items (i)–(iv) follow from the v20 synchronization and hyper-Ext sections. Items (v)–(ix) follow from the stable, sheaf, multiparameter, descent, and stack sections. Items (x)–(xii) follow from the higher-cell, representative, and defect sections. Items (xiii)–(xv) follow from the finite-to-infinite, ledger, and import sections. Item (xvi) is Theorem 7.14.1. $\square$

7.15 Explicit exclusions

Appendix N asserts none of the following.

- No derived, dg, A_∞ , sheaf, cosheaf, stack, higher-stack, stable, or multiparameter enhancement is Core by default.
- No higher-enhancement statement redefines the v20 standard degree- n eligibility or edge calculus.
- No field-coefficient Ext identity is extended to arbitrary rings or coefficient objects without the displayed hyper-Ext hypotheses.
- No module-coefficient hyper-Ext E_2 page is silently reused for an arbitrary coefficient complex.
- No spectral sequence is assumed to degenerate by existence alone.
- No stable fiber or cofiber is silently identified with a heart kernel or cokernel.
- No derived equivalence is assumed t -exact or heart-preserving by equivalence alone.
- No quasi-equivalence identifies filtrations, implementations, numerical environments, or artifacts by default.
- No sheaf, cosheaf, or stack object is Core-readable without a typed one-parameter extraction.
- No one-parameter slice or finite collection of slices is a complete multiparameter invariant without a coverage theorem.
- No pairwise overlap agreement is treated as higher descent.
- No object descent, stack descent, or hyperdescent is treated as certificate descent without a verifier-record descent theorem.
- No finite display of higher cells is assumed complete without a finite coherence reduction theorem.
- No zero-cost cell is exact without zero-reflecting cost semantics.
- No object equivalence or object distance controls morphism defects without an actual comparison square.
- No higher Type IV proxy replaces the actual comparison morphism, kernel/cokernel profiles, or canonical (μ, ν) diagnostics.
- No terminal-zero verdict is read as absence of transient defect, and no transient-zero verdict is read as terminal cleanliness.
- No categorical limit, colimit, metric limit, or effective descent is identified with the declared apex without an apex agreement theorem.
- No status, coherence, identity, schema, or replay defect is repaired by numerical metric slack.
- No enhancement artifact, manifest, build, or registration proves mathematical truth by itself.
- No AK certificate plus conceptual equivalence proves an external theorem without the complete Bridge and Return chain.
- No Technical Extension result enters the Core merely by appearing in Appendix N.

7.16 Provenance and final status

Appendix N depends on the v20 finite-certificate, higher-edge, transient-defect, finite-to-infinite, and Return disciplines of Main Chapters 1–8 and Foundation Appendices A–G; on the reviewed CM and TB mathematical statements at their individually verified strengths, without inferring joint artifact synchronization or

promotion; and on the frozen Technical Extension Appendices H–M. Its role is to close Part III without status inflation and to expose the precise obligations that any later Problem Interface must discharge.

Status classification. The principles and governance definitions are operational Technical Extension rules. The displayed algebraic, stable, metric, and categorical statements are proved at their stated hypotheses. Conditional extraction, descent, and transport statements remain conditional on every displayed interface hypothesis. Problem-specific conclusions and unproved realization/Return arrows remain outside theorem evidence.

Release closure. The Main-cross-audited integrated Part III TeX source, H–N integration body, PDF projection, audit report, correction patch, and checksum sidecar are the authoritative Technical Extension release binding. Earlier per-appendix packages remain immutable repair ancestry and are not used to override the integrated text. Formal CR promotion is deferred to the final v20 closure pass after all downstream appendices and the Companion are fixed.